People's Democratic Republic of Algeria
Ministry of Higher Education and Scientific Research
BADJI Mokhtar University Annaba
Faculty of Medicine – Skikda Annex
Anatomy Laboratory

Thoracic limb bones

Prof. A. ABDALLAH
Prof. S. BOUKOFFA
General Anatomy Laboratory
Faculty of Medicine – Annaba
Email: abdeldjalil.abdallah@univ-annaba.dz

Course outline

- **Introduction**
- **Clavicle**
- **Scapula**
- **Humerus**
- **Forearm skeleton: Ulna, Radius**
- **Hand skeleton**

Introduction

The skeleton of the thoracic limb includes the following paired bones: the Humerus (arm skeleton), the Radius and the ulna (forearm skeleton), and the Hand skeleton.
This skeleton is attached to the thorax by the shoulder girdle, formed of two paired bones, the scapula and the clavicle.

Clavicle

It is a long bone, elongated in an extended "S" shape, extending between the Scapula and the Sternum.

Set up :

- At the bottom: the face with a groove running through it;
- Outside: the flattened end;
- Forward: the concave border of this end.

Body :

Present to be described: 2 faces and 2 borders.

1. **Superior surface:**
   It is subcutaneous and palpable.
2. **Inferior surface :**
   Present:
   - In the middle, the groove for the subclavius muscle, which gives insertion to the subclavius muscle;
   - Internally, the costal tuberosity, which provides attachment for the costoclavicular ligament;
   - Outside, the trapezoid line and the conoid tubercle, which give attachment, respectively, to the trapezoid and conoid ligaments.
3. **Anterior border:**
   It is concave in its lateral 1/3, convex in its medial 2/3.
4. **Posterior border:**

It is convex in its lateral 1/3, concave in its medial 2/3.

Extremities:

1. **Acromial end:**
   It is lateral and flattened, it bears the acromial articular surface, and articulates with the acromion of the scapula.
2. **Sternal end:**
   It is medial and rounded, it bears the sternal articular surface, and articulates with the sternum and the first costal cartilage.

Scapula

It is a flat, triangular, dorsal bone, lying against the rib cage. The scapula projects onto the 2nd rib above and the 7th rib below. It articulates with the humerus and the clavicle.

Set up :

- Ahead: the concave face of the bone;
- Outside: the oval articular surface;
- At the top: the notched border.

Faces:

1. **Anterior (costal) surface:**
   It is excavated, forming the subscapular fossa, traversed by 2 or 3 oblique ridges, and gives insertion to the subscapularis muscle.
2. **Posterior (dorsal) surface:**
   It is divided by the scapular spine into two fields, the supraspinous fossa and the infraspinous fossa.
   - Scapular spine: it is a triangular bony plate, perpendicular on the posterior face, and extends outwards into the Acromion, the latter bearing the clavicular articular surface which articulates with the clavicle.
   - Supraspinous fossa: provides insertion for the supraspinatus muscle.
   - Infraspinous fossa: provides insertion for the infraspinatus, teres minor and teres major muscles.

Borders:

- Upper border (cervical): presents the scapular notch, which is surmounted by the transverse ligament of the scapula, and where the suprascapular artery and nerve pass.
- Medial (spinal) border.
- Lateral (axillary) border.

Angles:

- Upper angle.
- Lower angle (tip of the scapula).
- Lateral angle, presents three training programs:
  - *Coracoid process*: it has a shape like a half-flexed finger, it gives insertion to the biceps brachii (short head), coracobrachialis and pectoralis minor muscles.
  - *Glenoid cavity*: the glenoid cavity is an oval-shaped articular surface, notched anteriorly by the glenoid notch, and has the glenoid tubercle in the center. The cavity articulates with the humeral head.
  - *Neck of the scapula*: it is circular and narrowed, it supports the glenoid cavity. The neck has two tubercles, the supraglenoid tubercle, which gives insertion to the biceps brachii muscle (long head), and the infraglenoid tubercle, which gives insertion to the triceps brachii muscle (long head).

Humerus

It is a long bone, which alone forms the skeleton of the arm; it articulates with the scapula at the top and the bones of the forearm at the bottom. It is made up of a diaphysis and two epiphyses.

Set up :

- Up :L'rounded end of the bone;
- Inside:therearticular surface of this extremity;
- Ahead :THEgroove which runs from top to bottom of this extremity.

Diaphysis:

It is cylindrical at the top and triangular and flattened at the bottom. It has three faces and three borders to describe.

1. **Anteromedial surface**: presents the crest of the lesser tubercle which gives insertion to the latissimus dorsi muscles.
2. **Anterolateral view**: presents the deltoid tuberosity (the deltoid "V"), where the deltoid muscle inserts.
3. **Back view**: is crossed by the radial nerve groove, through which the radial nerve passes. It gives insertion to the triceps brachii muscle, the lateral head above the groove, and the medial head below the groove.
4. **Borders**:
   - The anterior one corresponds to the crest of the major tubercle, and is where the major pectoral muscle inserts.
   - The medial
   - The full-back

Proximal epiphysis:

It contains six anatomical structures.

- **Humeral head**: it represents 1/3 of a sphere, with a radius of 30 mm. The head articulates with the glenoid cavity of the scapula.
- **Major tubercle**: is a large lateral protrusion.
- **Minor tuber**: is a small anterior projection.
- **Intertubercular groove**: it separates the two tubercles, major and minor; covered by the transverse humeral ligament, and provides passage for the tendon of the biceps brachii muscle (long head).
- **Anatomical neck**: it separates the humeral head from the two tubercles.
- **Surgical neck**: it separates the proximal epiphysis from the diaphysis, and it is vulnerable to fractures.

Distal epiphysis:

It is flattened, and has three anatomical structures.

1. **Humeral condyle**: it is a joint projection that comprises three articular surfaces:
   - *Chapter*: rounded lateral projection, not visible from the back, surmounted by the radial fossa. It is articulated with the radial head.
   - *Trochlea*: located internally, it is a pulley formed by two cheeks separated by the groove. It is surmounted by two fossae, the coronoid fossa anteriorly and the olecranon fossa posteriorly. It articulates with the trochlear notch of the ulna.
   - *Capitular-trochlear zone* (conoid zone): located between the capitulum and the trochlea.
2. **Medial epicondyle**: it is a very developed and palpable protrusion; it provides attachment for the ulnar collateral ligament of the elbow. It is crossed posteriorly by the groove for the ulnar nerve.

3. **Lateral epicondyle**: it is a poorly developed projection, it provides attachment for the radial collateral ligament of the elbow.

Forearm skeleton

It comprises two long bones, the Ulna and the Radius. These two bones are joined by the proximal radioulnar joint, the distal radioulnar joint and the interosseous forearm membrane.

(1) Ulna

It is the medial bone of the forearm; it articulates with the humeral trochlea above, the radius laterally, and the radioulnar articular disc below. It is formed of a diaphysis and two epiphyses.

Set up :

- Up :L'the largest end;
- Ahead :therehook-shaped articular surface of this end;
- Outside:theresmall articular facet of this extremity.

Diaphysis:

Present to describe three faces and three borders.

- **Faces**: anterior, medial and posterior.
- **Borders**: anterior, posterior and lateral, the latter is called interosseous, where the antebrachial interosseous membrane attaches.

Proximal epiphysis:

It presents two processes, the olecranon and the coronoid process, which together form a "hook" that carries the trochlear notch of the ulna, articular with the humeral trochlea.

1. **Olecranon**: it is a vertical and posterior, subcutaneous process, carries the upper part of the trochlear notch, and gives insertion to the triceps brachii muscle.
2. **Coronoid process**: it is a horizontal and prior process; it presents:
   - Above, the lower part of the trochlear notch;
   - Outside, the radial notch of the ulna, articular with the radial articular circumference;
   - In front, the ulnar tuberosity, where the brachialis muscle inserts.

Distal epiphysis:

It presents, to be described, the ulnar head and the ulnar styloid process.

1. Ulnar head: articulates with the ulnar notch of the radius and the radioulnar articular disc of the wrist.
2. Ulnar styloid process: it is a small posteromedial projection, it gives attachment to the radioulnar articular disc of the wrist and to the ulnar collateral ligament of the carpus.

(2) Radius

It is the lateral bone of the forearm; it articulates with the capitulum above, the ulna medially, and the carpus below. It is formed of a diaphysis and two epiphyses.

Set up :

- Inside: the sharpest border of the bone;
- At the bottom: the largest end;
- Backwards: thereface of this end, which is furrowed.

Diaphysis:

Present to be described, 3 faces and 3 borders.

- **Faces**: anterior, lateral and posterior.
- **Borders**: anterior, posterior and medial, the latter is called interosseous, where the antebrachial interosseous membrane attaches.

Proximal epiphysis:

It is cylindrical and has a radial head, radial neck and radial tuberosity.

- **Radial head**, is made up of:
  - **Radial articular fossa**: superior and concave, articulates with the humeral capitulum and the capitulo-trochlear zone.
  - **Radial joint circumference**: peripheral, articulates with the radial notch of the ulna and the annular ligament.
- **Radial neck**;
- **Radial tuberosity**: ovoid protrusion, where the biceps brachii muscle ends.

Distal epiphysis:

It is a quadrangular pyramid with a bottom base and 4 faces.

- **Side view**: extends downwards by the radial styloid process, the latter being delimited by two grooves through which three muscle tendons pass;
- **Back view**: hollowed out with two grooves, through which seven (7) muscle tendons pass;
- **Medial face**: carries the ulnar notch of the radius which articulates with the ulnar head;
- **Base**: carries the carpal articular surface, which articulates with the radial articular surface of the carpus.

Hand skeleton

It is made up of three parts: the Carpus which corresponds to the wrist, the Metacarpus which forms the skeleton of the palm, and the Phalanges which form the skeleton of the fingers.

Carpus:

It is made up of eight (8) bones, arranged in 2 rows:

- Proximal row: formed from outside in, of the Scaphoid, Lunate, Triquetrum and Pisiform bones;
- Distal row: formed from outside in, of the Trapezium, Trapezoid, Capitate and Hamate bones.

Note: Carpal Tunnel

The anterior face of the carpus is concave, forming the carpal groove, which is completed by the flexor retinaculum forming the carpal tunnel, which provides passage for the median nerve and muscle tendons.

Metacarpus:

It is composed of five (5) metacarpal bones, numbered from "I" to "V", starting from the outside in.

Phalanges:

There are fourteen (14) of them, three phalanges per finger and two phalanges of the thumb. Numbered as follows:

- For the thumb: proximal phalanx (P1) and distal phalanx (P2);
- For the last four fingers: proximal phalanx (P1), intermediate phalanx (P2) and distal phalanx (P3).

Each phalanx is a long bone, having a body, a proximal end, called the base, and a distal end, called the head.

Three points to note:

- **Sesamoid bones**: these are small bones located on the palmar surface of the digital joints.
- **Bone development**: this is done by membranous or cartilaginous ossification. After birth, the bone undergoes two types of growth, in width (at the level of the periosteum) and in length (at the level of the growth plate, also called the conjugate plate).
- **Bone age** (in children and adolescents): It is assessed on an X-ray of the left wrist and hand. It is related to the ossification of the carpal bones and the epiphyses of the long bones of the wrist and hand.



Pr. A. ABDALLAH
Pr. S. BOUKOFFA
Laboratoire d'Anatomie Générale
Faculté de Médecine – Annaba
Email: abdeldjalil.abdallah@univ-annaba.dz

Os du membre thoracique

Tout **un membre thoracique** pour **une main** qui...

... **prend** et **manipule**



Plan du cours

- Introduction
- Clavicule
- Scapula
- Humérus
- Squelette de l'avant-bras: Ulna, Radius
- Os de la main

Introduction

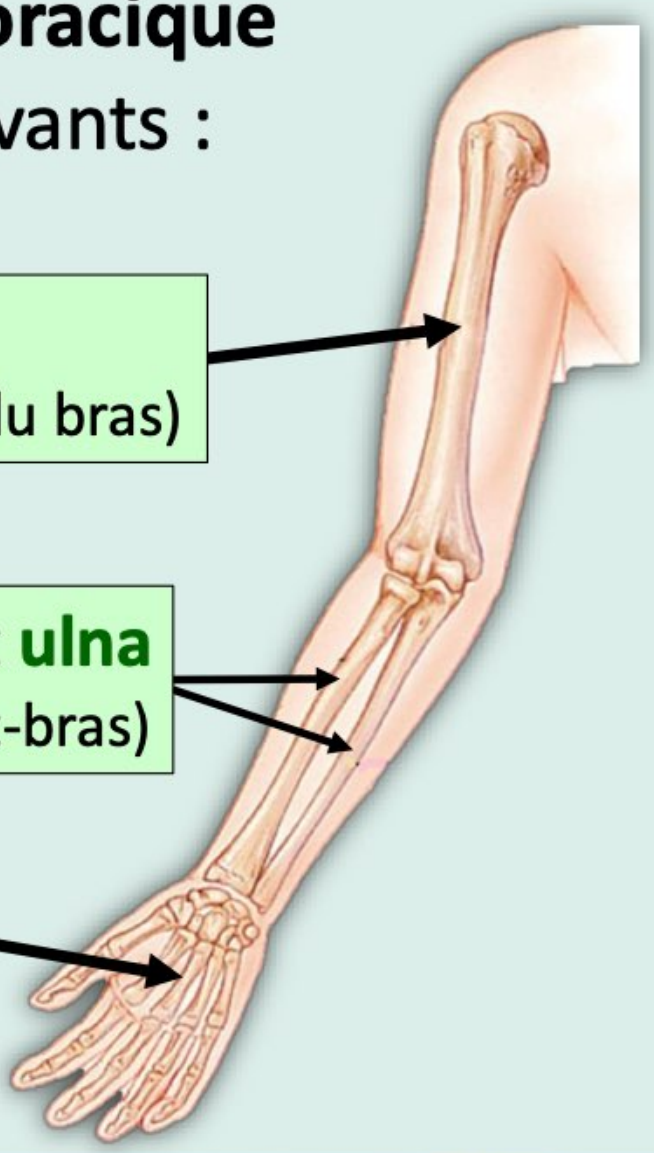
Squelette du membre thoracique

comprend **les os pairs** suivants :

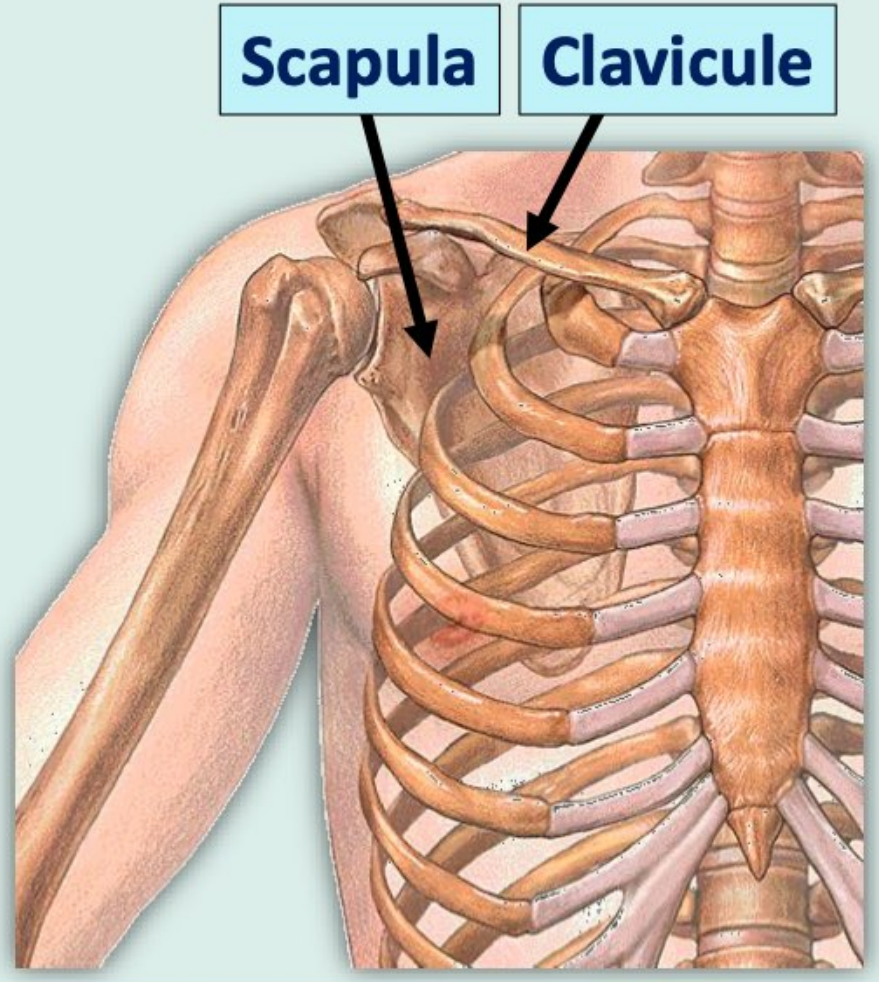
Humérus
(squelette du bras)

Radius et **ulna**
(squelette de l'avant-bras)

Squelette de la main

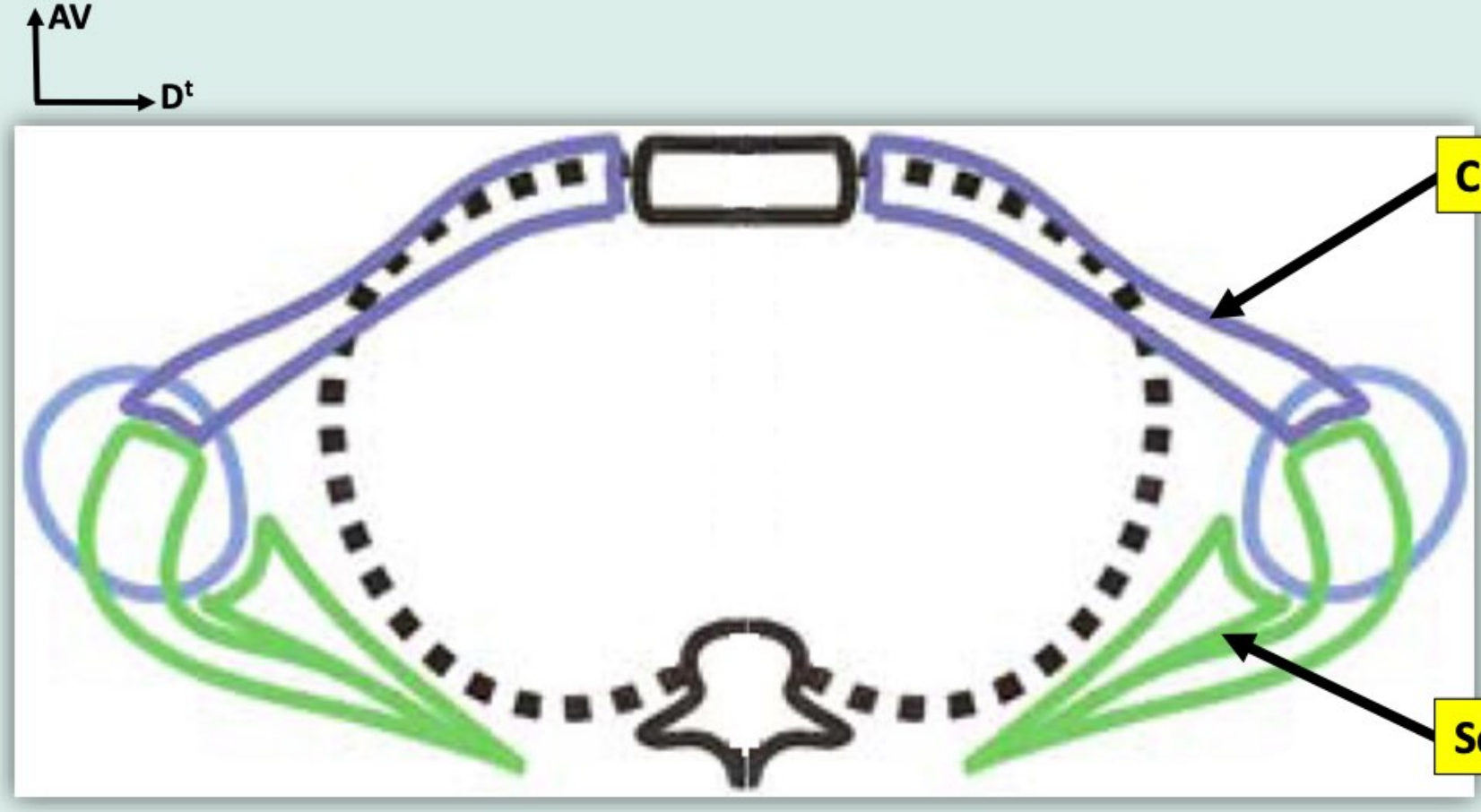


- Ce **squelette** est attaché au **thorax** par: **ceinture scapulaire**
- Formée de **2 os pairs** :



Ceinture scapulaire

vue de haut

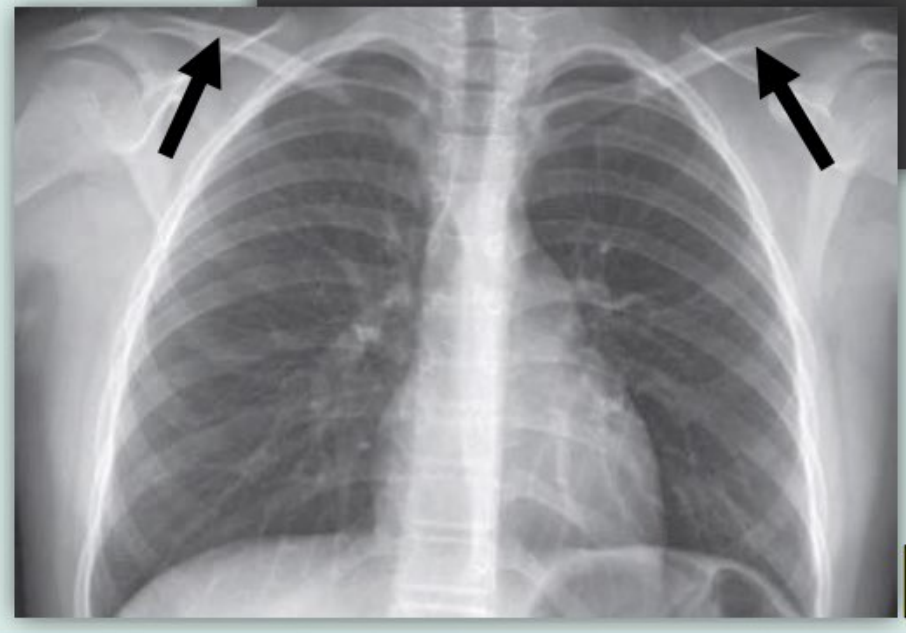


Clavicule

Os long

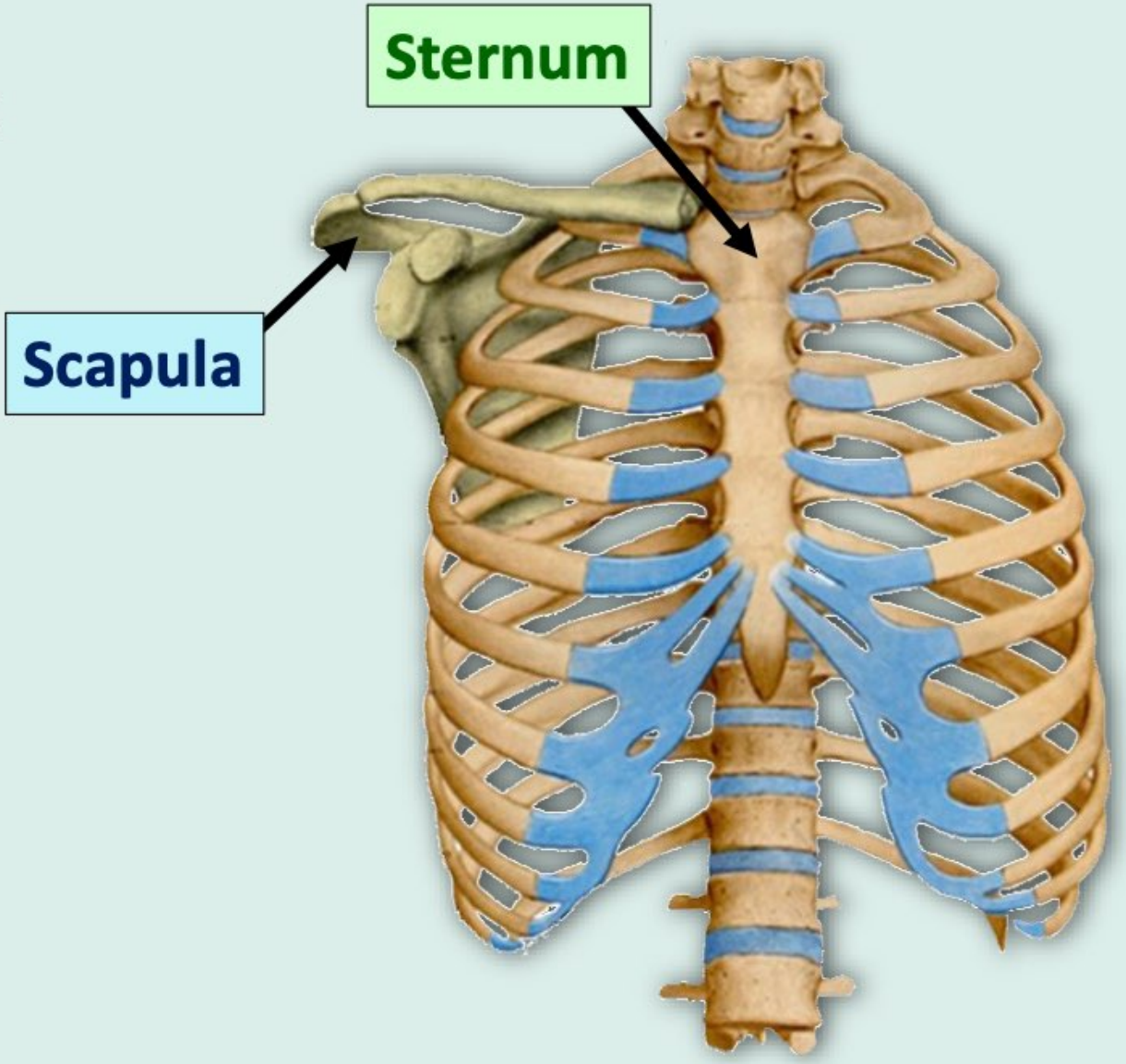
Allongé en « S » étiré





Téléthorax

S’étend entre:



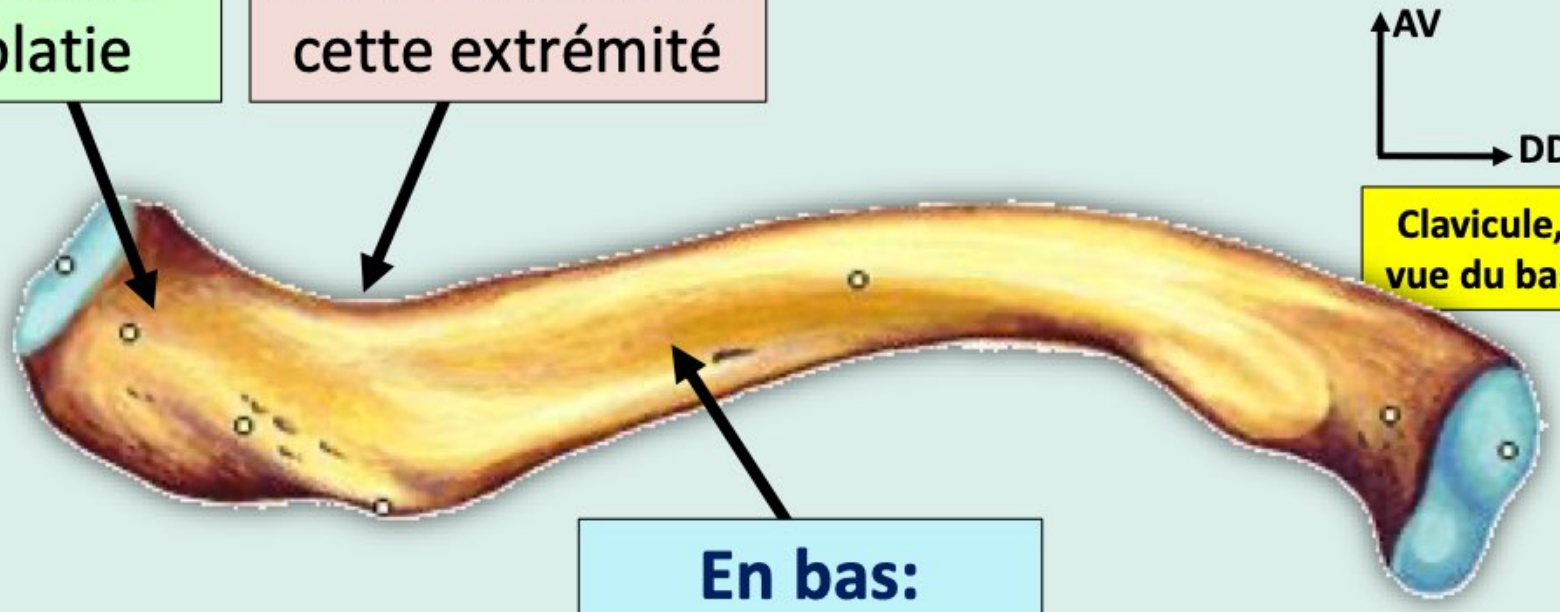
Mise en place

Corps

2 Extrémités

En dehors: extrémité aplatie

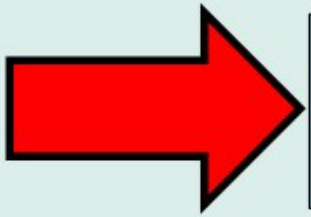
En avant: bord concave de cette extrémité



Clavicule, vue du bas

En bas: face parcourue d'un sillon

Mise en place



Corps

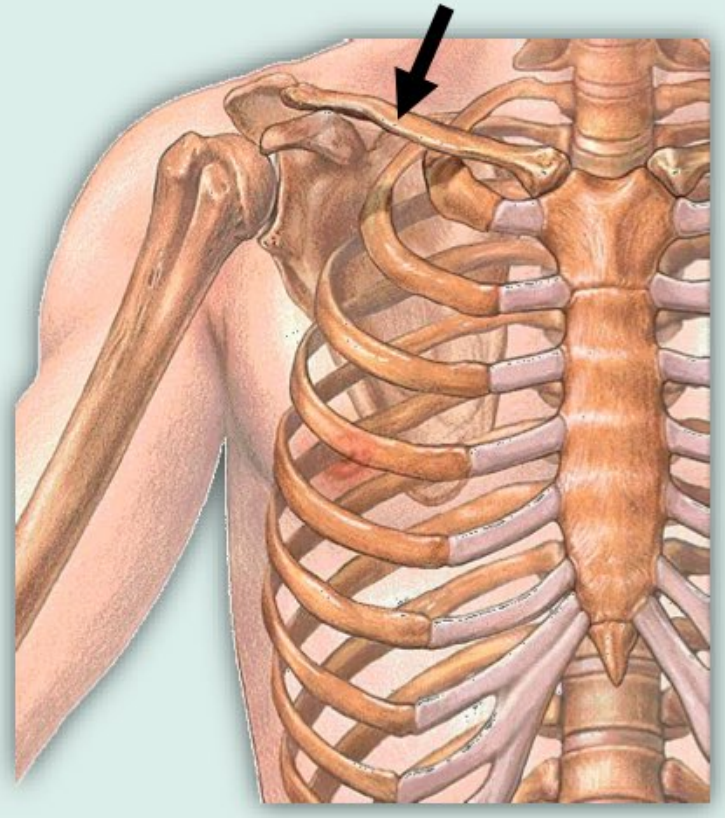
2 Extrémités

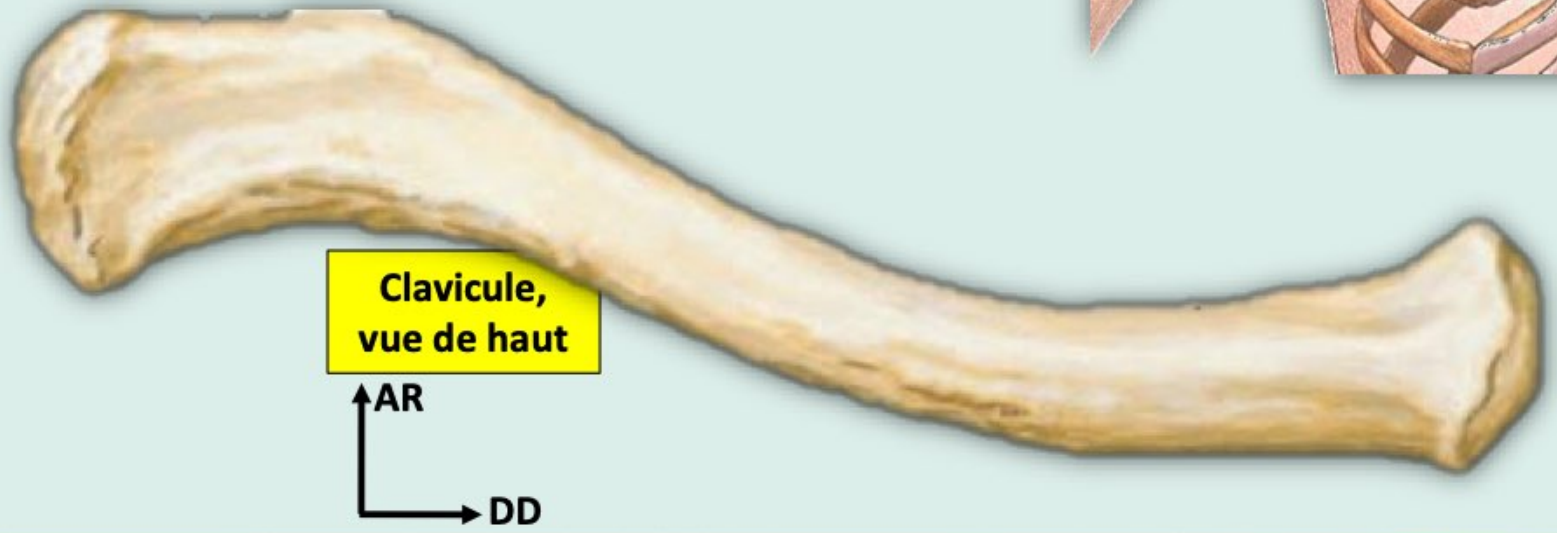
Présente à décrire:

2 faces et **2 bords**

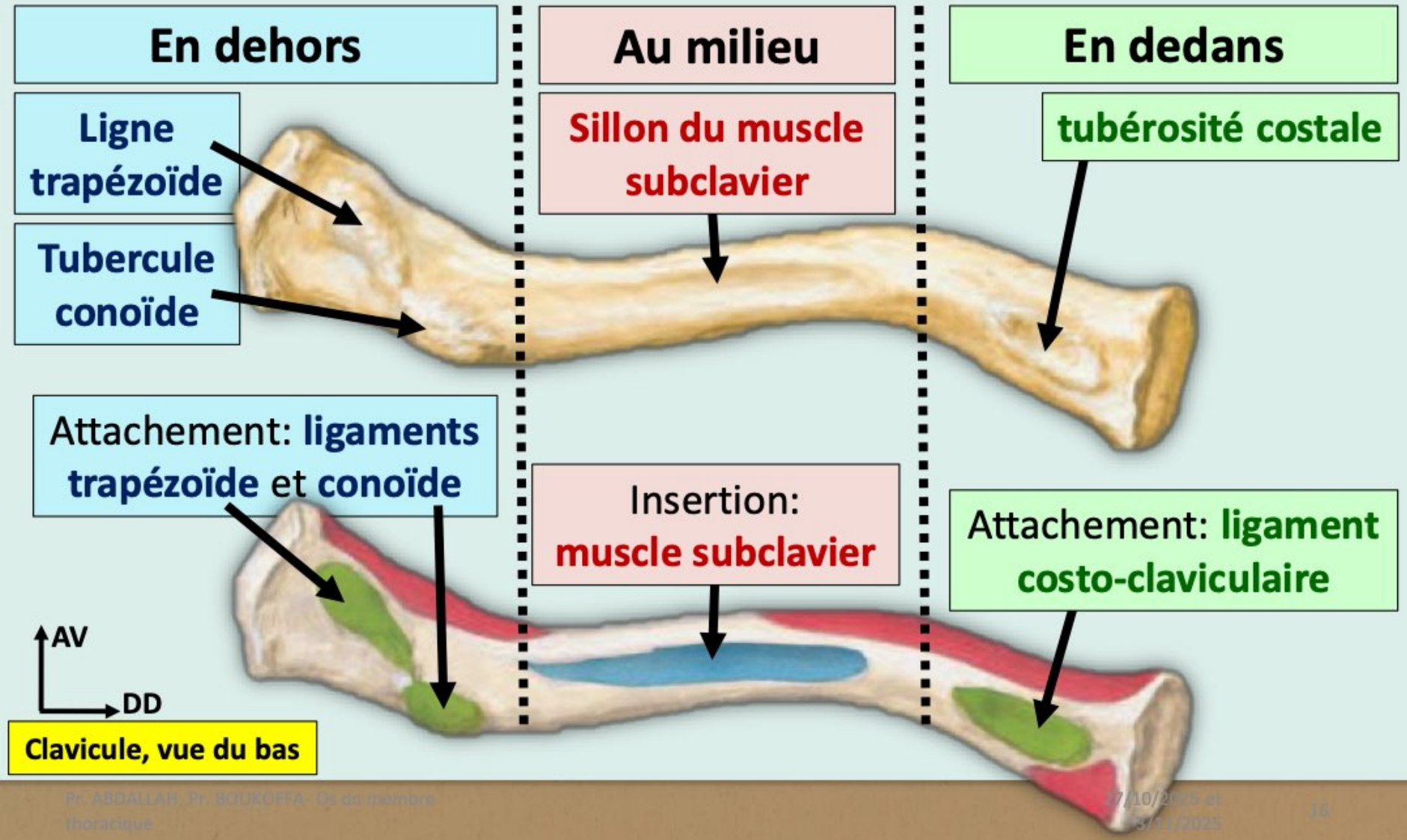
Face supérieure

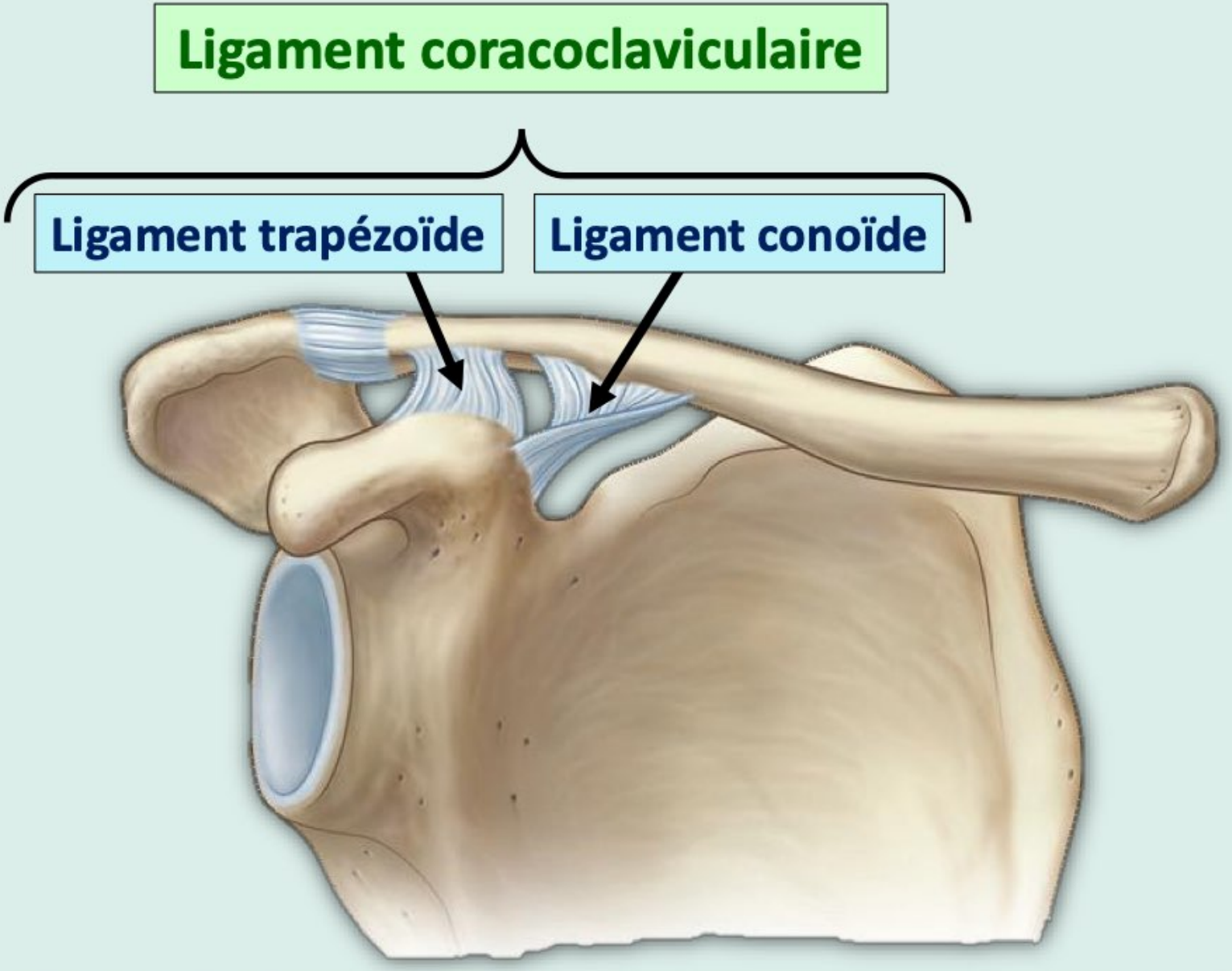
- **Sous-cutanée** et **palpable**





Face inférieure




Ligament coracoclaviculaire
Ligament trapézoïde
Ligament conoïde

Bord postérieur

1/3 latéral : **convexe**

2/3 médiaux: **concave**



Clavicule, vue de haut

1/3 latéral: **concave**

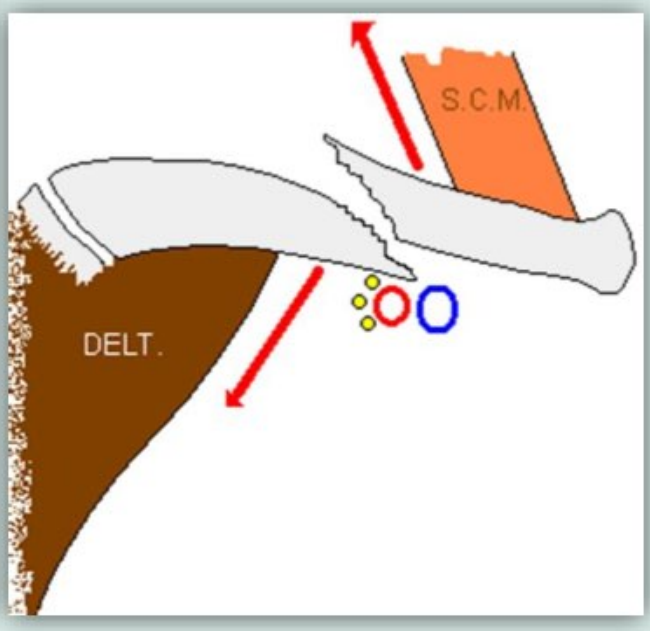
2/3 médiaux: **convexe**

Bord antérieur

Fracture de la clavicule









Bandage en 8

Mise en place

Corps

2 Extrémités

Extrémité acromiale

Latérale et **aplatie**
Porte:
surface articulaire acromiale



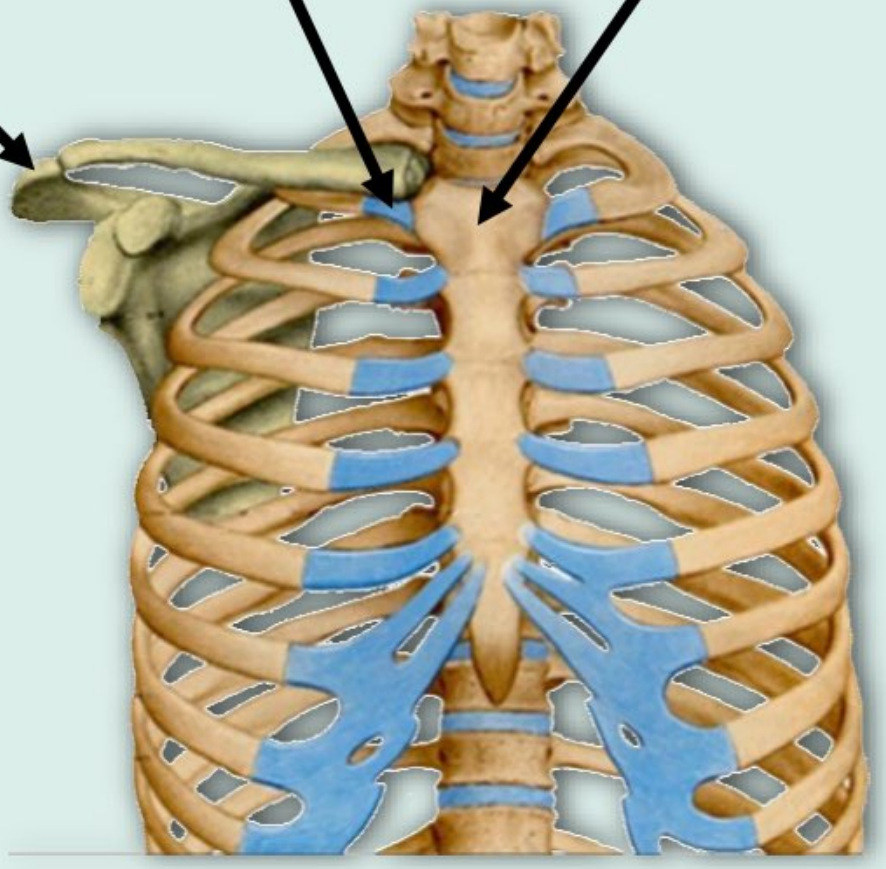
Clavicule, vue du bas

Médiale et **arrondie**
Porte:
surface articulaire sternale

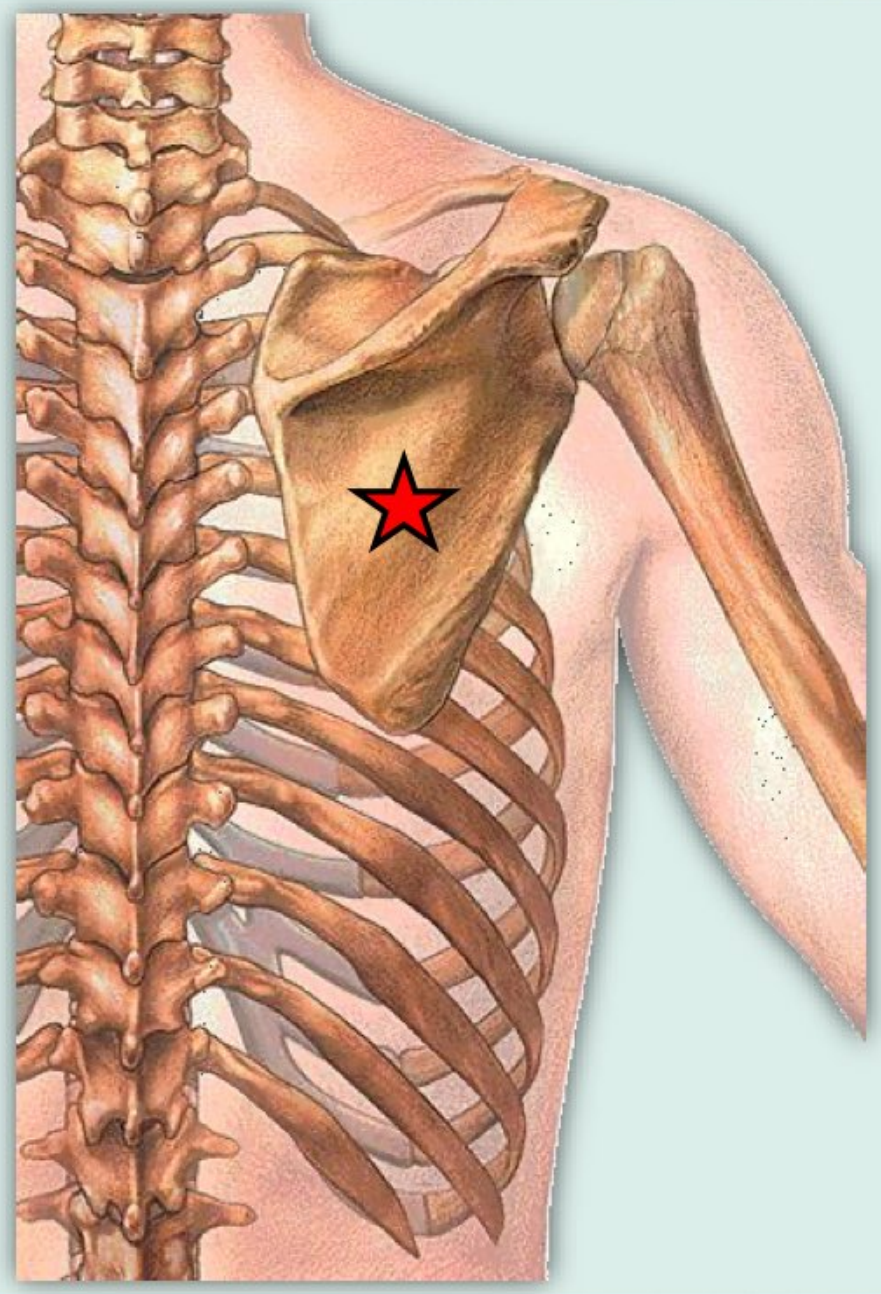
Extrémité sternale

Extrémité acromiale s'articule avec: **acromion** (scapula)

Extrémité sternale s'articule avec: **1er cartilage costal** et **sternum**

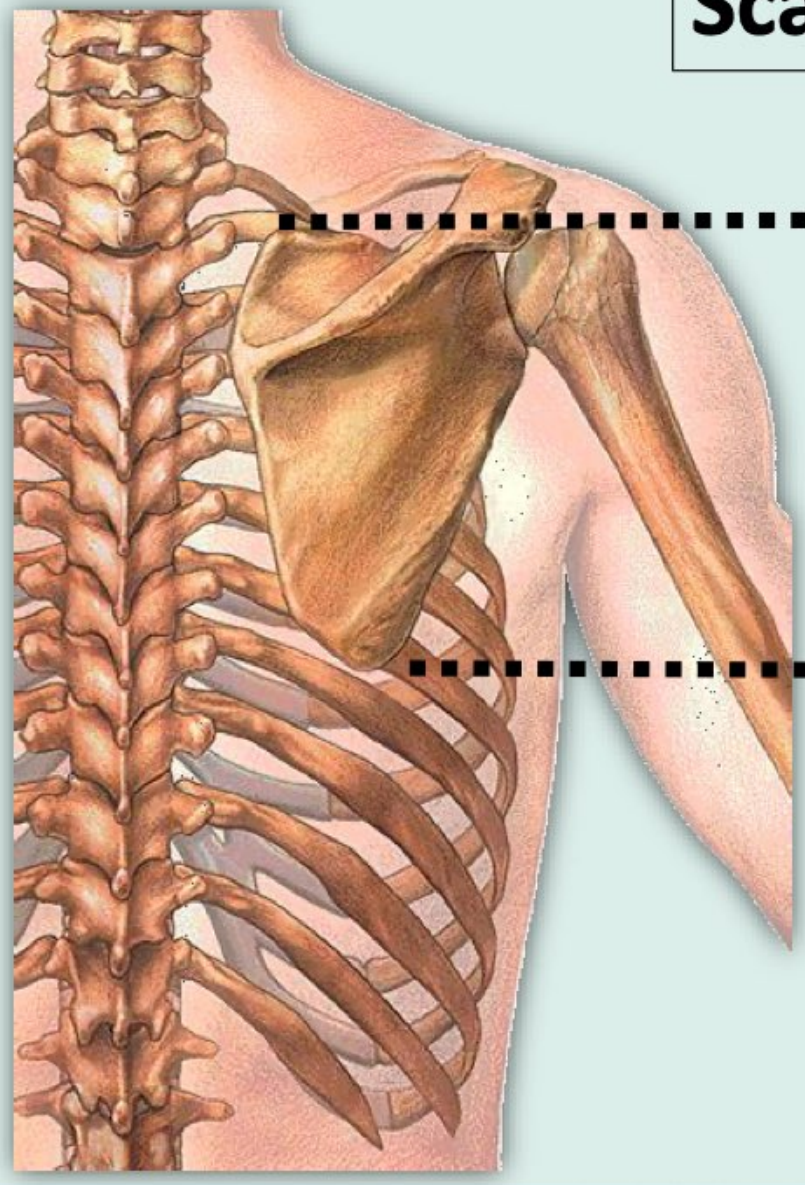


Scapula



- **Os plat, triangulaire** et **dorsal**
- Plaqué contre **la cage thoracique**

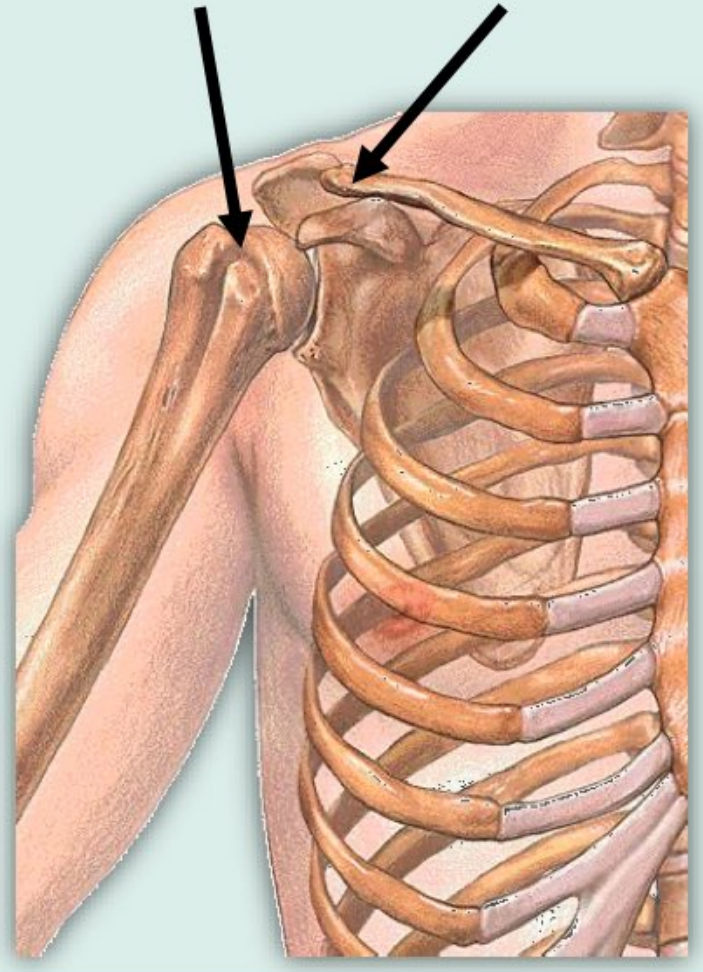
Scapula se projette entre:

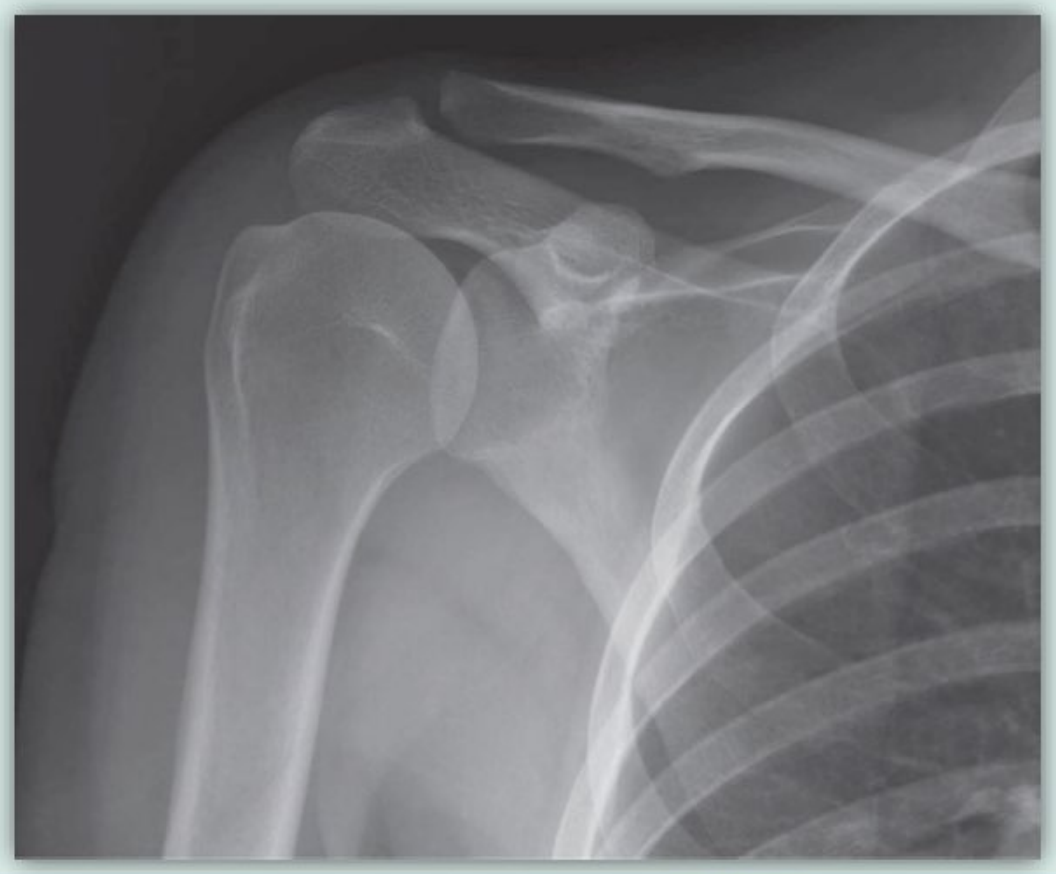


2ème côte

7ème côte

- S’articule avec:
  humérus et **clavicule**





Mise en place

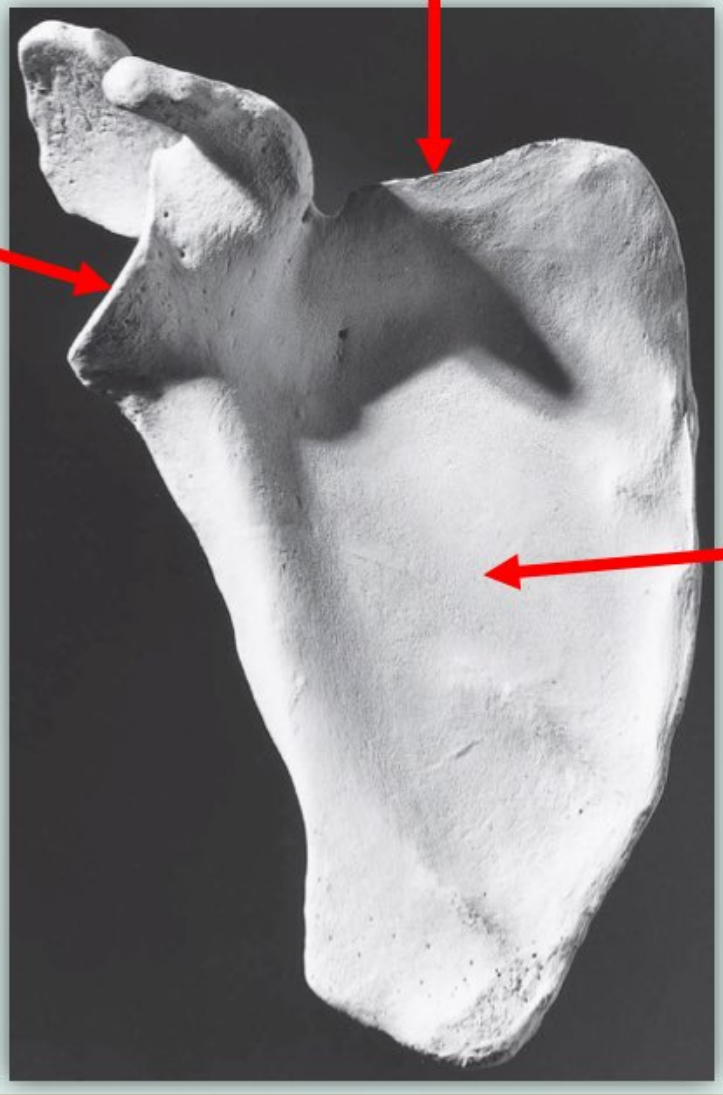
2 Faces

3 Bords

3 Angles

En haut: **bord échancré**

En dehors: **surface articulaire ovalaire**



En avant: **face concave**

H^t

DD

Scapula droite, vue antérieure

Mise en place

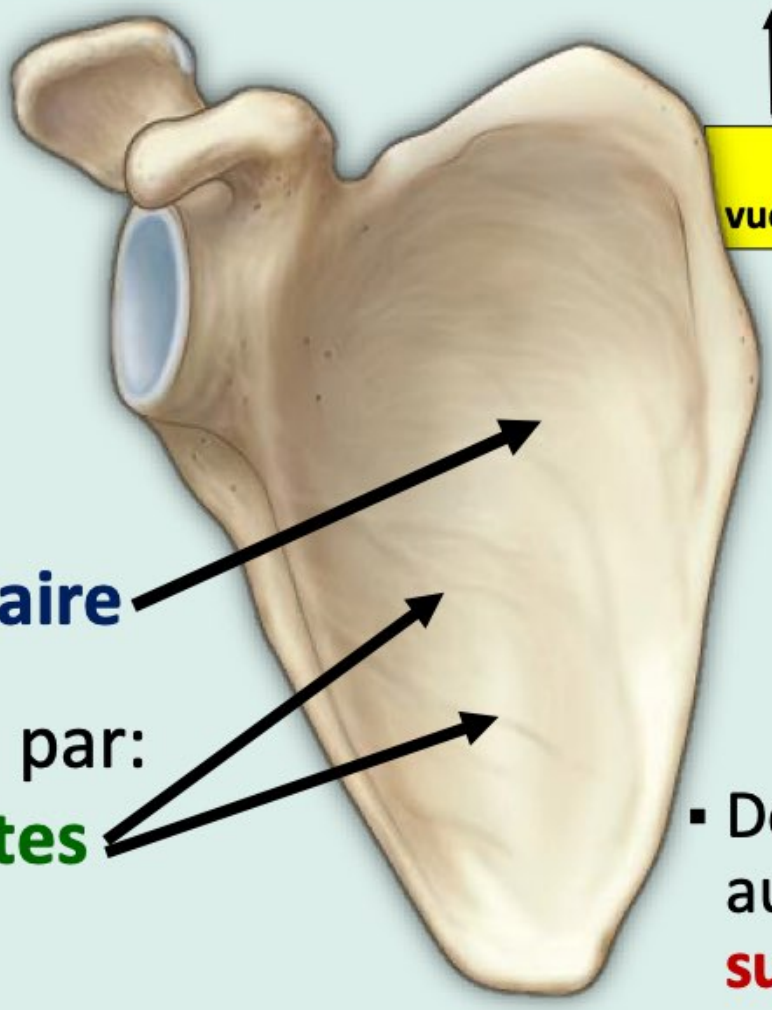
2 Faces

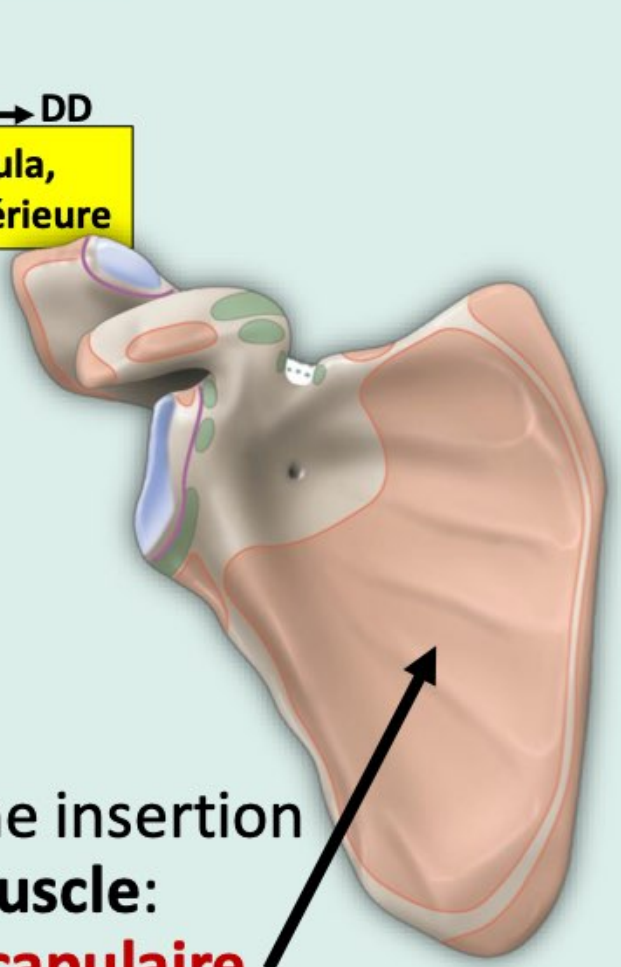
3 Bords

3 Angles

Face antérieure (costale)

- Excavée
- Forme: **fosse subscapulaire**
- Parcourue par: **2** ou **3 crêtes obliques**





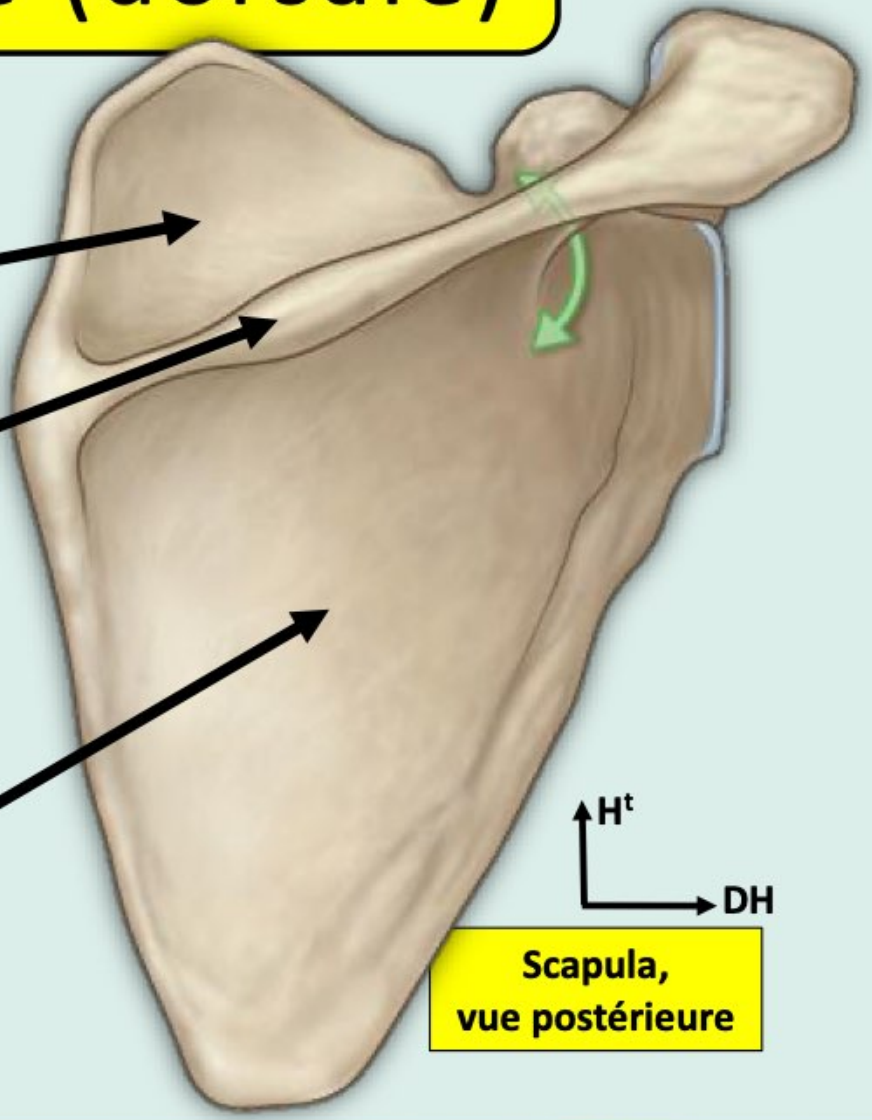
- Donne insertion au **muscle**: **subscapulaire**

Face postérieure (dorsale)

Fosse supra-épineuse

Divisée par:
épine scapulaire
en **2 champs:**

Fosse infra-épineuse



Scapula, vue postérieure

Épine scapulaire

= lame osseuse

Triangulaire



Perpendiculaire
sur face postérieure



Se prolonge **en dehors**
par: **acromion**



Acromion porte:
surface articulaire claviculaire

qui s'articule avec:
clavicule





Donne insertion au muscle:
supra-épineux

Fosse supra-épineuse

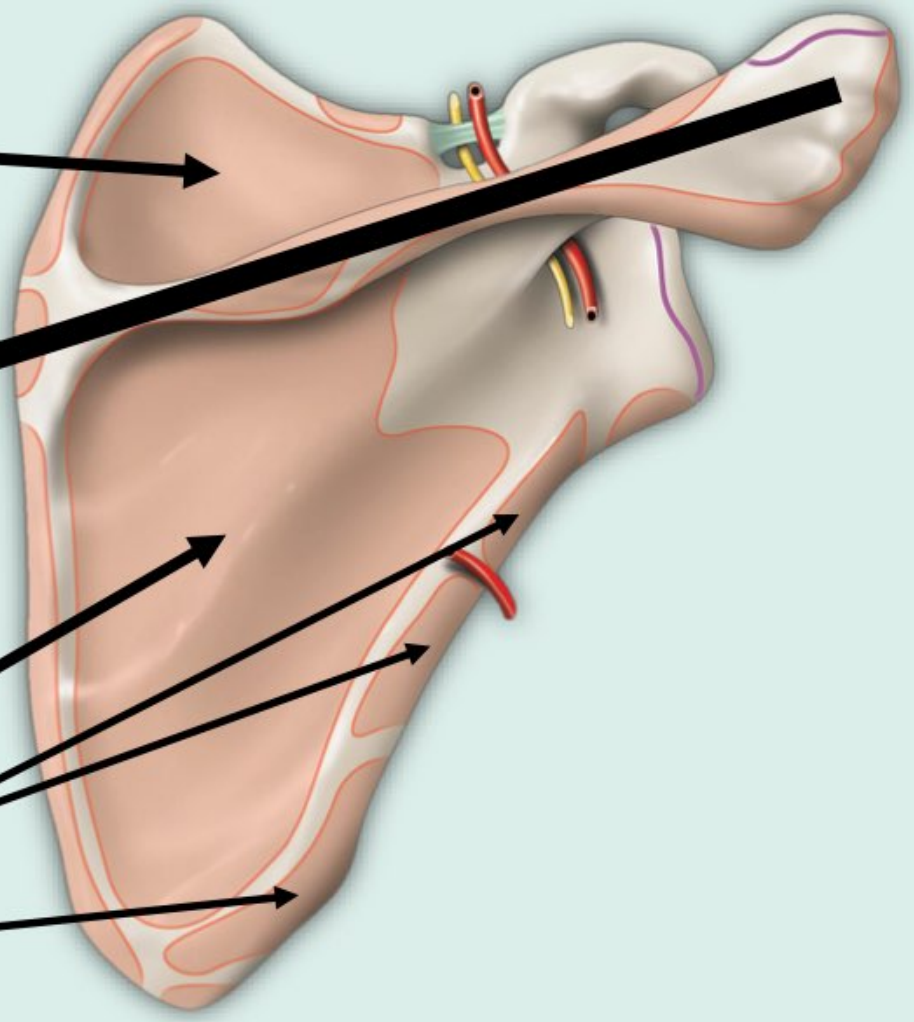
Fosse infra-épineuse

Donne insertion aux **muscles**:

infra-épineux

petit rond

grand rond

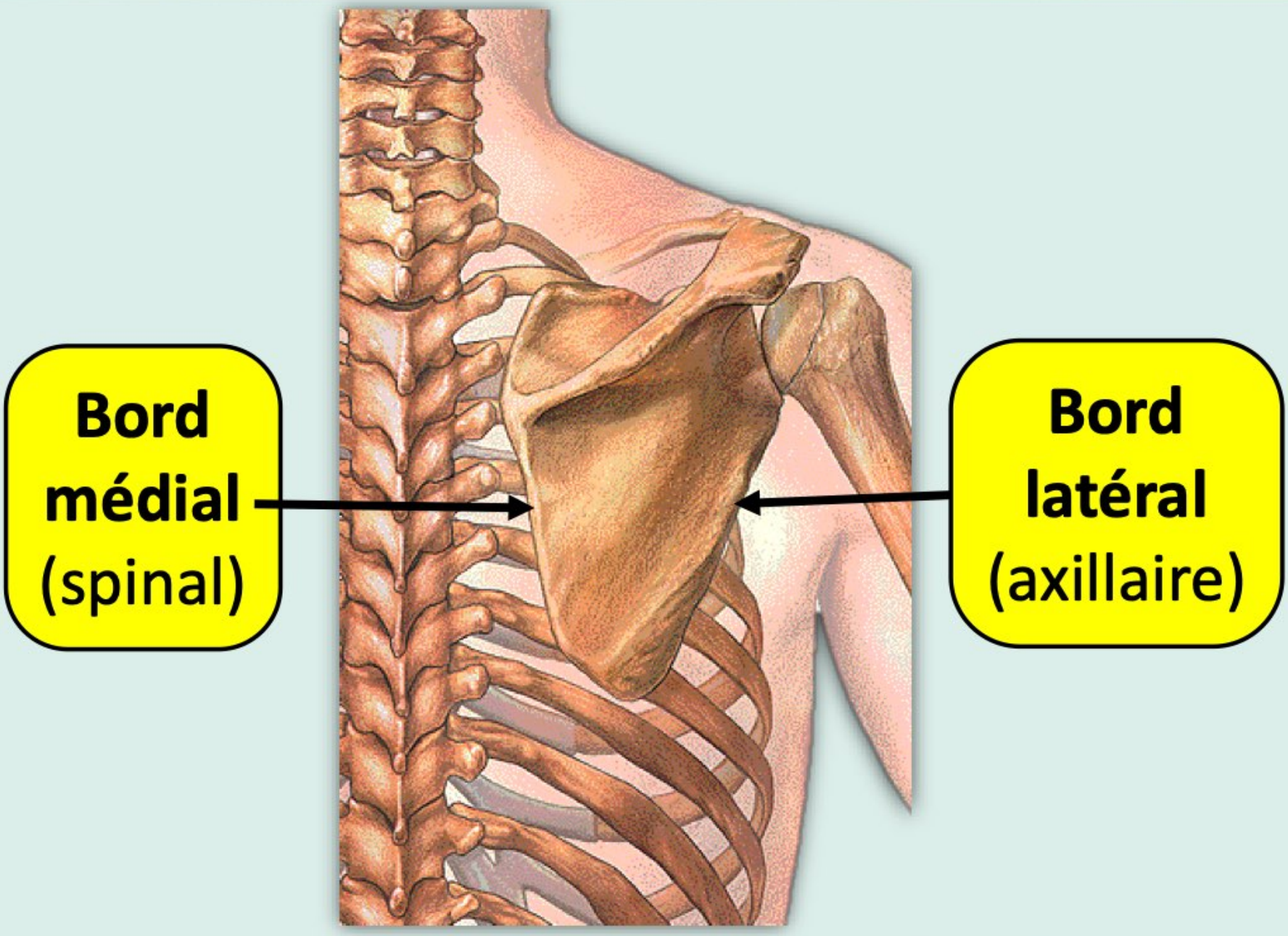


Mise en place

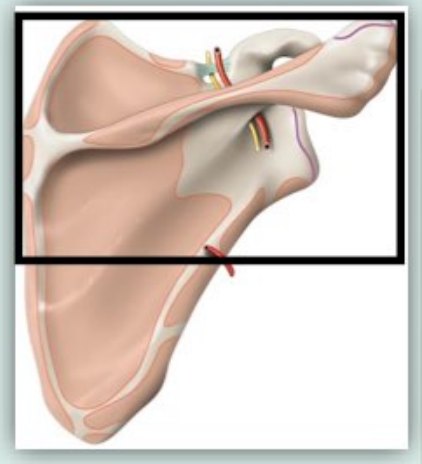
2 Faces

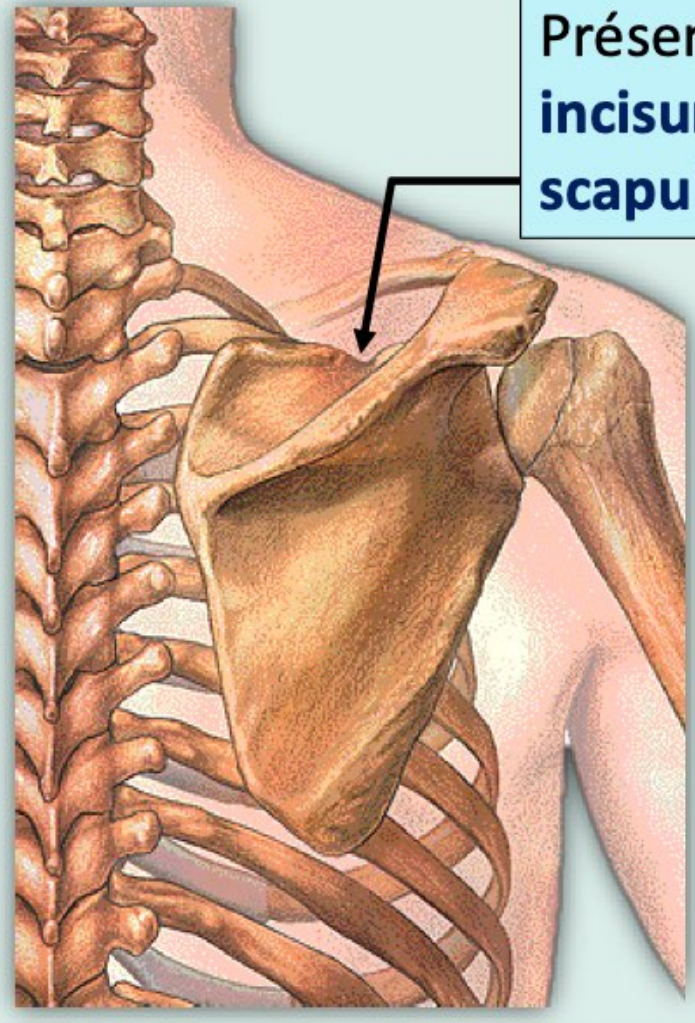
3 Bords

3 Angles


Bord médial (spinal)
Bord latéral (axillaire)

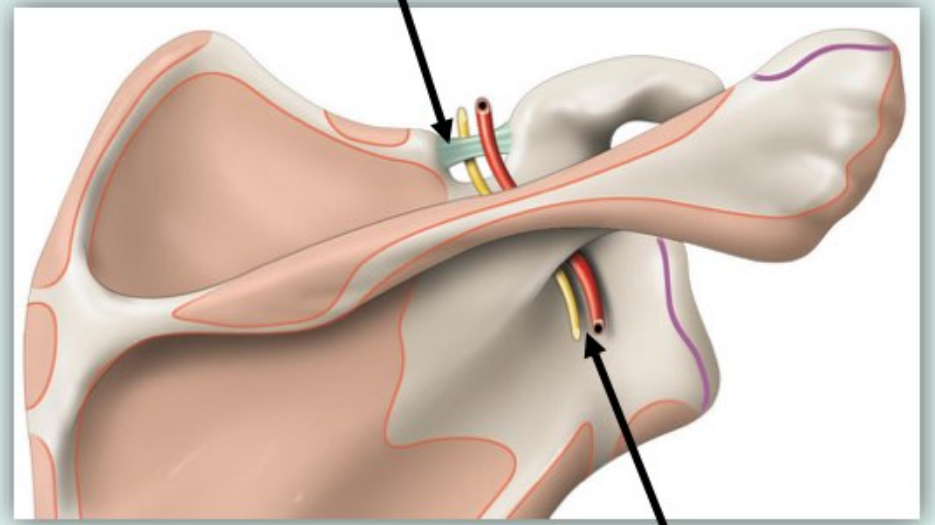
Bord supérieur (cervical)





Présente:
incisure scapulaire

Surmontée par:
ligament transverse de la scapula



Où passent:
artère et **nerf suprascapulaires**

Mise en place

2 Faces

3 Bords

3 Angles

Angle supérieur

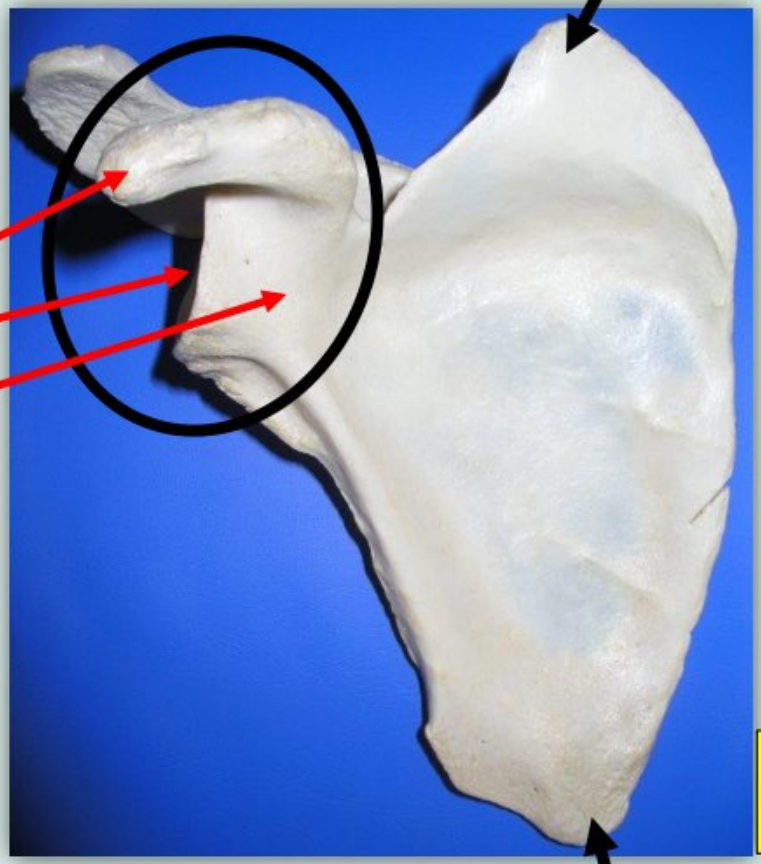
Angle latéral

Présente:

(1) **Processus coracoïde**

(2) **Cavité glénoïdale**

(3) **Col de la scapula**



H^t

DD

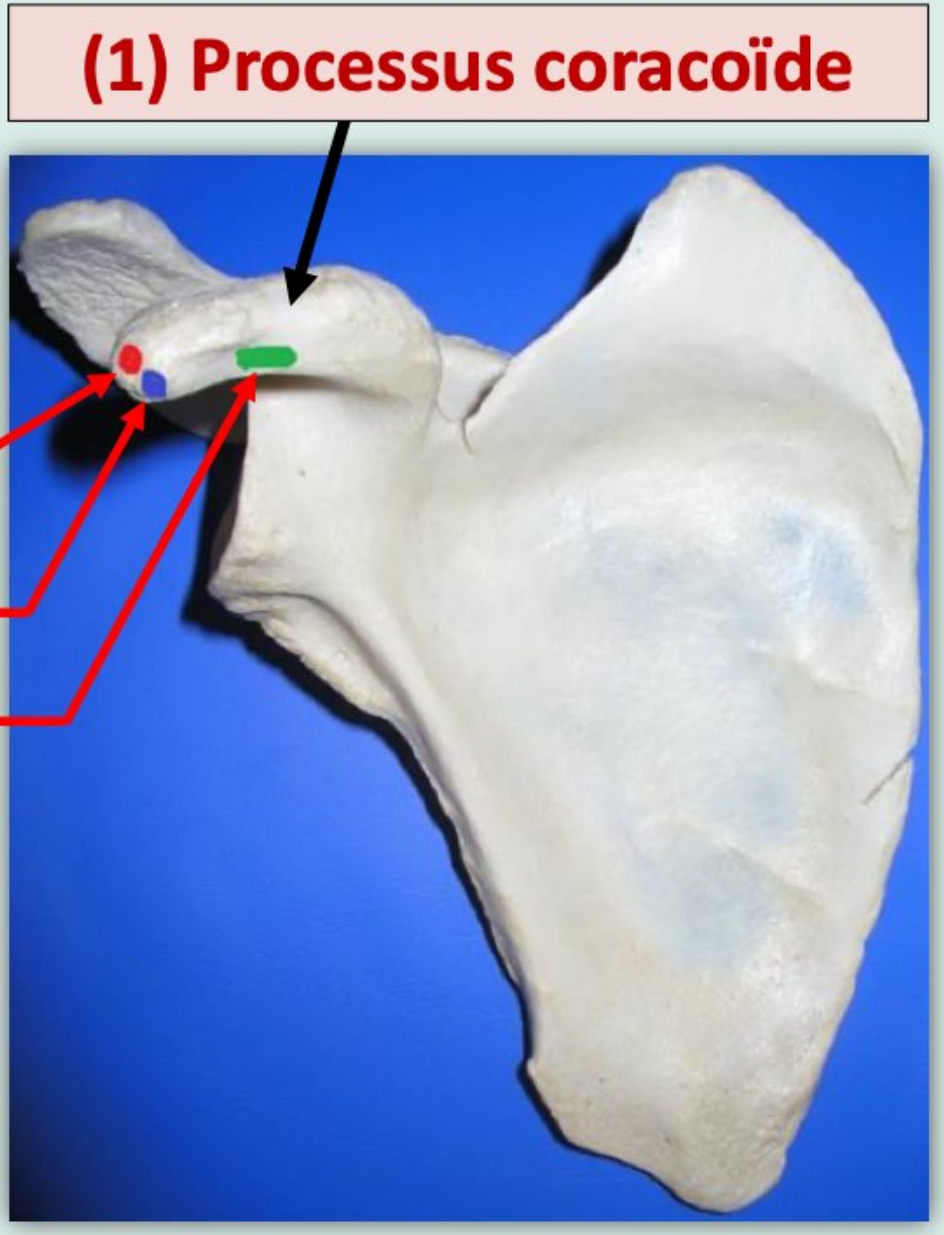
Scapula, vue antérieure

Angle inférieur
(pointe de la scapula)

Forme = **doigt demi-fléchit**

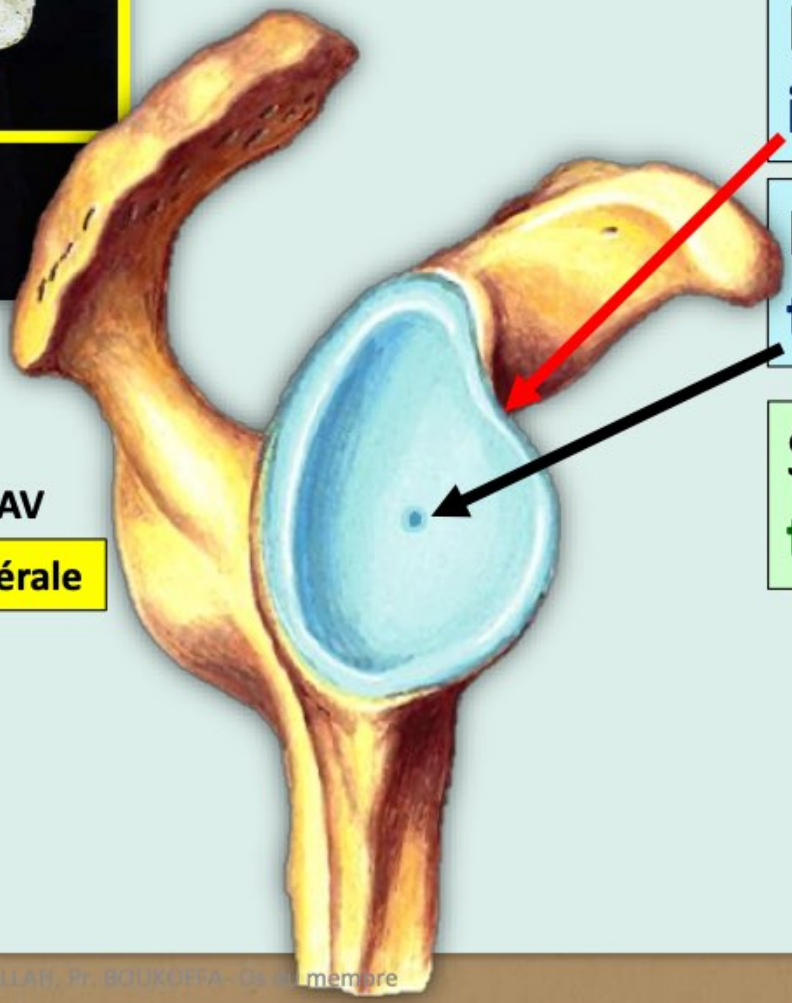
Donne insertion aux **muscles:**

- **biceps brachial** (chef court)
- **coracobrachial**
- **pectoral mineur**



(2) Cavité glénoïdale



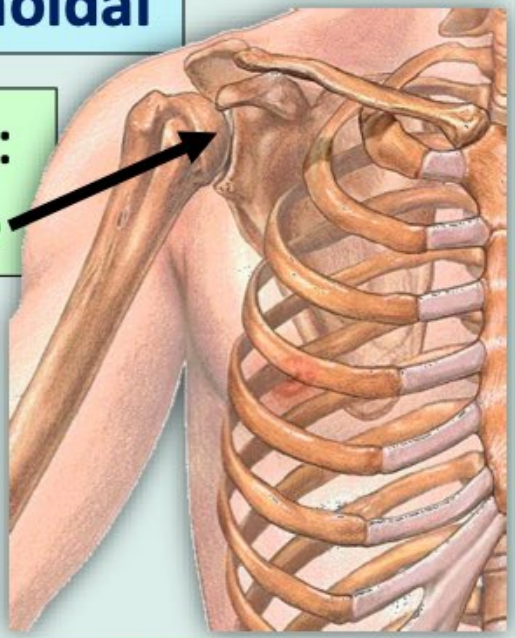


Surface articulaire ovalaire

Échancrée en avant par:
incisure glénoïdale

Présente au centre:
tubercule glénoïdal

S'articule avec:
tête humérale



H^t

AV

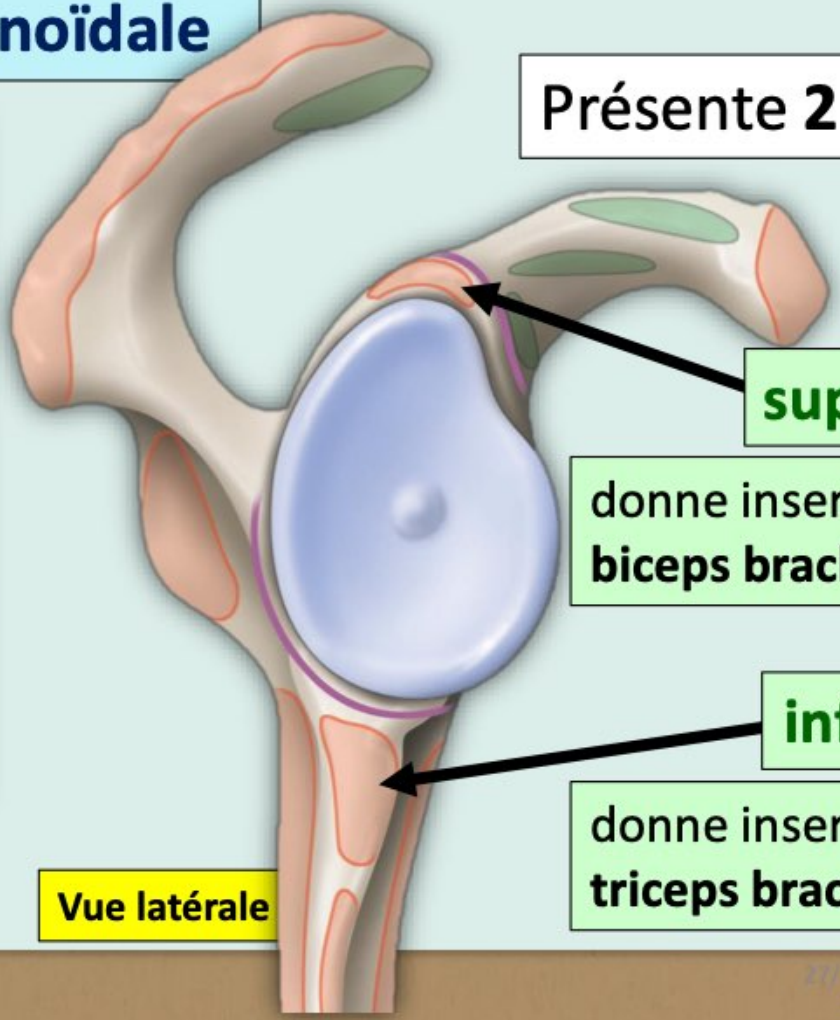
Vue latérale

(3) Col de la scapula

- **Circulaire** et **rétrécie**
- Supporte: **cavité glénoïdale**



Présente **2 tubercules**:



supra-glénoïdal

donne insertion au muscle **biceps brachial** (chef long)

infra-glénoïdal

donne insertion au muscle **triceps brachial** (chef long)

Vue latérale

Humérus

Os long

Forme à lui seul: squelette du bras





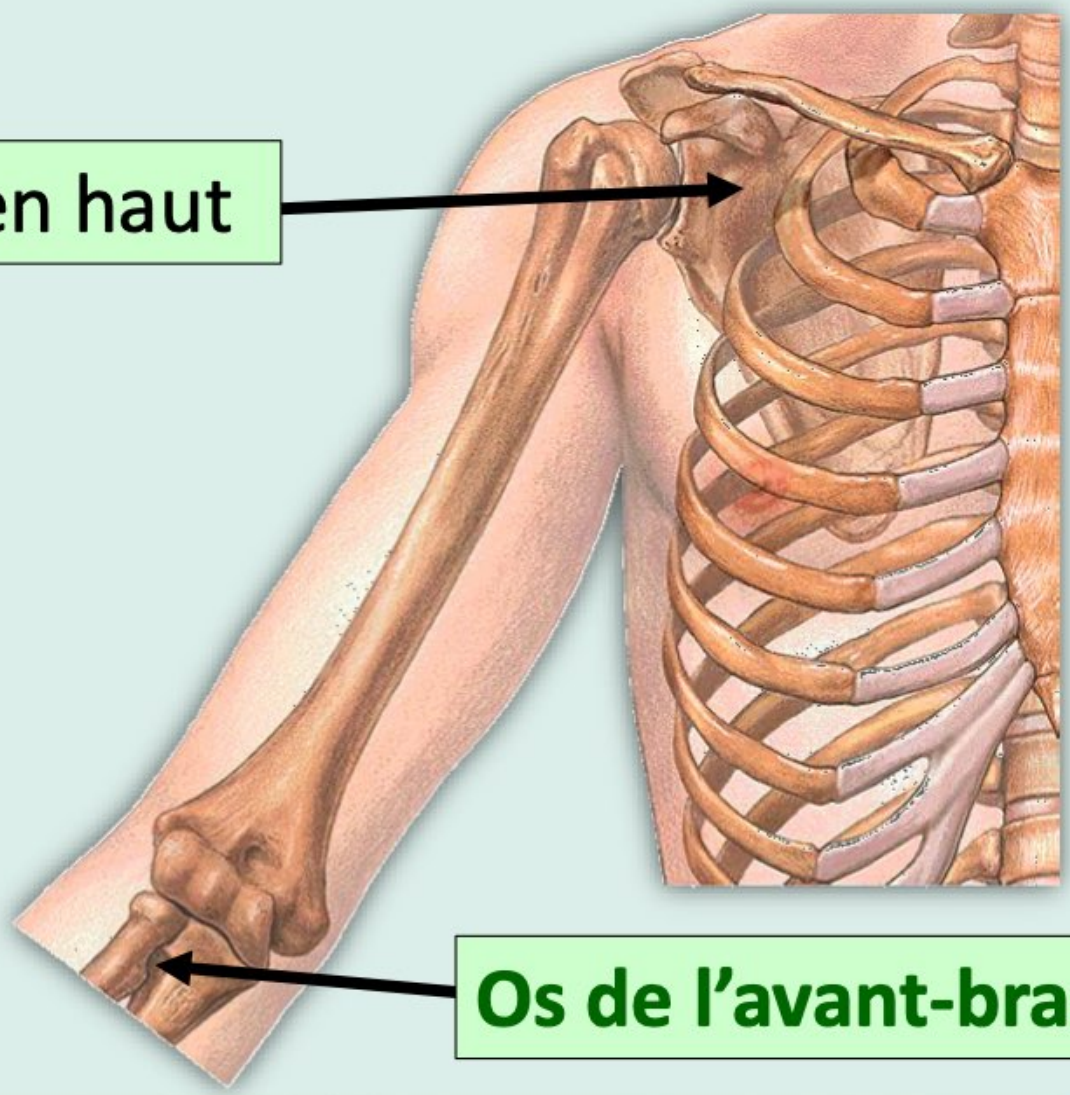
Humérus droit, vue antérieure



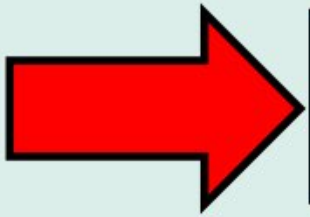
Humérus droit, vue postérieure

S’articule avec:

Scapula, en haut



Os de l’avant-bras, en bas



Mise en place

Diaphyse

Epiphyse proximale

Epiphyse distale





En haut :
extrémité arrondie de l'os

En dedans :
surface articulaire
de cette extrémité

En avant :
sillon qui parcourt de haut en bas cette extrémité

Mise en place

Diaphyse

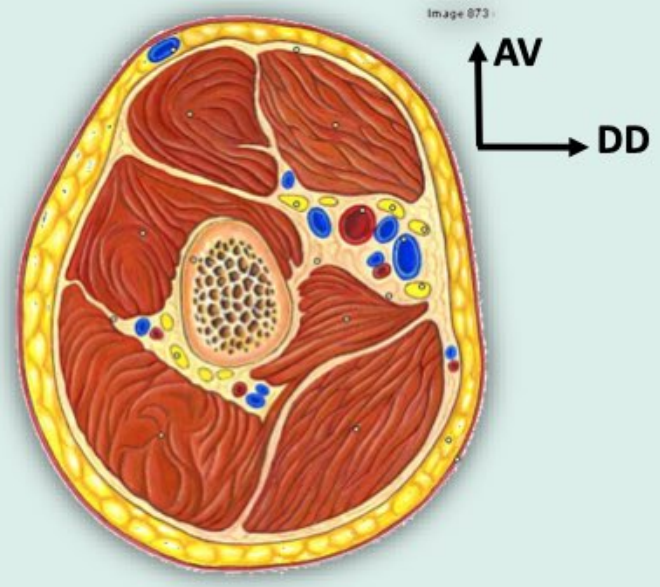
Epiphyse proximale

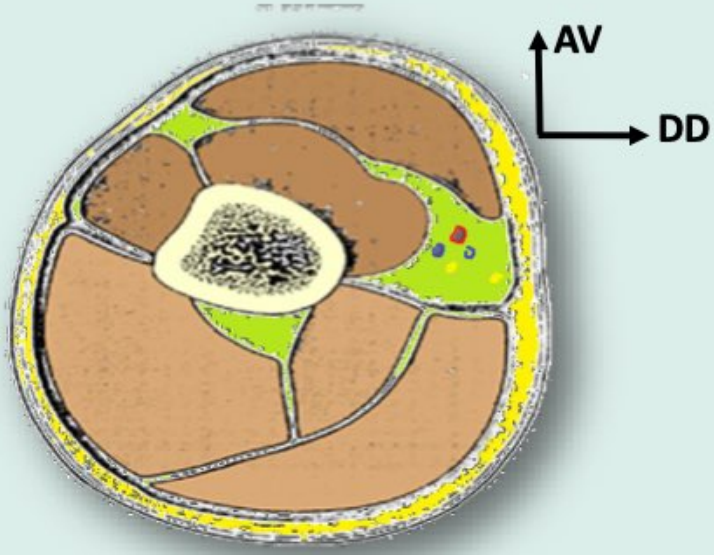
Epiphyse distale

Cylindrique, en haut

Triangulaire et **aplatie**, en bas

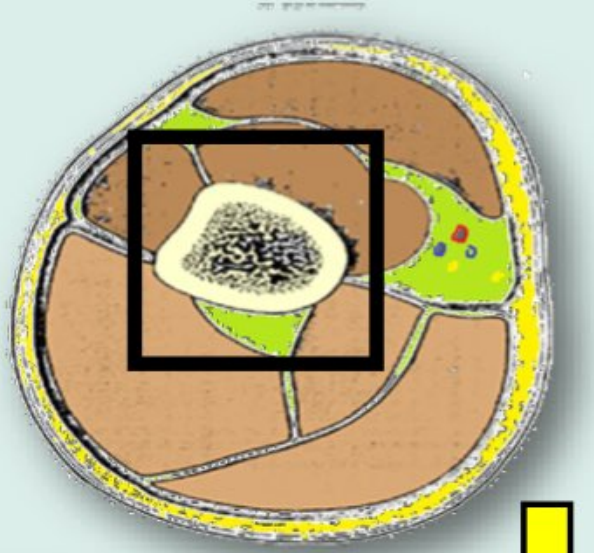






Présente à décrire:

- 3 faces
- 3 bords



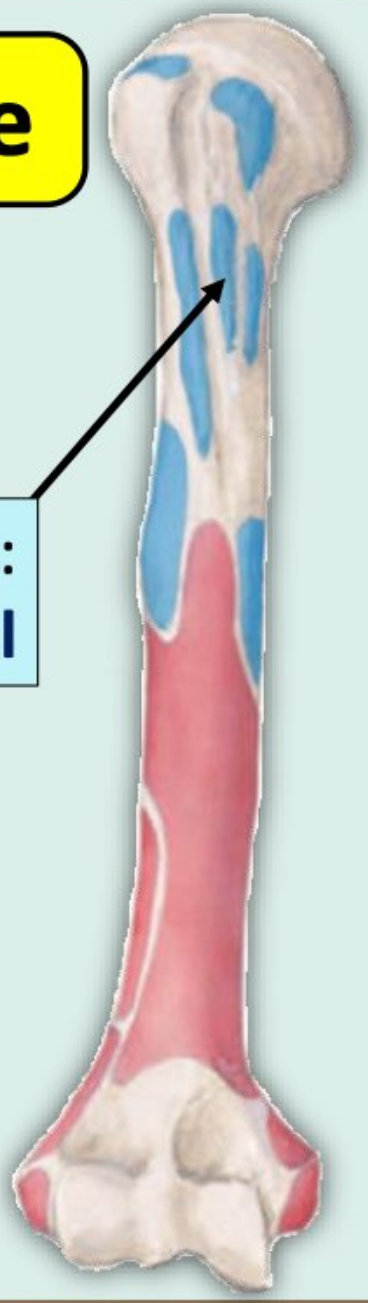


Face antéro-médiale

Présente:
Crête du tubercule mineur

qui donne insertion au **muscle**:
grand dorsal

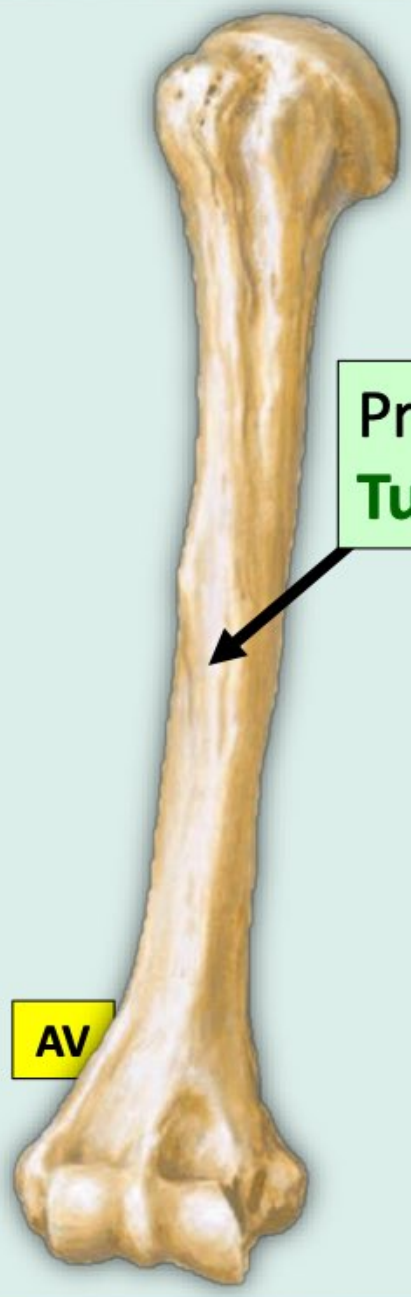


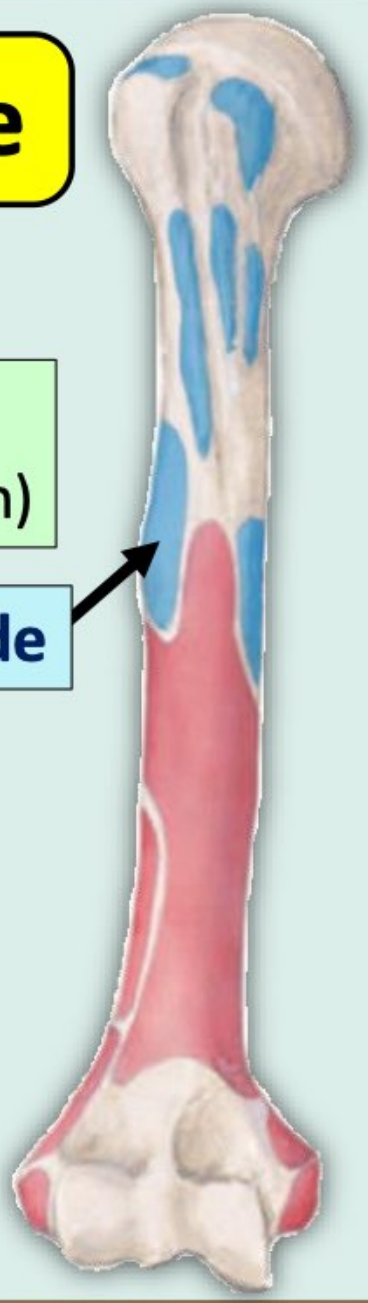


Face antéro-latérale

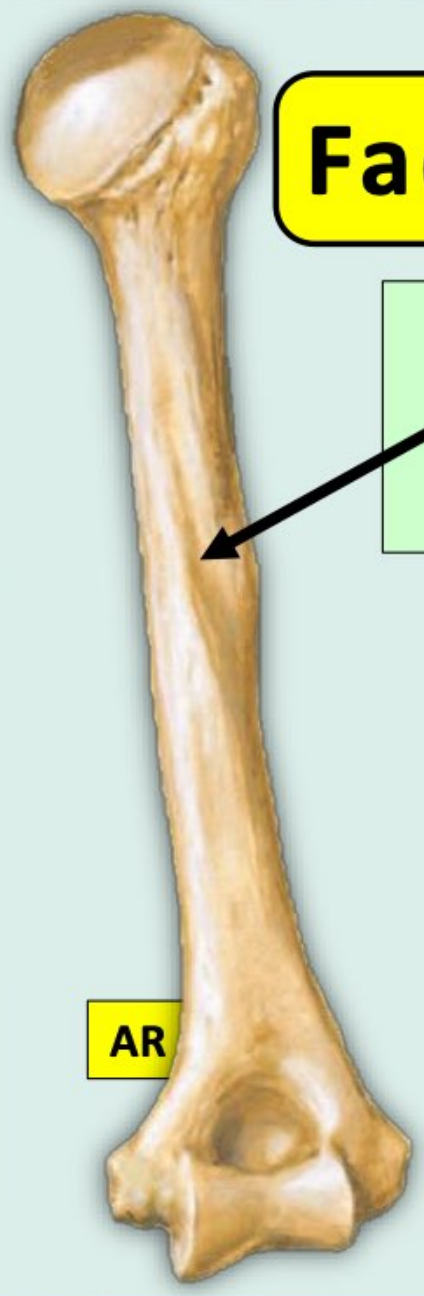
Présente:
Tubérosité deltoïdienne (« V » deltoïdien)

où s'insère: **muscle deltoïde**





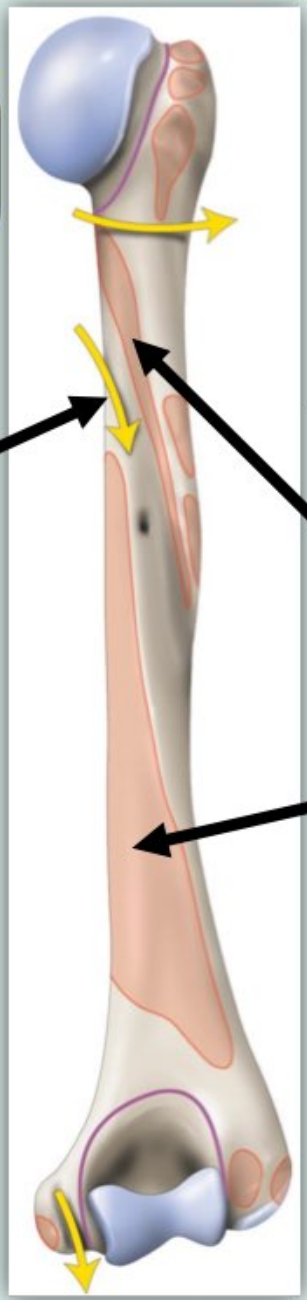
Face postérieure



Traversée par:
sillon du nerf radial
(où passe: **nerf radial**)



Lésion du nerf radial



Elle donne insertion au **muscle triceps brachial**:

Chef latéral, au-dessus du sillon

Chef médial, au-dessous du sillon



Bord antérieur:

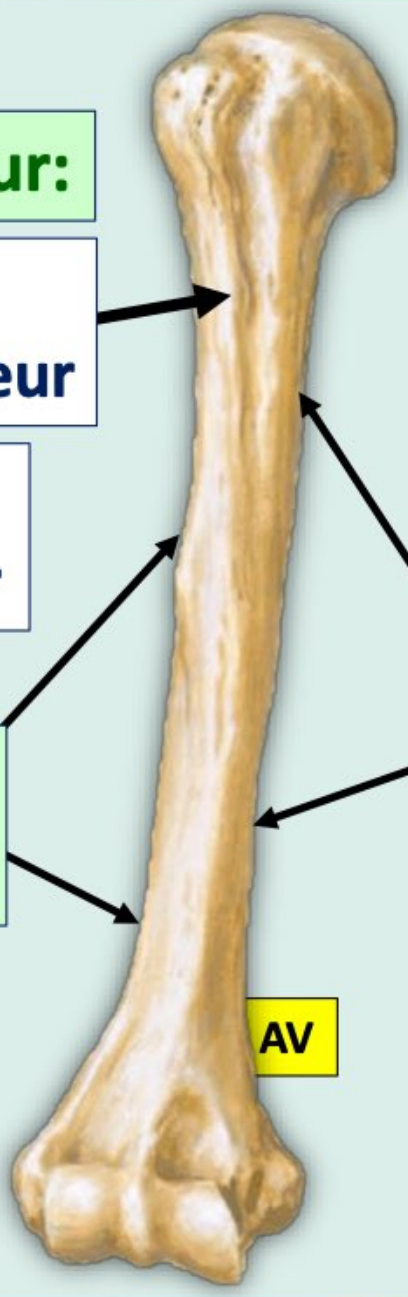
répond à:
crête du tubercule majeur

où s'insère:
muscle pectoral majeur

Bord médial

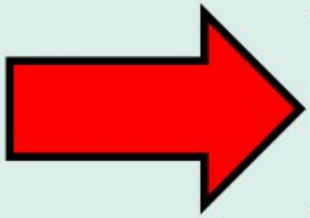
Bord latéral

3 Bords



Mise en place

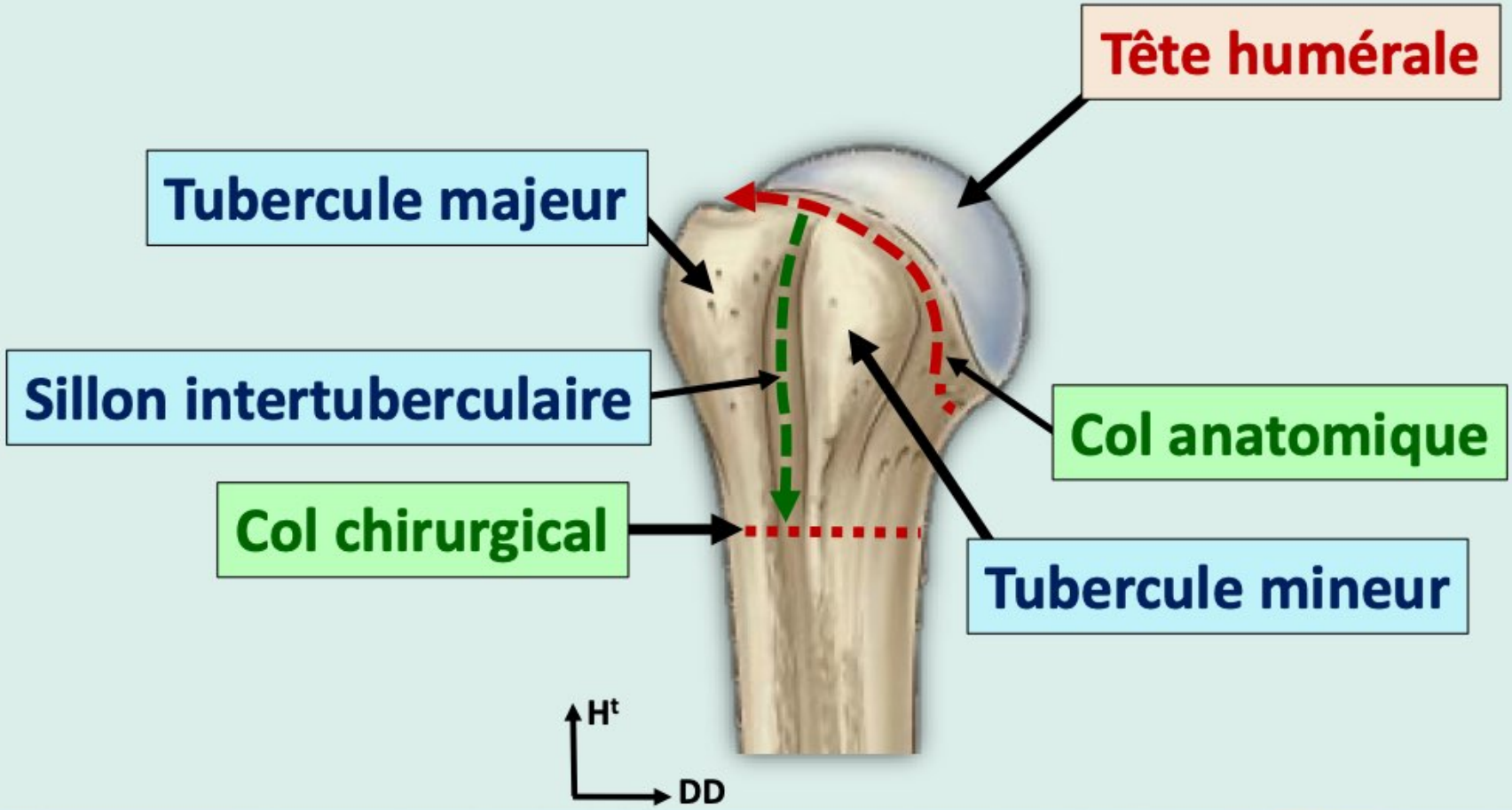
Diaphyse



Epiphyse proximale

Epiphyse distale

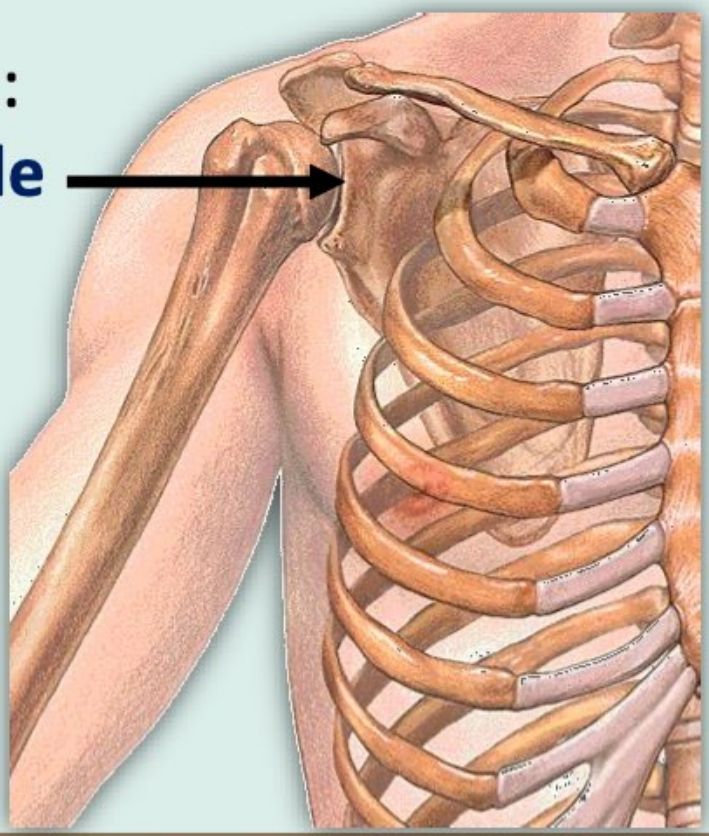
Porte: 6 structures anatomiques





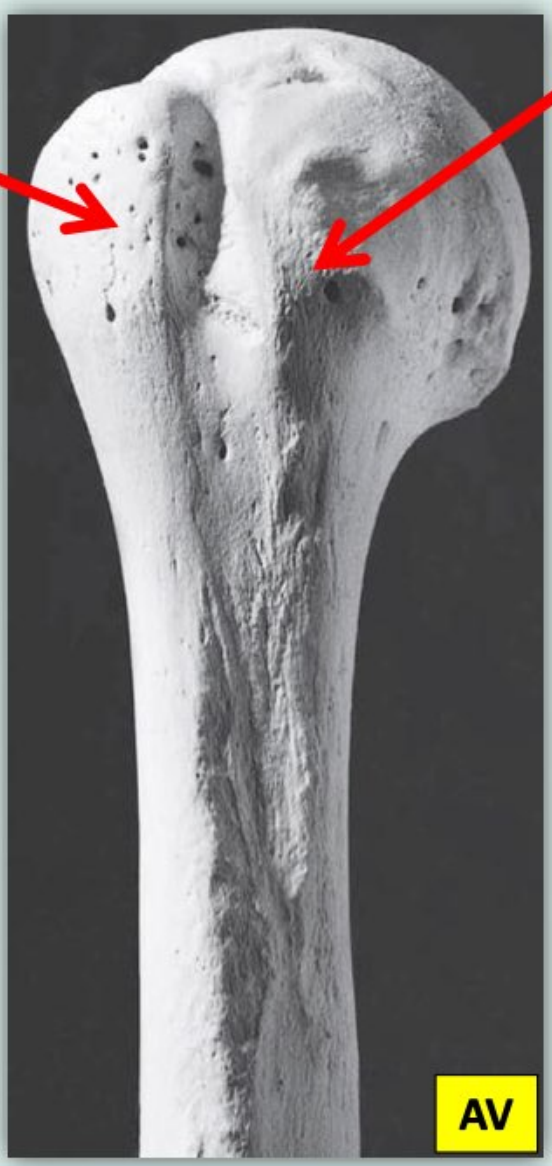
1- Tête humérale

- **1/3** d'une **sphère** (rayon = 30 mm)
- **Articulaire** avec:
  cavité glénoïdale
  de la scapula



2- Tubercule majeur

Grosse saillie latérale

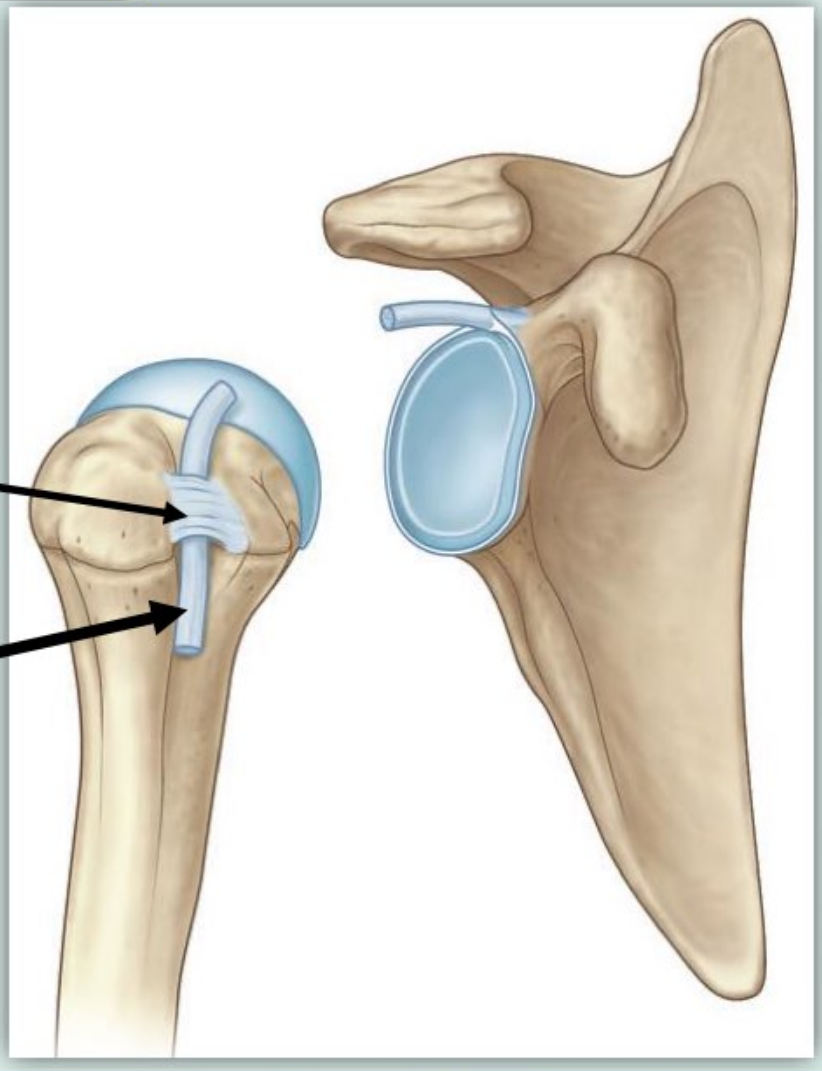


3- Tubercule mineur

Petite saillie antérieure

4- Sillon intertuberculaire

- Sépare: **2 tubercules**
- Recouvert par: **ligament huméral transverse**
- Livre passage au: **tendon** du **muscle biceps brachial** (chef long)



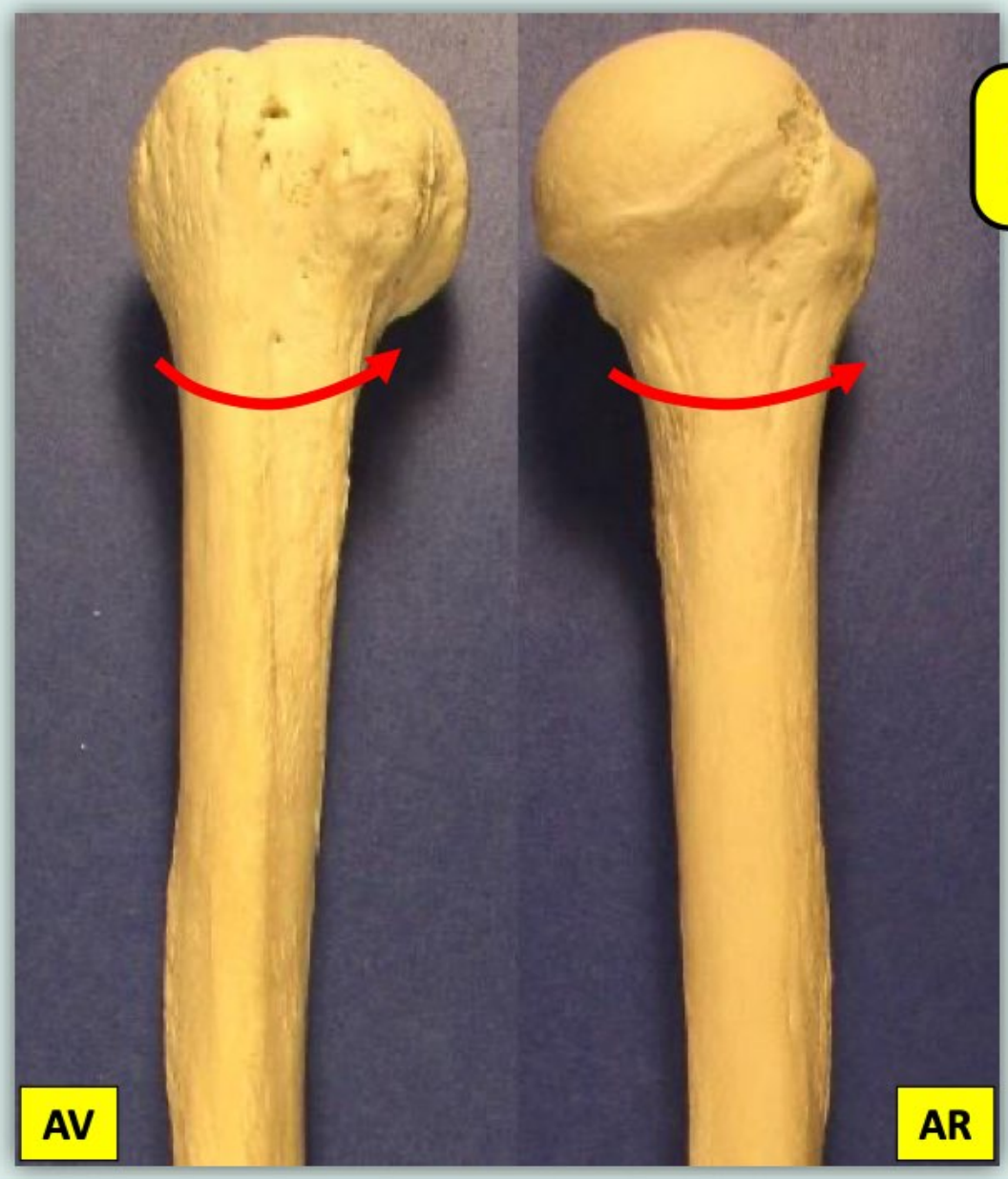




5- Col anatomique

Sépare:

tête humérale
des **2 tubercules**



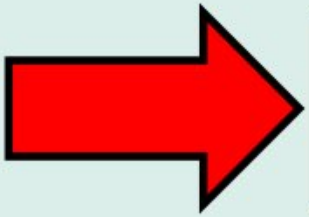
6- Col chirurgical

- Sépare: **épiphyse proximale** et **diaphyse**
- **Vulnérable** aux **fractures**

Mise en place

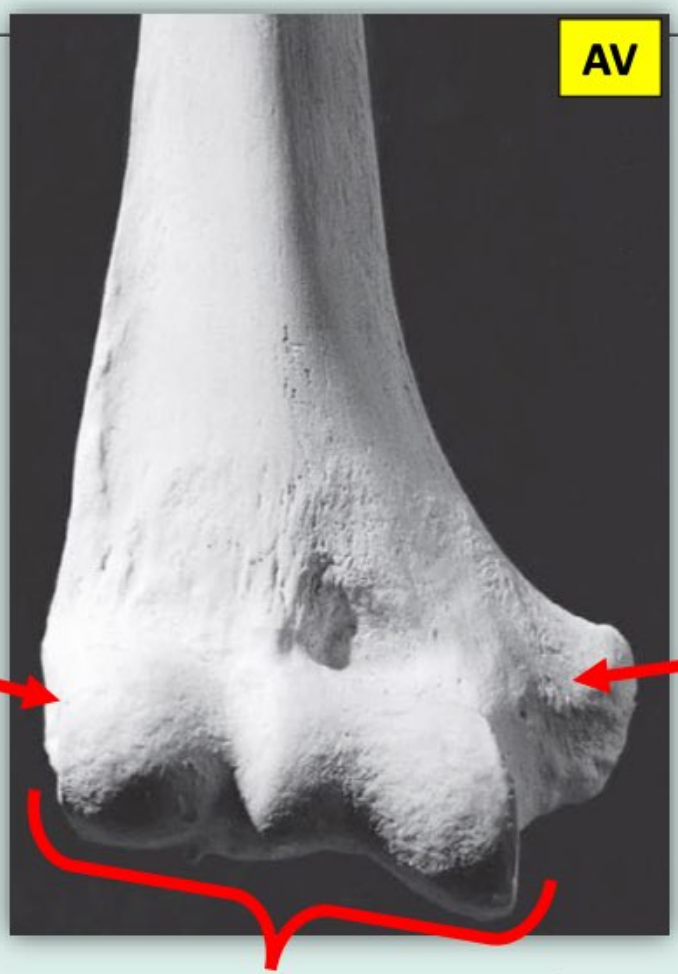
Diaphyse

Epiphyse proximale



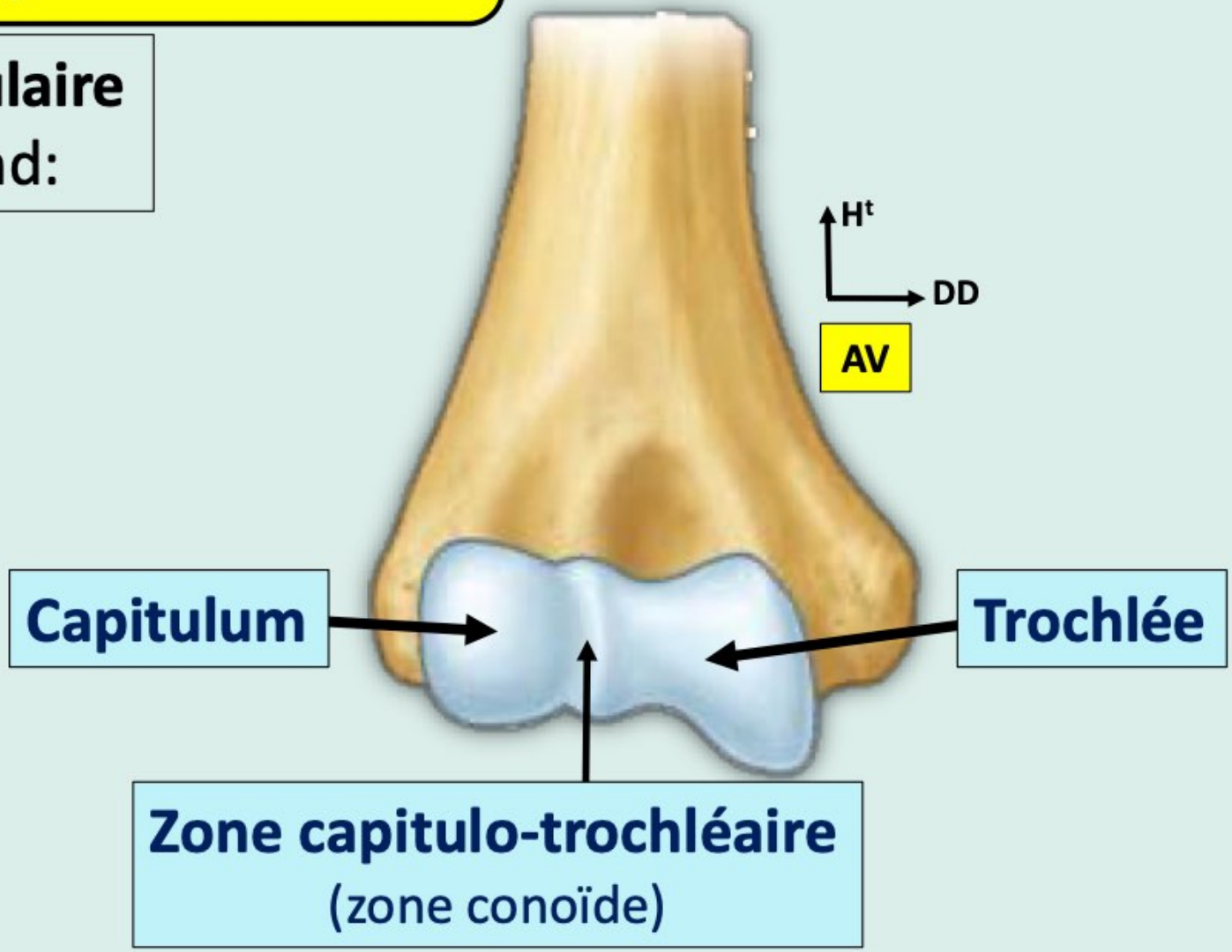
Epiphyse distale

Aplatie, avec: 3 structures anatomiques



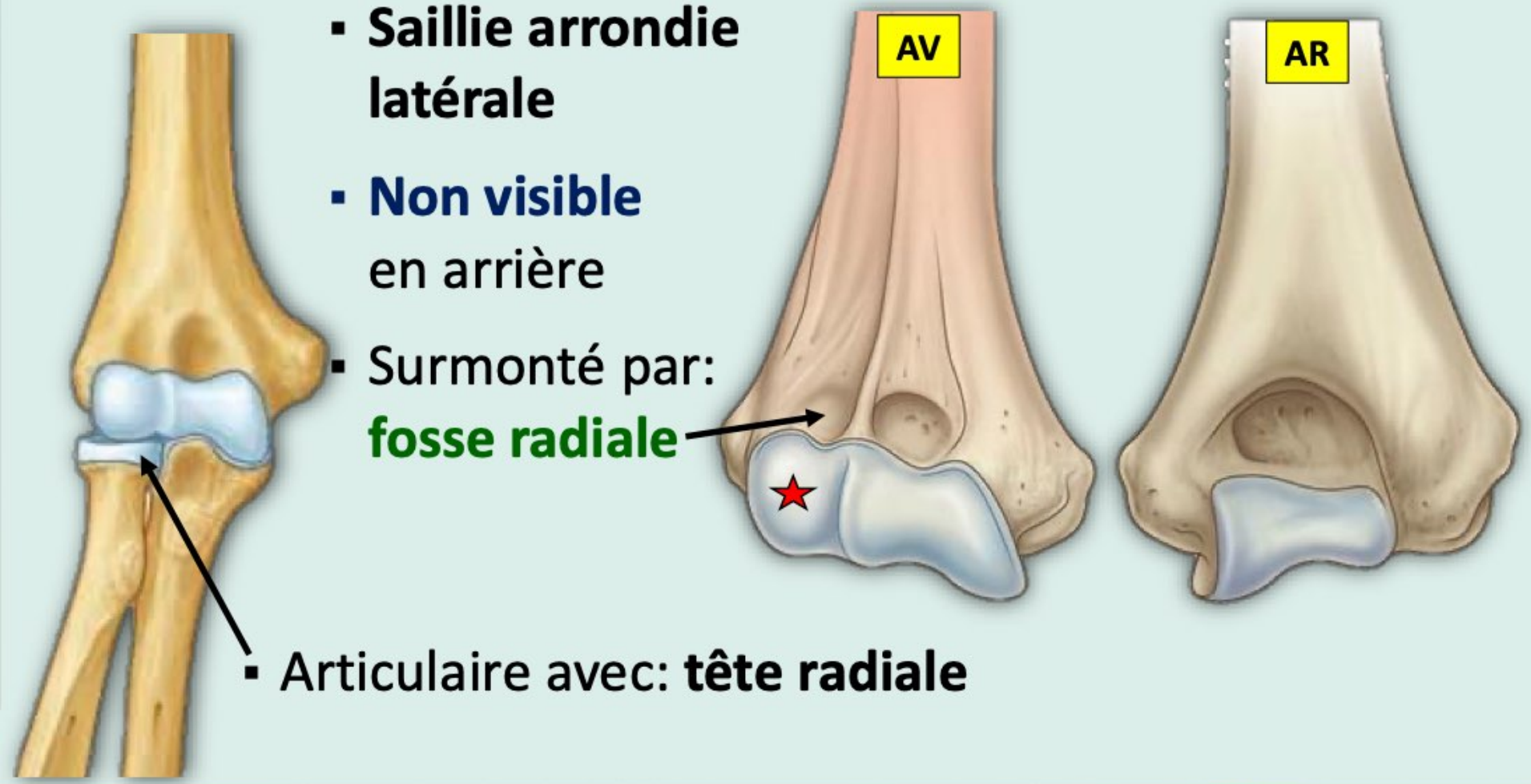
1- Condyle huméral

Saillie articulaire
qui comprend:



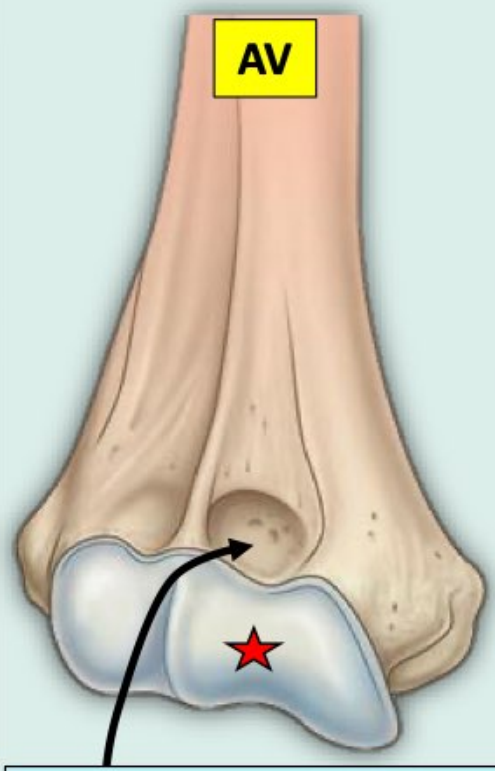
Capitulum huméral

- **Saillie arrondie latérale**
- **Non visible** en arrière
- Surmonté par: **fosse radiale**



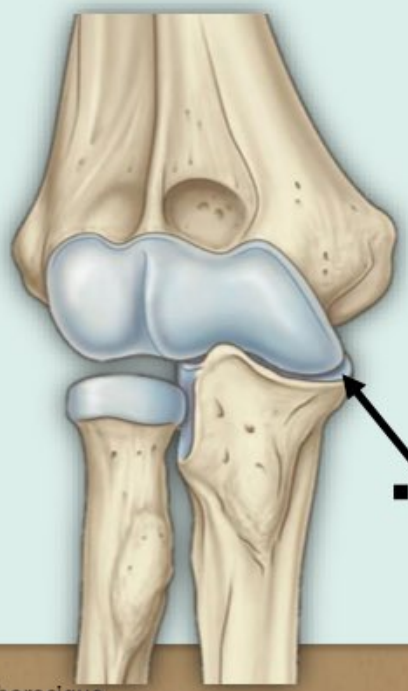
- Articulaire avec: **tête radiale**

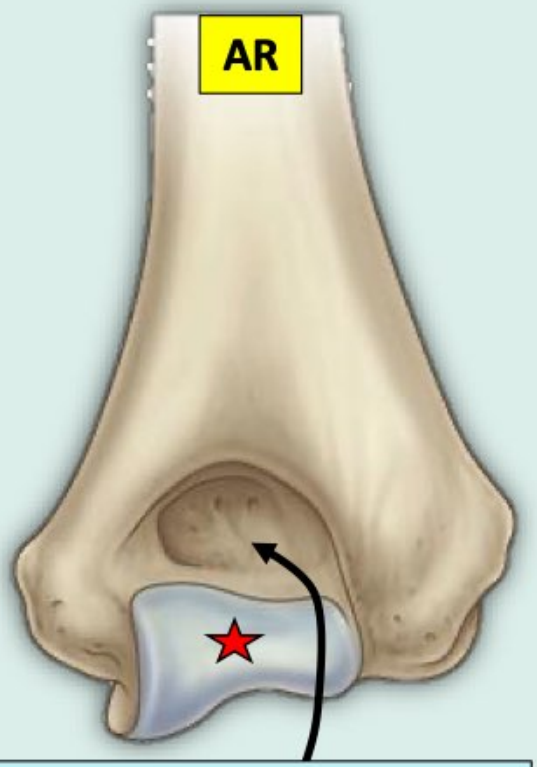
Trochlée humérale



- = **poulie** médiale
- formée de **2 joues** séparées par: **gorge**
- Surmontée par **2 fosses**:

Fosse coronoïdienne, en avant

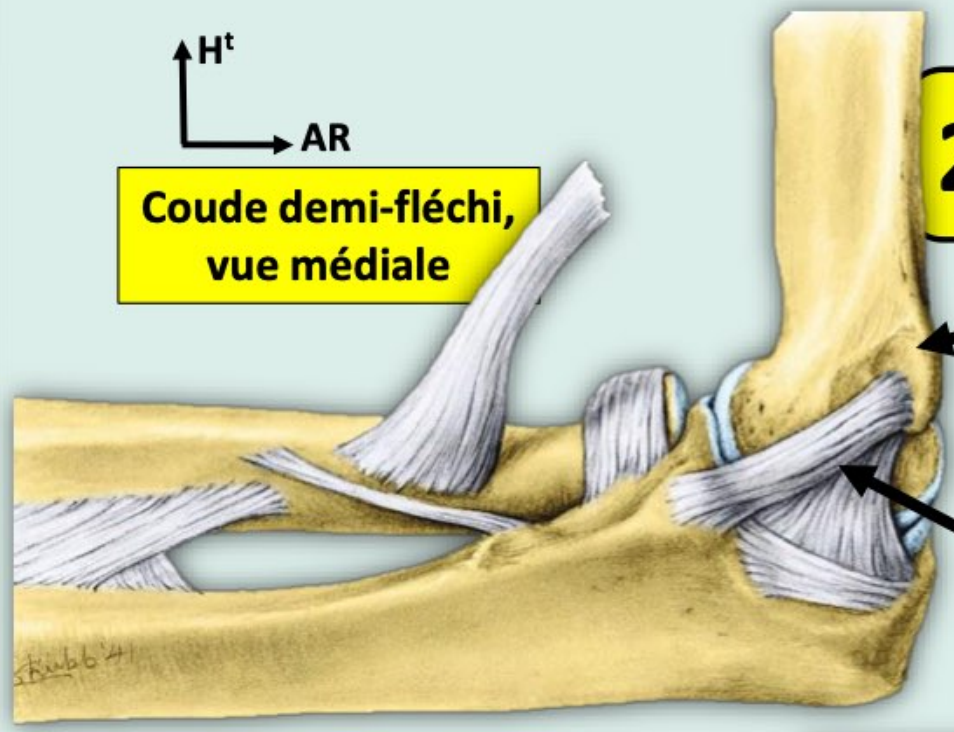




Fosse olécrânienne, en arrière

- S'articule avec: **incisure trochléaire de l'ulna**

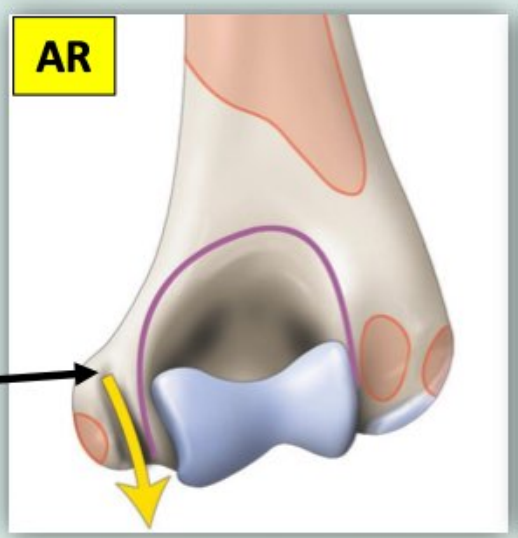
2- Epicondyle médial



Coude demi-fléchi, vue médiale

- = **saillie** très développée et palpable
- Donne attache au: **ligament collatéral ulnaire du coude**

- Traversé en arrière par: **sillon du nerf ulnaire**

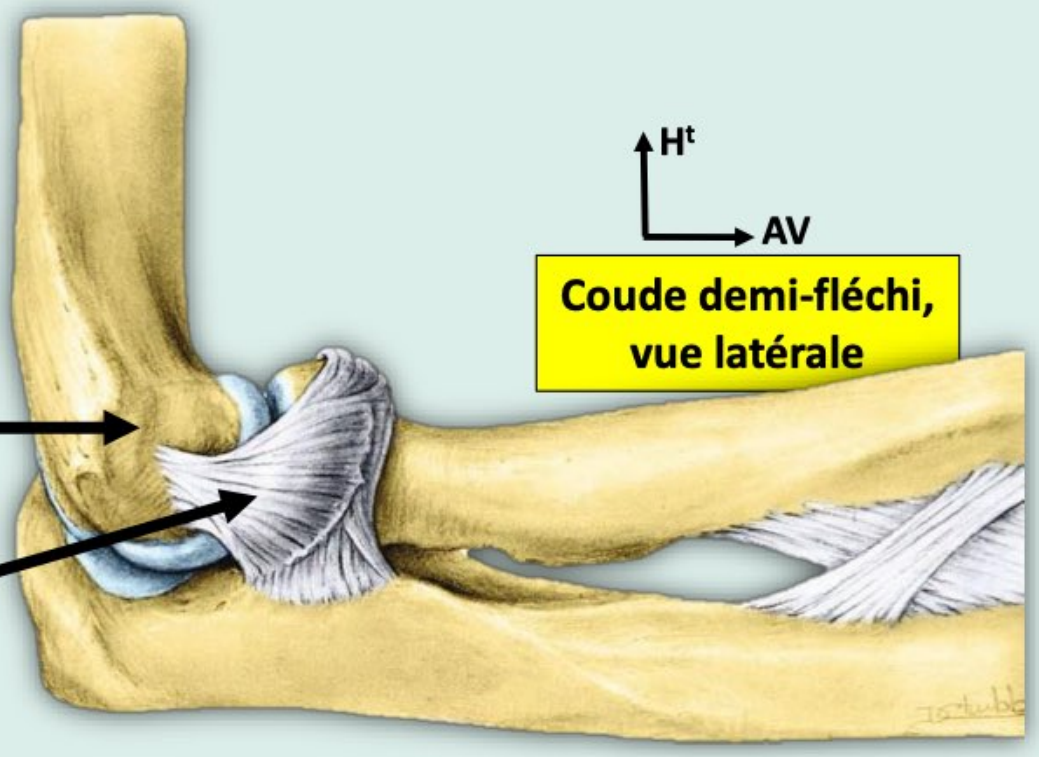




Lésion du nerf ulnaire

3- Epicondyle latéral

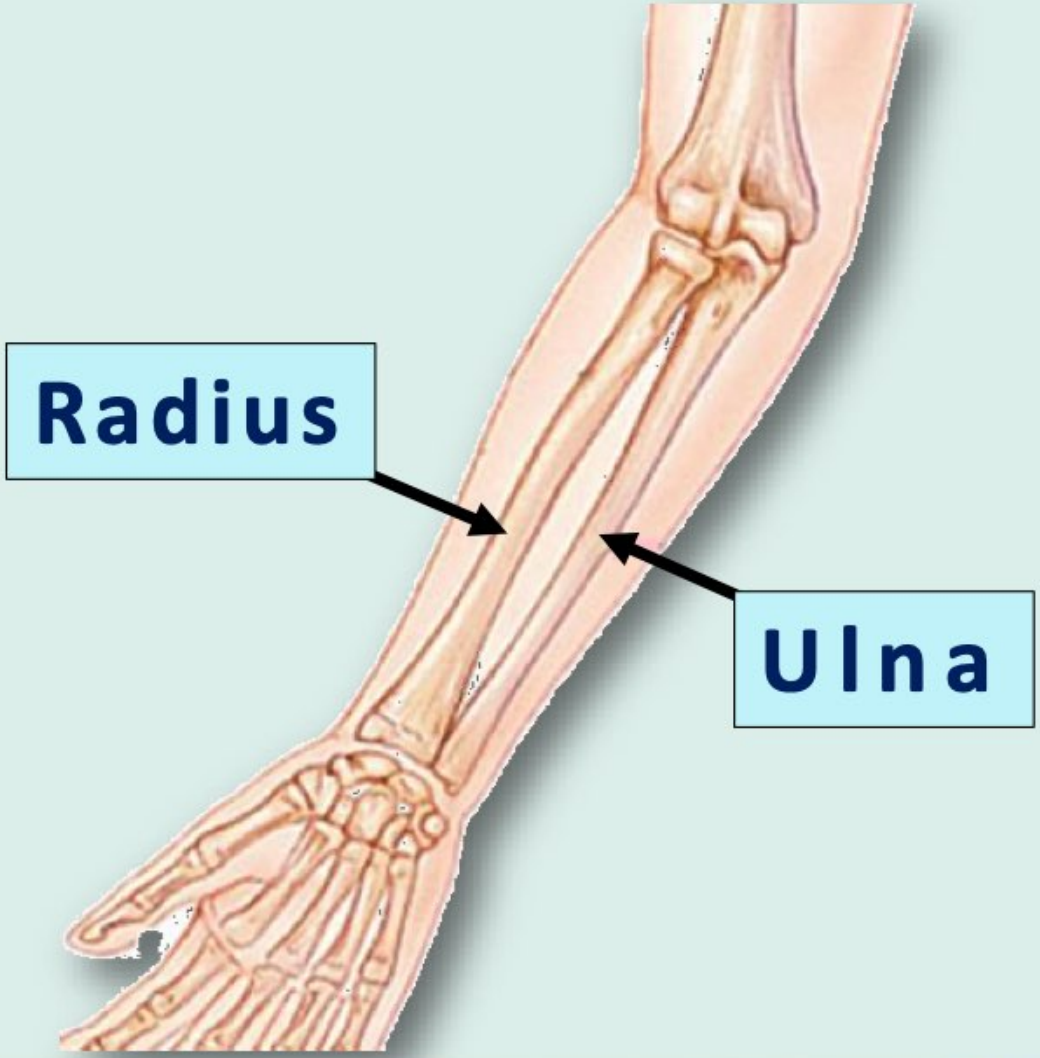
- = **saillie** peu développée
- Donne attache au:
  ligament collatéral radial du coude



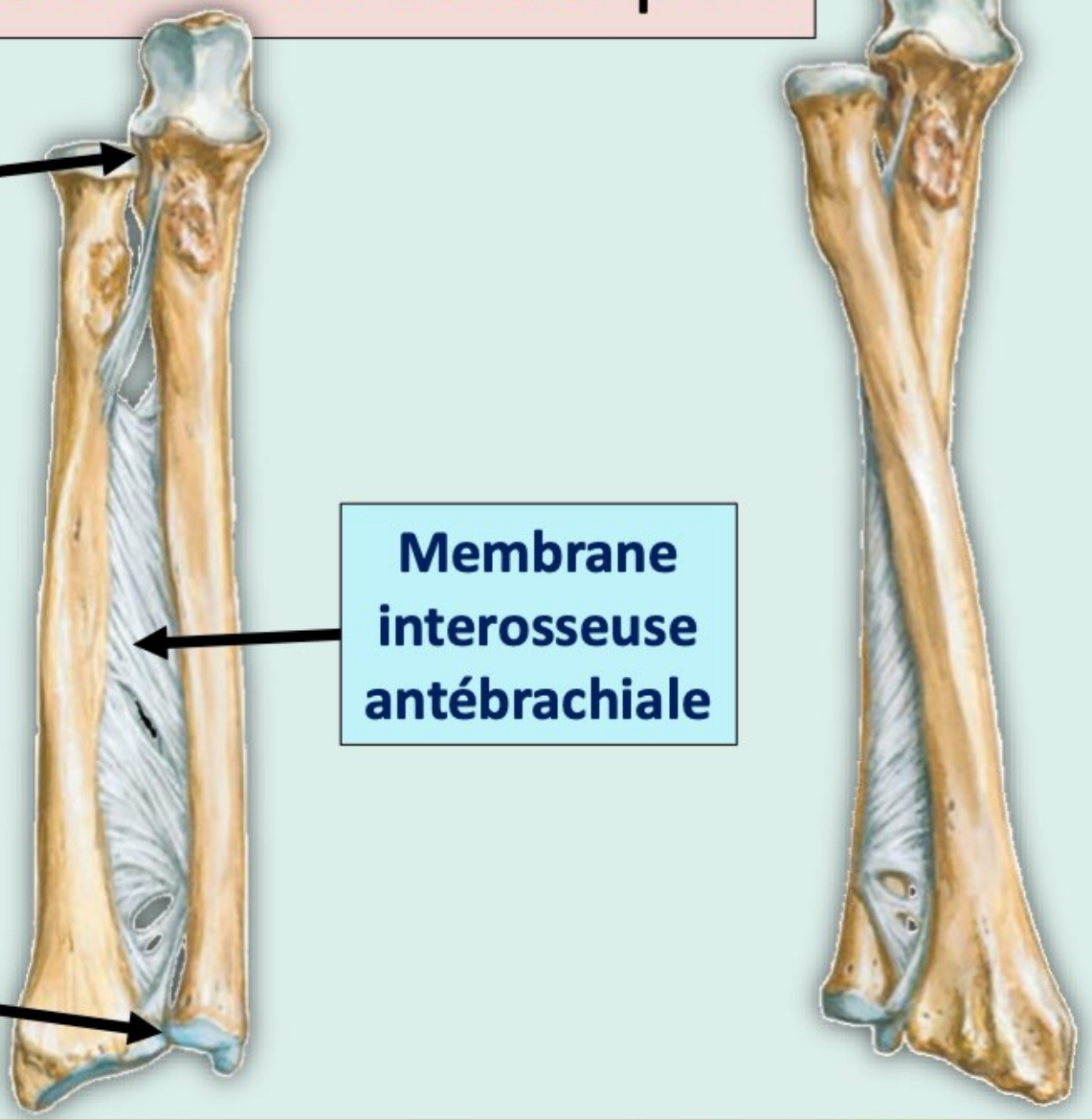
Squelette de l'avant-bras



Comprend: 2 os longs



Ces 2 os sont **solidarisés** par:



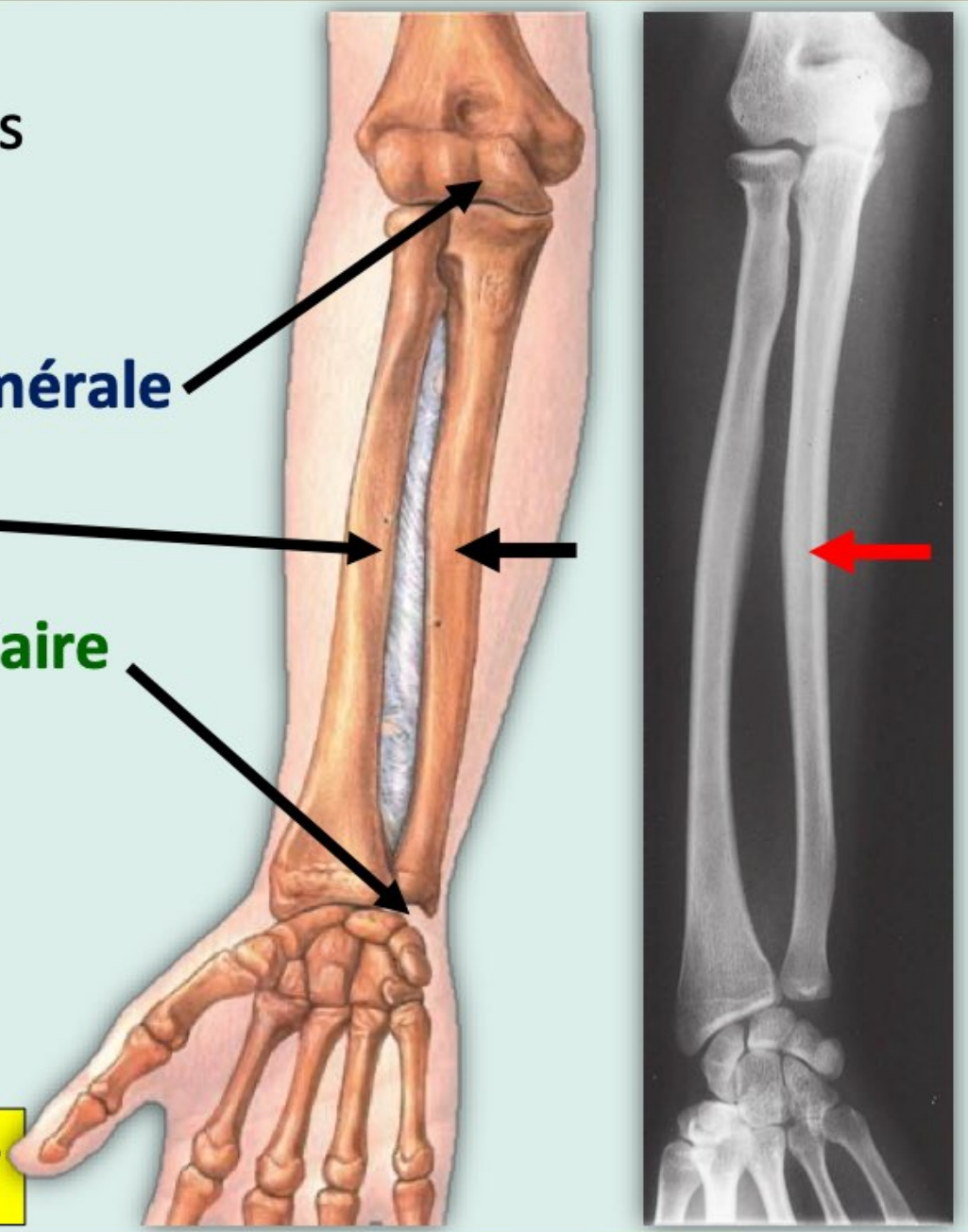
Articulation radio-ulnaire proximale

Membrane interosseuse antébrachiale

Articulation radio-ulnaire distale

(1) Ulna

- **Os médial** de l'avant-bras
- S'articule avec :
  - En haut, **trochlée humérale**
  - En dehors, **radius**
  - En bas, **disque articulaire radio-ulnaire**



Squelette avant-bras, vue antérieure

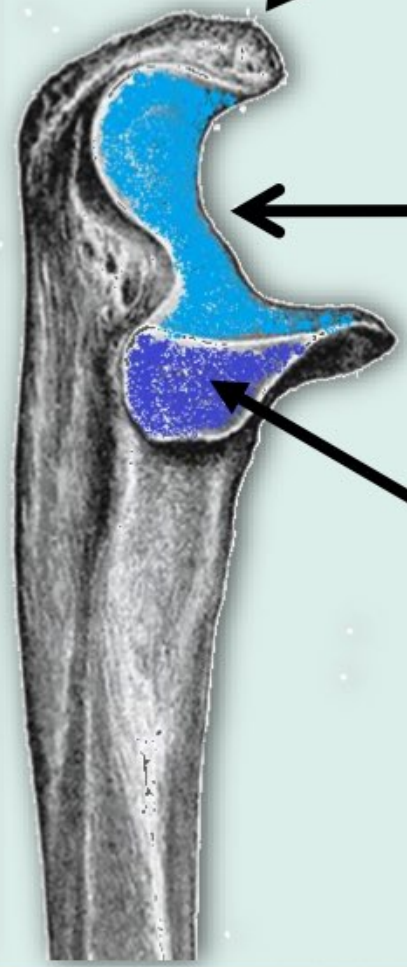
Mise en place

Diaphyse

Epiphyse proximale

Epiphyse distale





En haut :
extrémité la plus volumineuse

En avant :
surface articulaire en crochet de cette extrémité

En dehors :
petite facette articulaire de cette extrémité

Ht
AV

Vue latérale

Mise en place

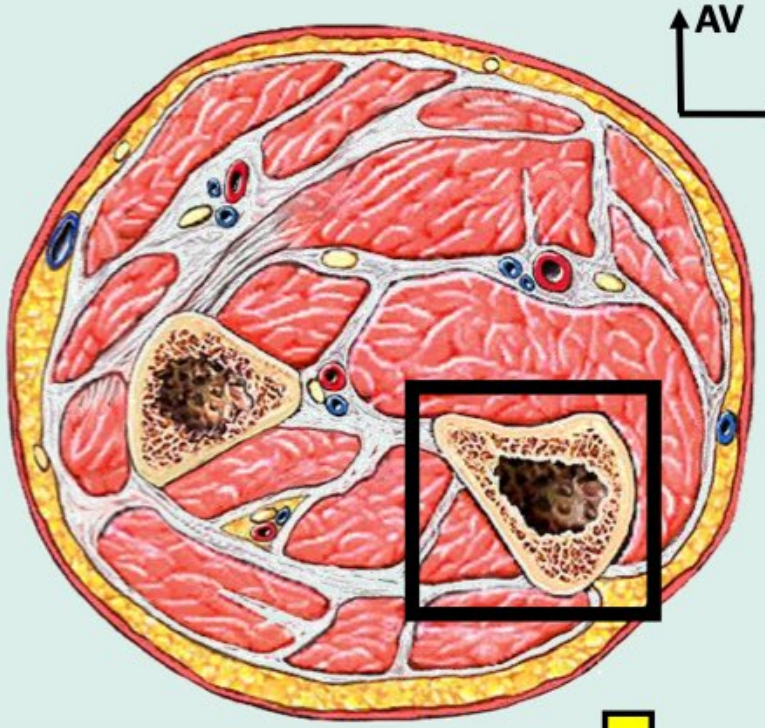
Diaphyse

Epiphyse proximale

Epiphyse distale

Présente à décrire:

- 3 faces
- 3 bords









Face antérieure

Face médiale

Face postérieure

Bord antérieur

Bord postérieur

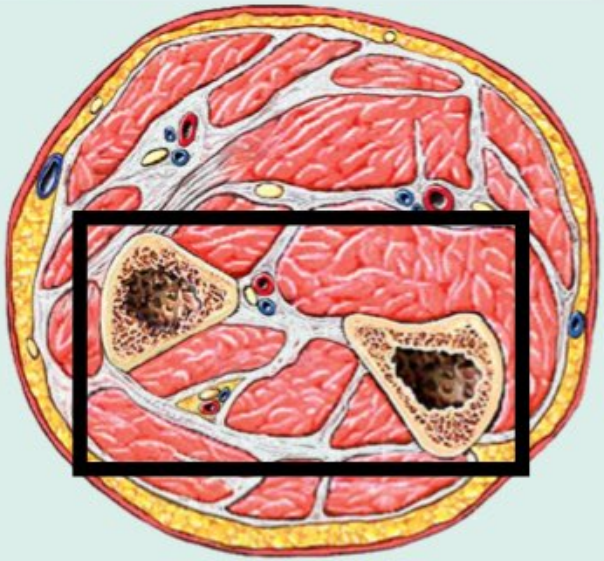
Bord latéral (interosseux)

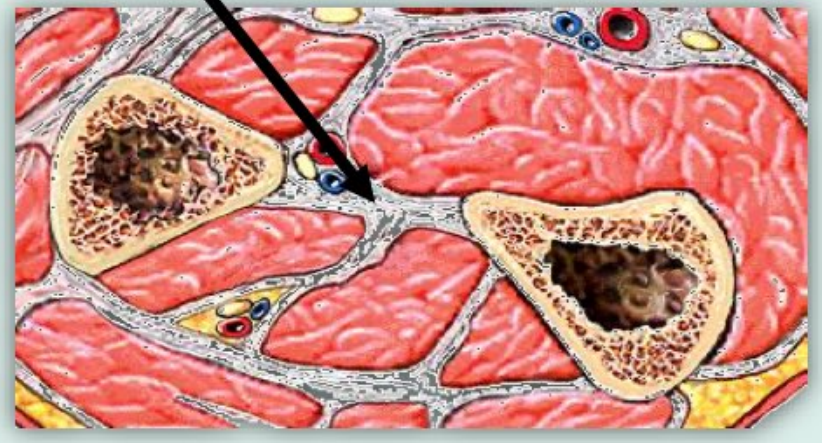
Bord latéral (interosseux)



Bord latéral
(interosseux)

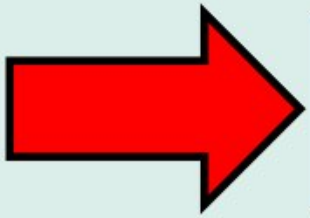
Où se fixe:
membrane interosseuse antébrachiale





Mise en place

Diaphyse



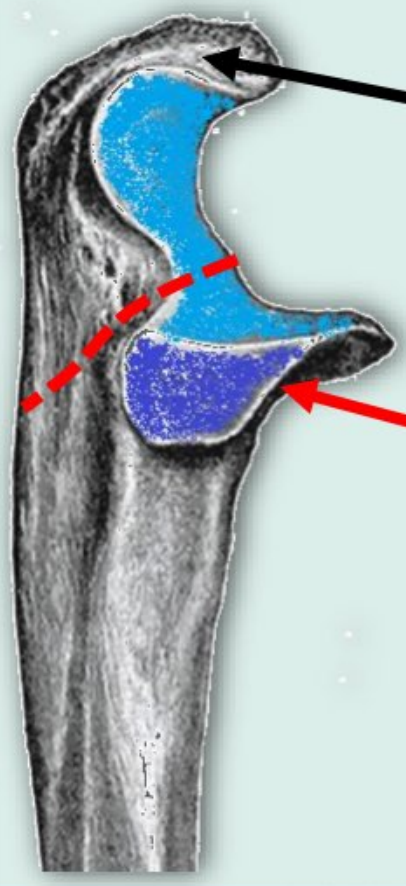
Epiphyse proximale

Epiphyse distale

Présente: **2 processus**



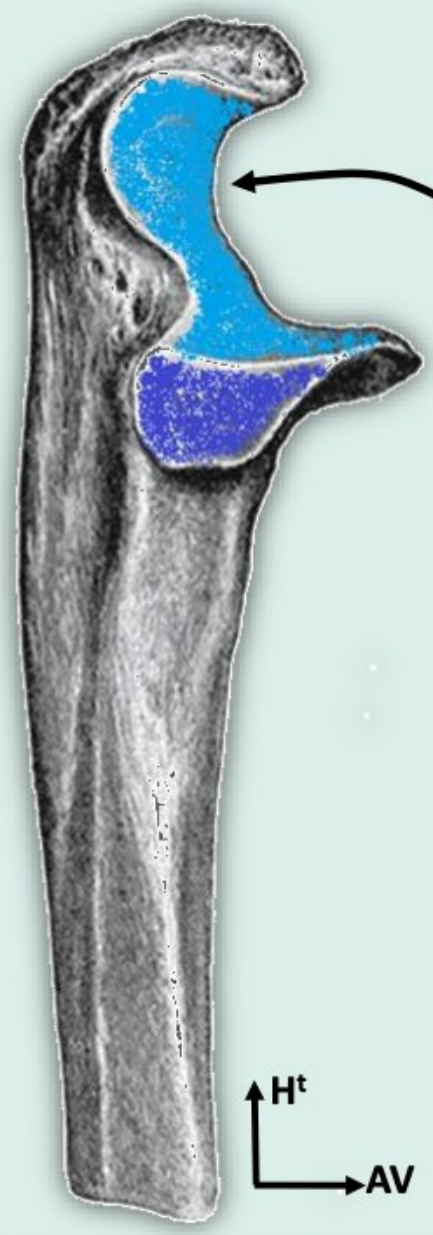
Vue latérale



Olécrâne

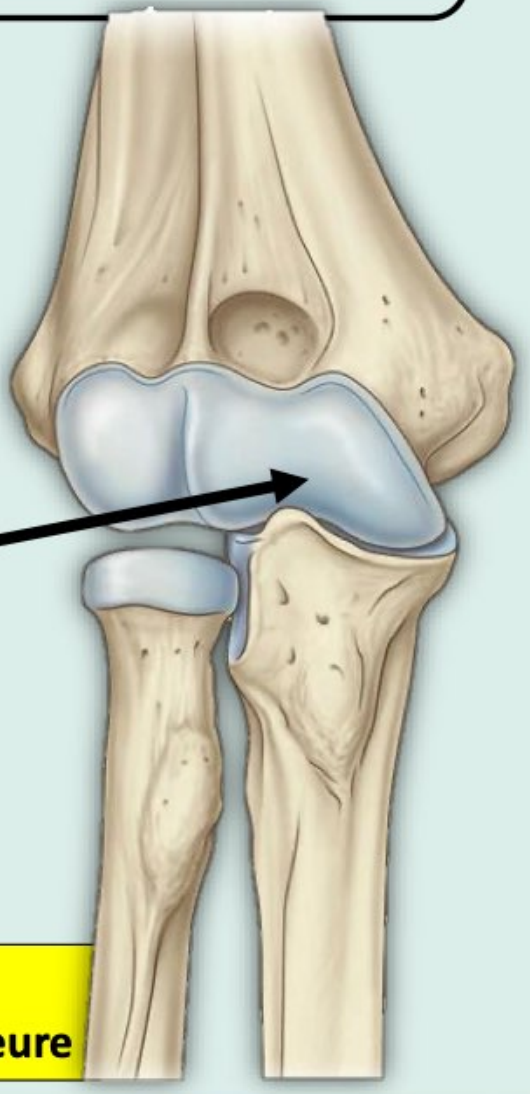
Processus coronoïde

Ces **2 processus ensemble:**



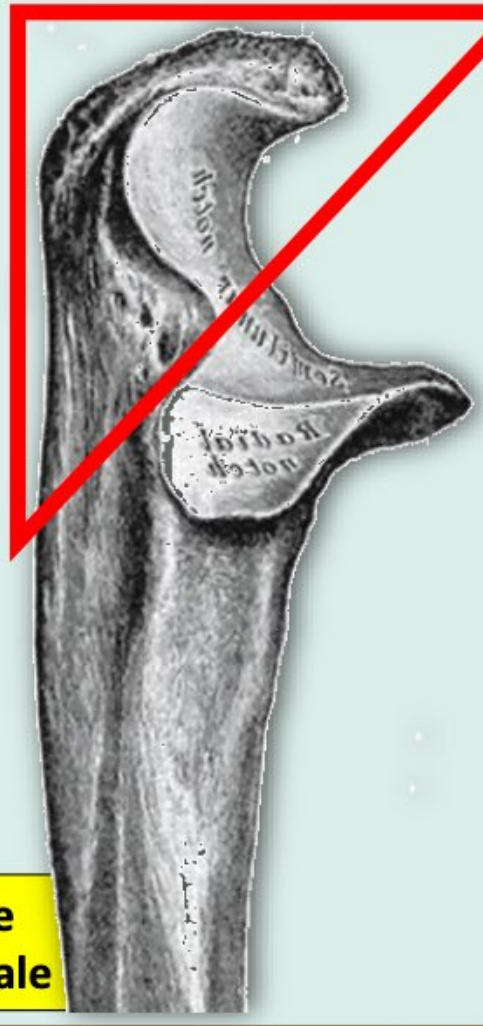
- Forment un « **crochet** » qui porte: **incisure trochléaire de l'ulna**

qui s'articule avec: **trochlée humérale**



Coude, vue antérieure

1 Olécrâne

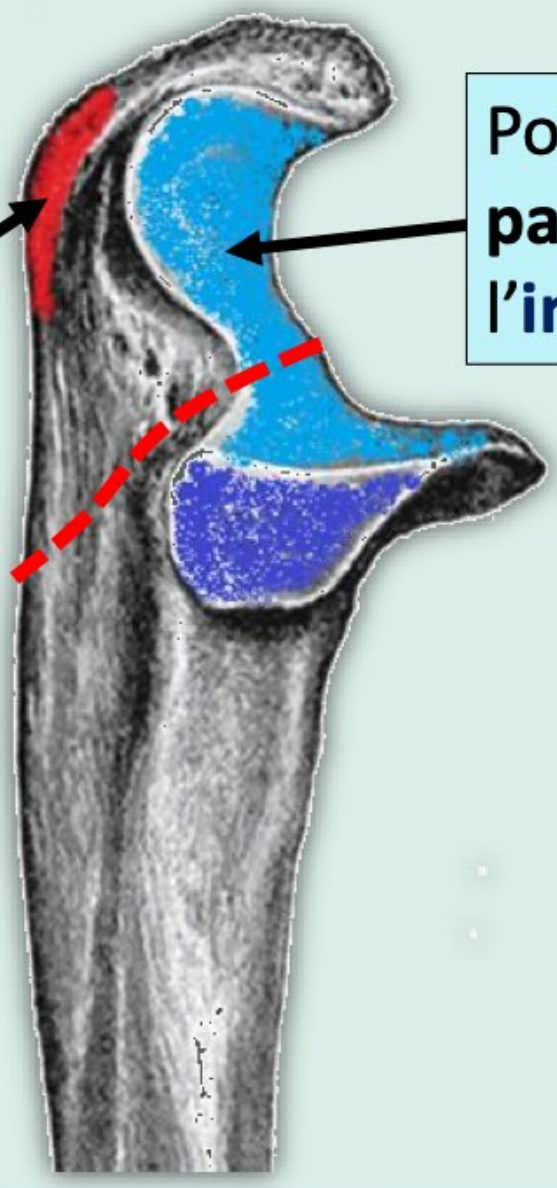


Vue latérale

- = **processus vertical** et **postérieur**
- Sous-cutané

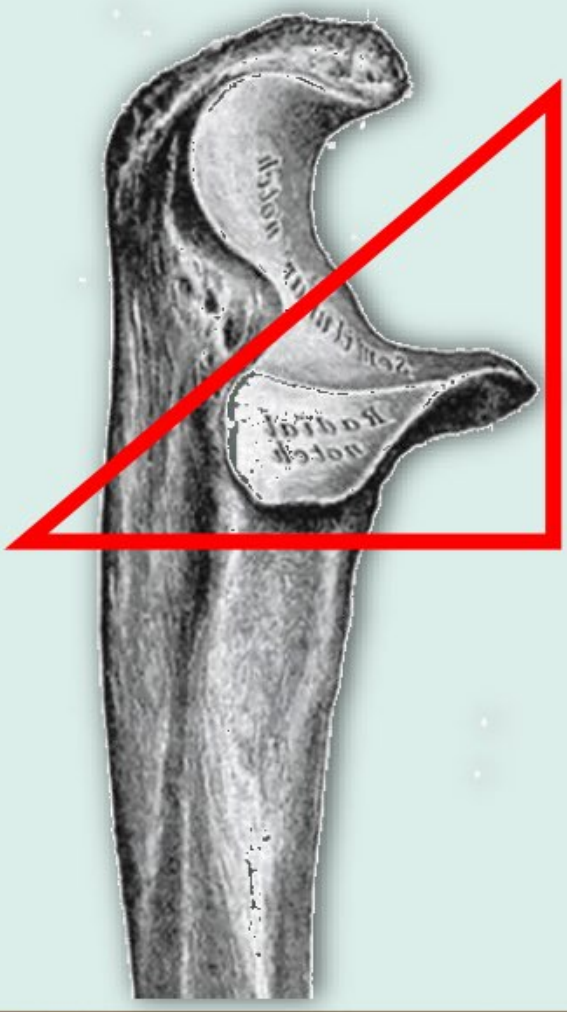


Insertion du **muscle**:
triceps brachial

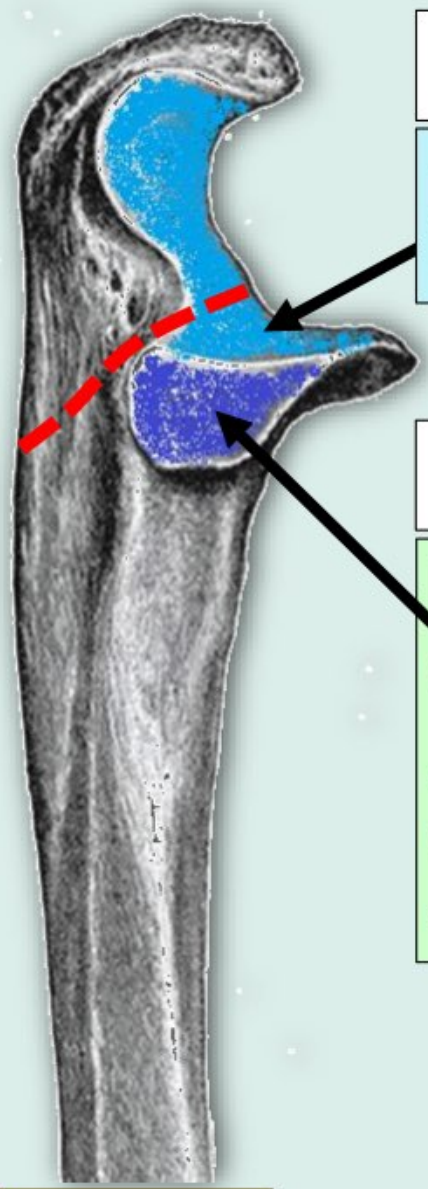


Porte:
partie supérieure de
l'**incisure trochléaire**

2 Processus coronoïde



- = **processus horizontal** et **antérieur**



Vue latérale

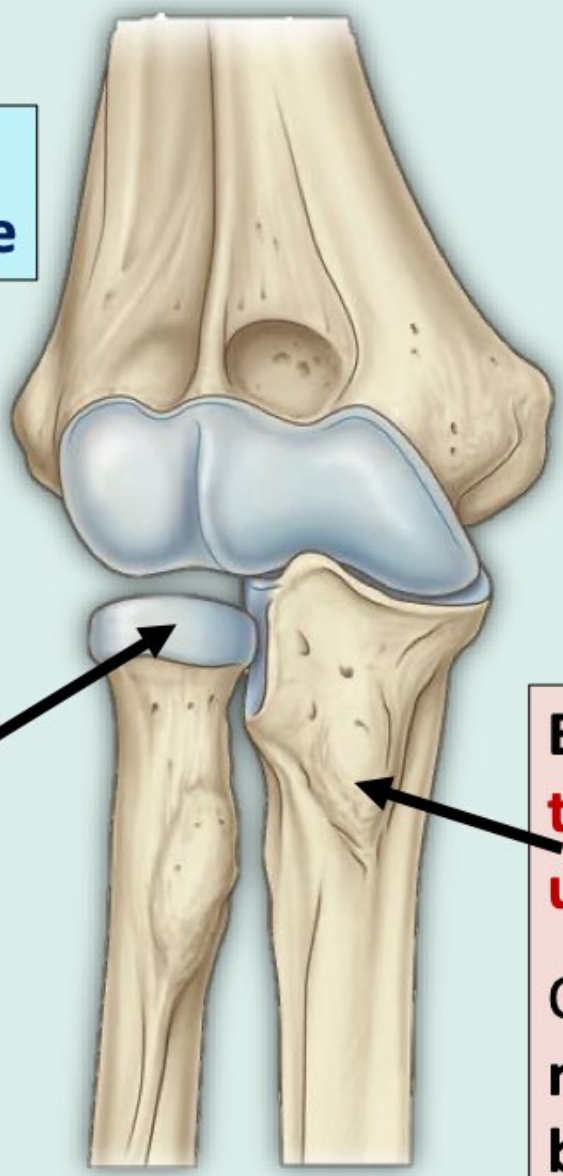
En haut

partie inférieure de l'**incisure trochléaire**

En dehors

Incisure radiale de l'ulna

Articulaire avec: **circonférence articulaire radiale**



En avant

En avant: tubérosité ulnaire

Où s'insère: **muscle brachial**

Mise en place

Diaphyse

Epiphyse proximale

Epiphyse distale



Présente à décrire:

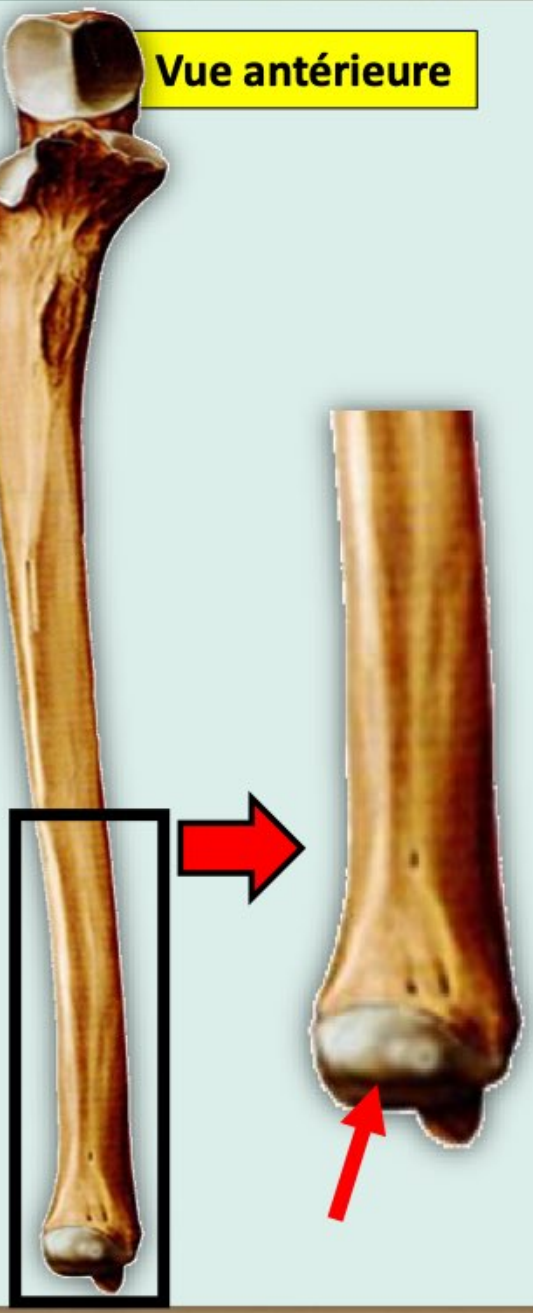


Tête ulnaire

Processus styloïde ulnaire

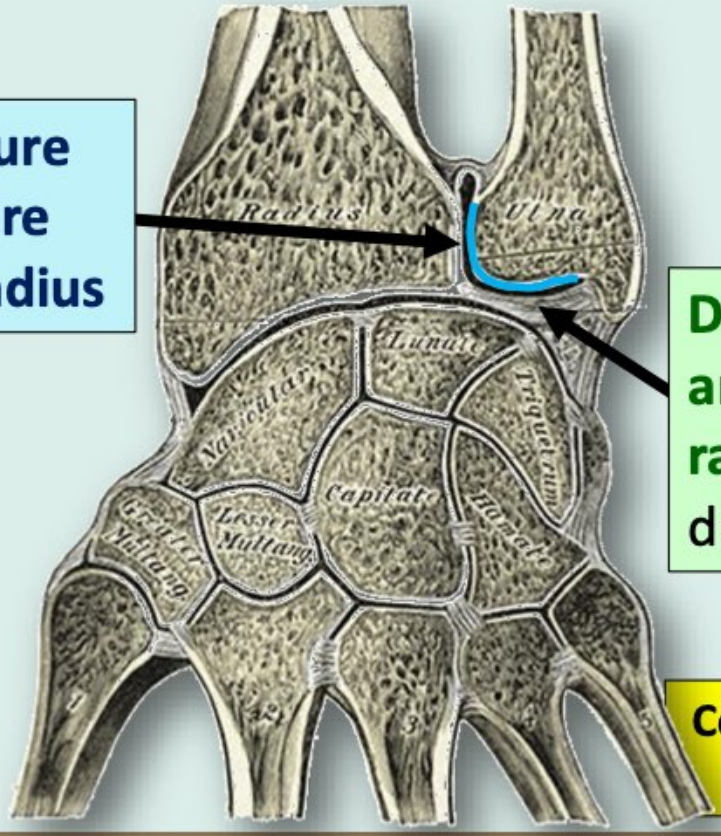
1 Tête ulnaire

Vue antérieure



S'articule avec:

Incisure ulnaire du radius



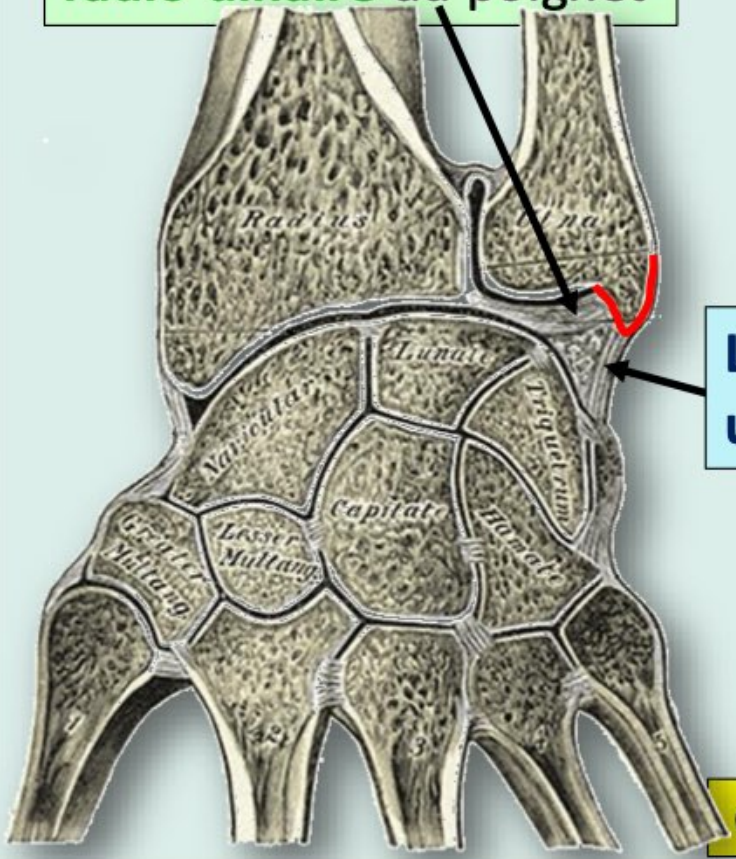
Disque articulaire radio-ulnaire du poignet

Coupe frontale du poignet

2 Processus styloïde ulnaire

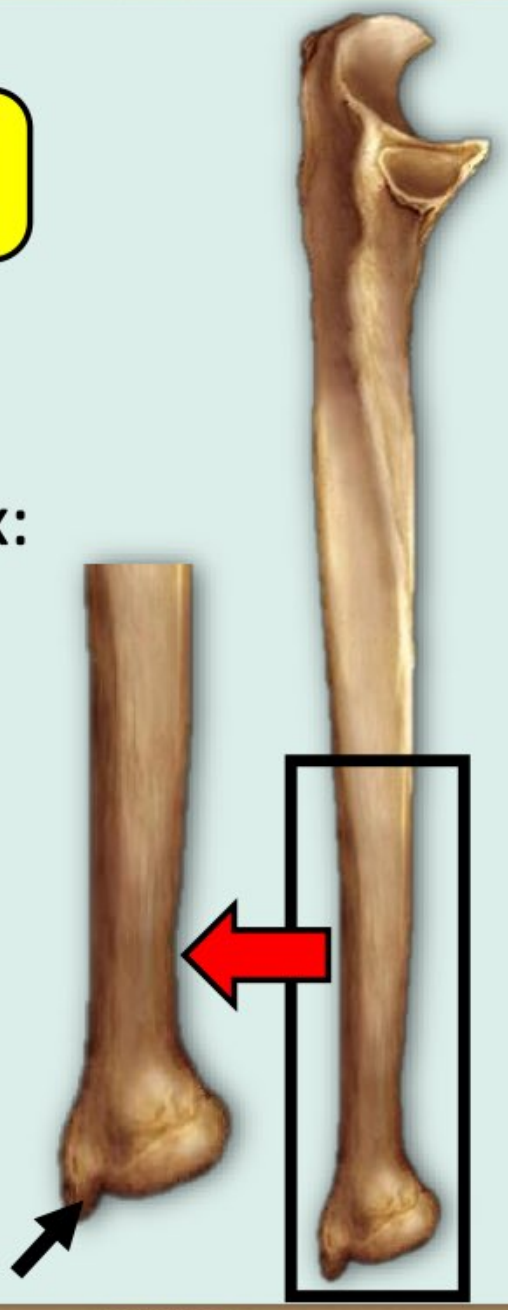
- Petite saillie postéro-médiale
- Donne attache aux:

Disque articulaire radio-ulnaire du poignet



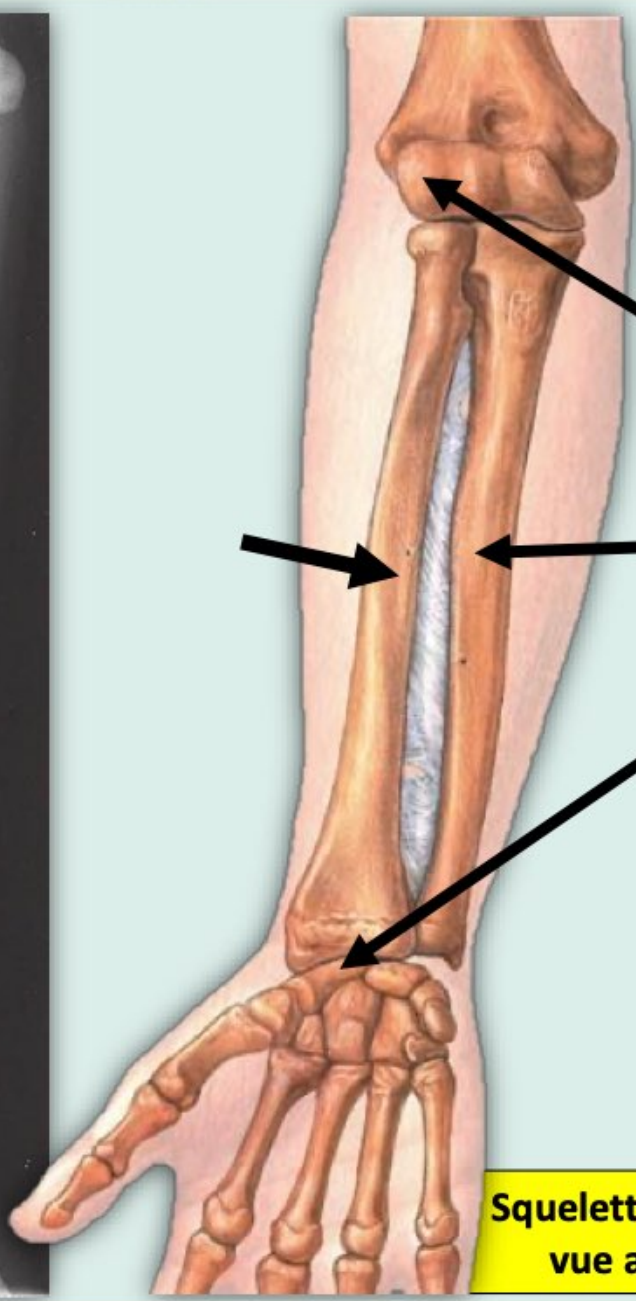
Ligament collatéral ulnaire du carpe

Coupe frontale du poignet



(2) Radius





- **Os latéral** de l'avant-bras
- **S'articule** avec :
  - En haut, **capitulum**
  - En dedans, **ulna**
  - En bas, **carpe**

Squelette avant-bras, vue antérieure

Mise en place

Diaphyse

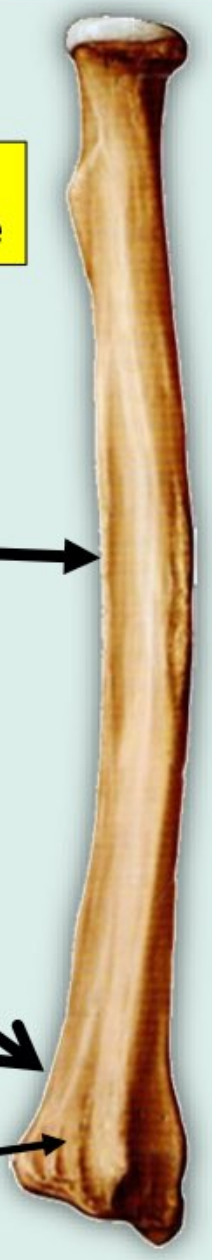
Epiphyse proximale

Epiphyse distale

Ht

DH

Radius droit, vue postérieure



En dedans:
bord le plus tranchant de l'os

En bas:
extrémité la plus volumineuse

En arrière:
face de cette extrémité **creusée de sillons**

Mise en place

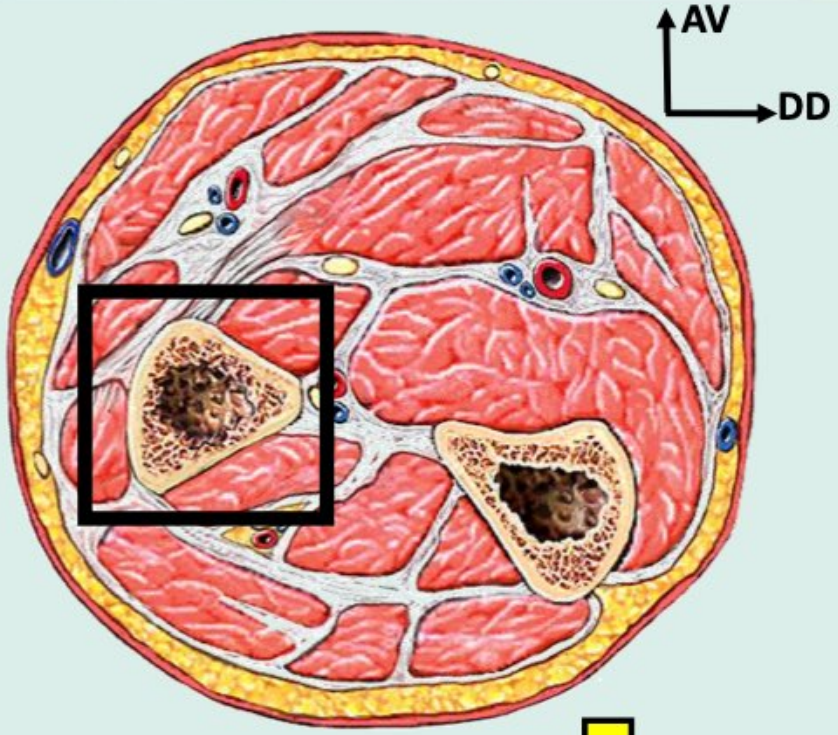
Diaphyse

Epiphyse proximale

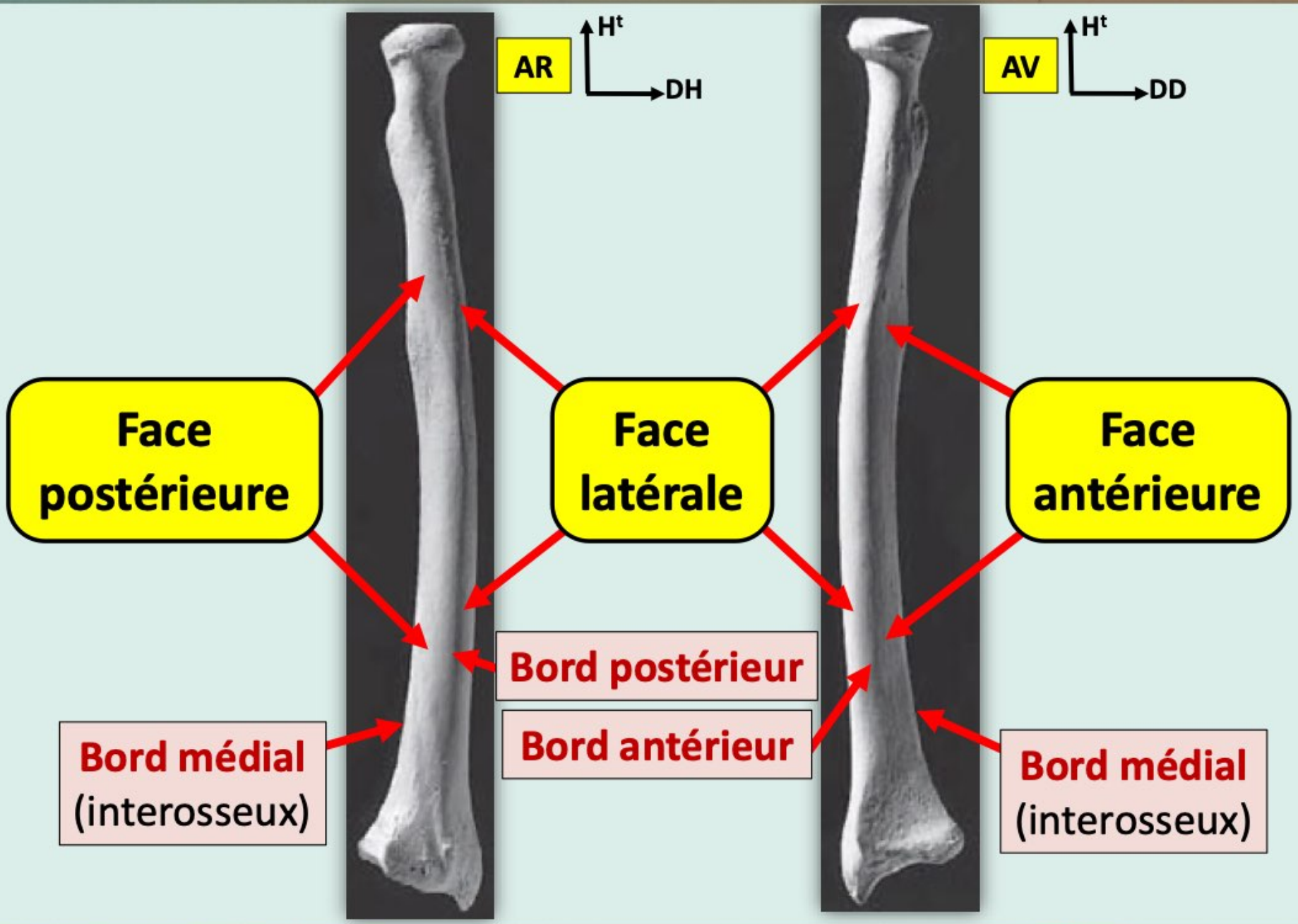
Epiphyse distale

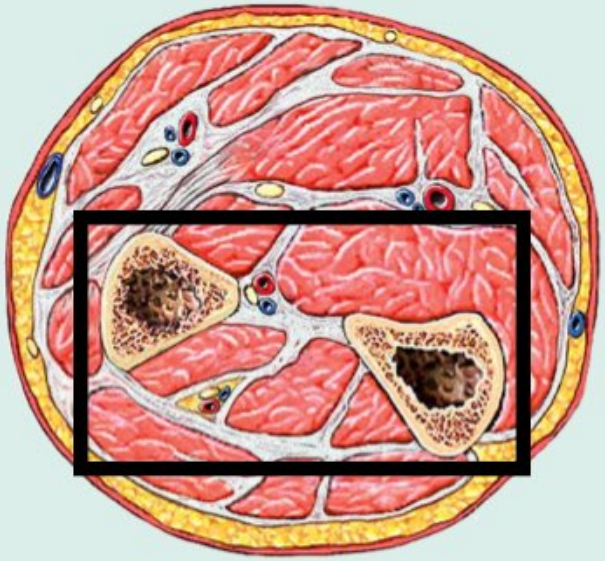
Présente à décrire:

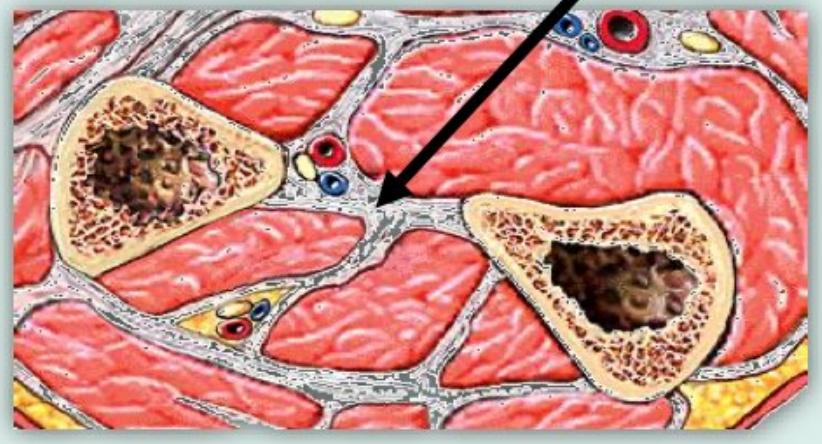
- 3 faces
- 3 bords






AR
Ht
DH
AV
Ht
DD
Face postérieure
Face latérale
Face antérieure
Bord postérieur
Bord antérieur
Bord médial (interosseux)
Bord médial (interosseux)





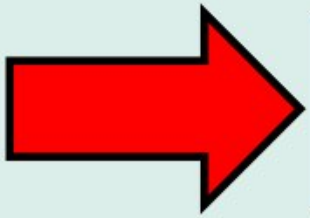
Bord médial
(interosseux)

Où se fixe:
membrane interosseuse antébrachiale



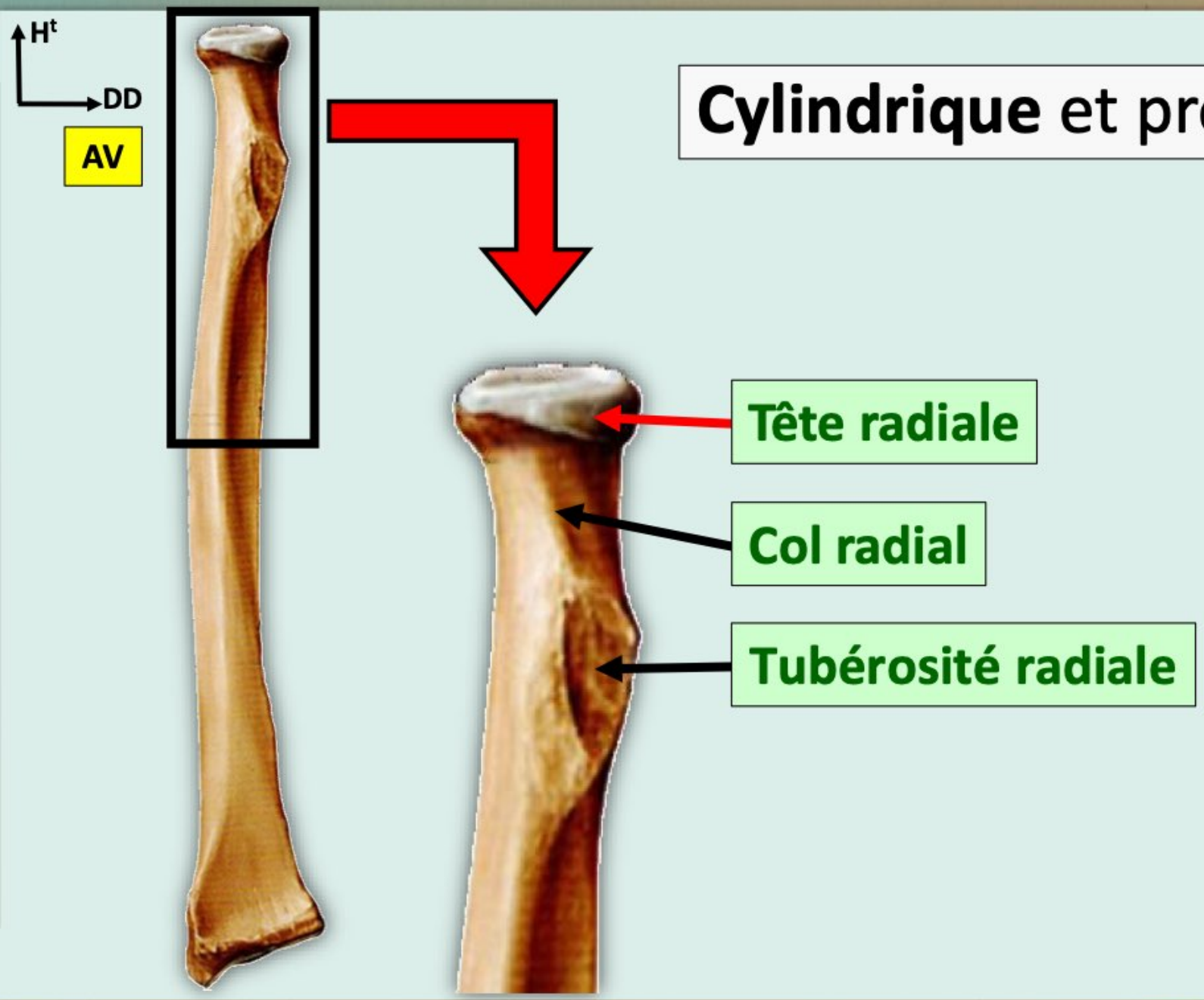
Mise en place

Diaphyse



Epiphyse proximale

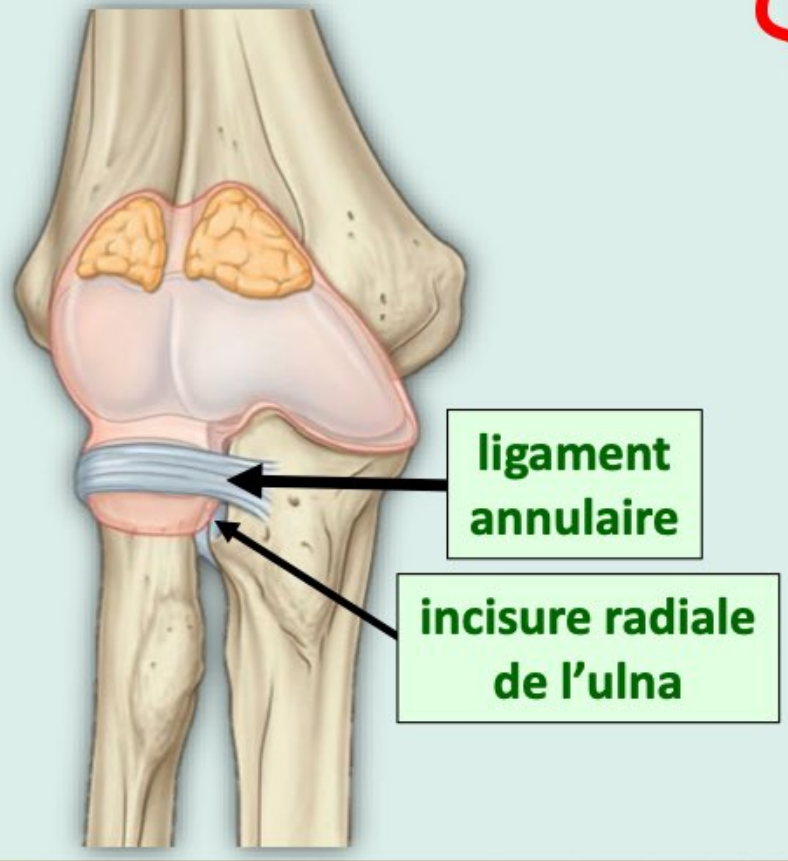
Epiphyse distale


Ht
DD
AV
Cylindrique et présente:
Tête radiale
Col radial
Tubérosité radiale

1- Tête radiale

Circonférence articulaire radiale

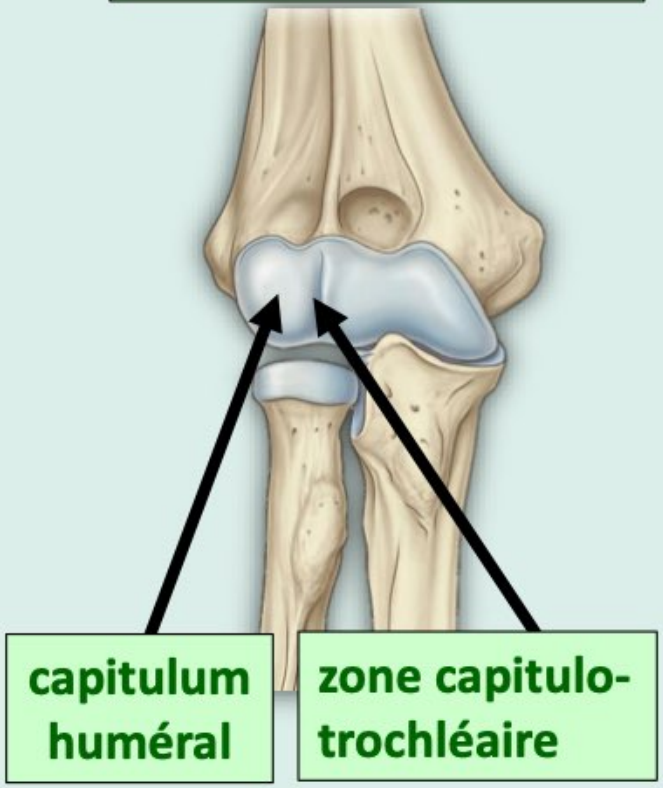
Périphérique, s'articule avec:





Fossette articulaire radiale

Supérieure et **concave**
S'articule avec:



2- Col radial

3- Tubérosité radiale



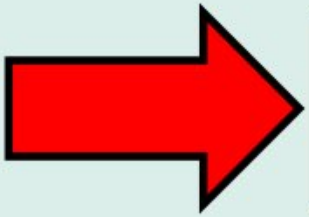
- Saillie ovoïde
- Où se termine: **muscle biceps brachial**



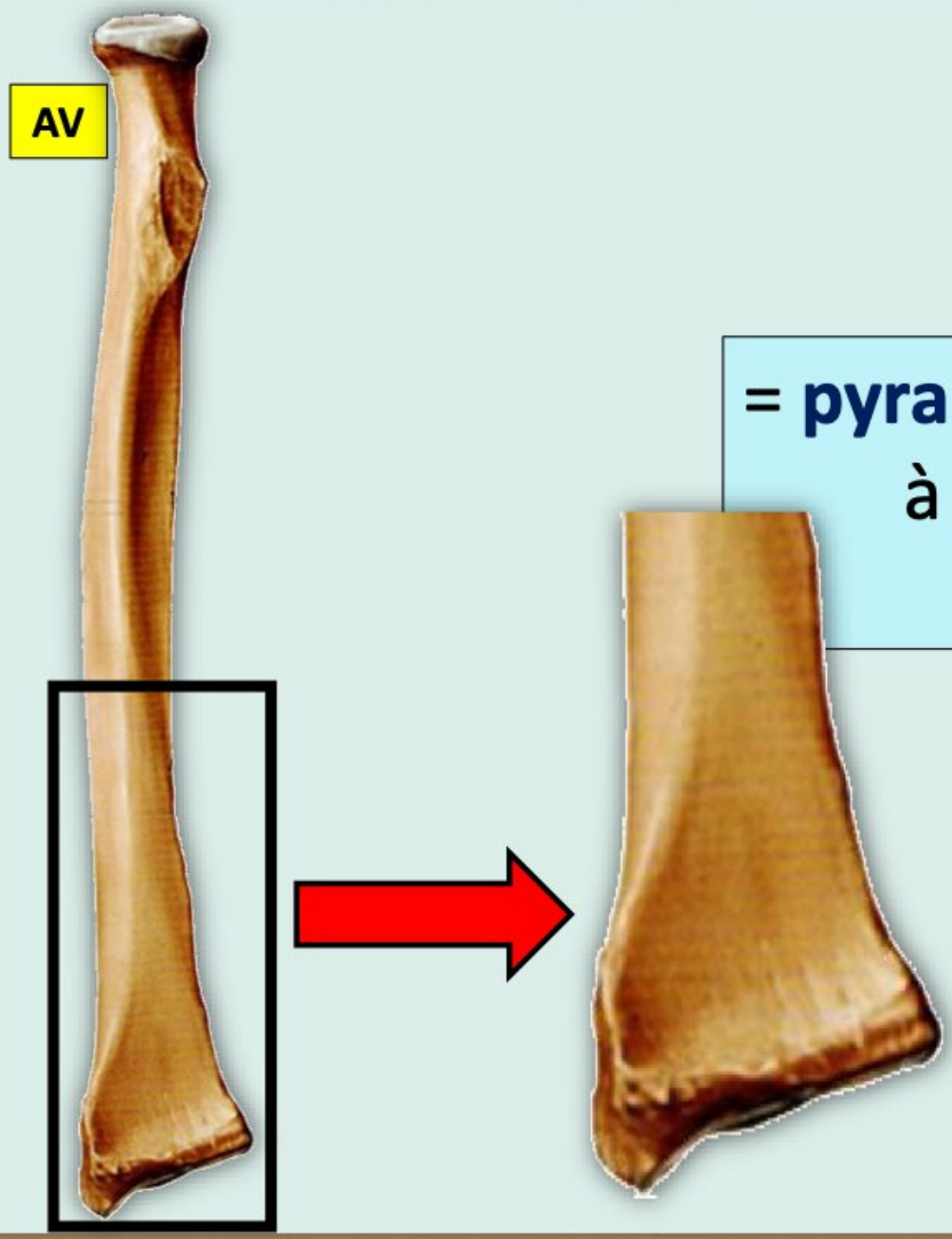
Mise en place

Diaphyse

Epiphyse proximale



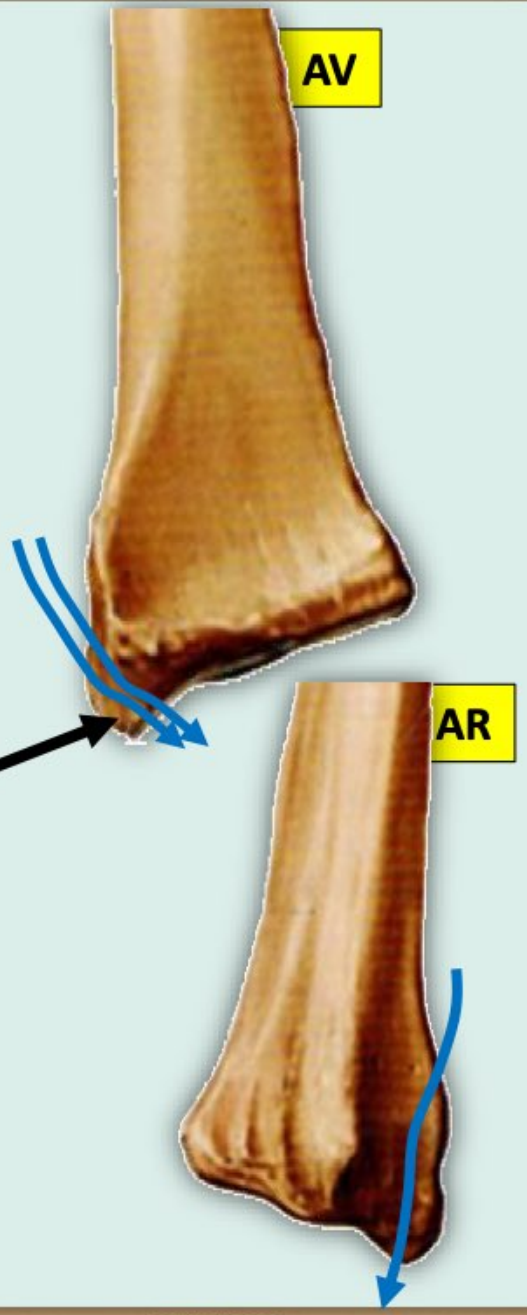
Epiphyse distale



= **pyramide quadrangulaire**
à **base inférieure,**
avec **4 faces…**

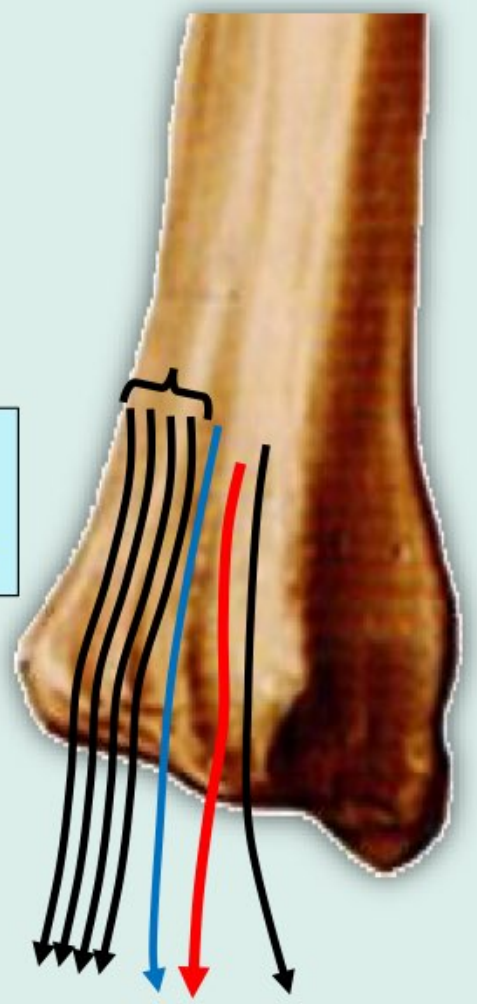
Sa face latérale:

- Se prolonge **en bas** par: **processus styloïde radial**
- Ce processus délimité par: **2 sillons** où passent: **3 tendons musculaires**



Sa face postérieure:

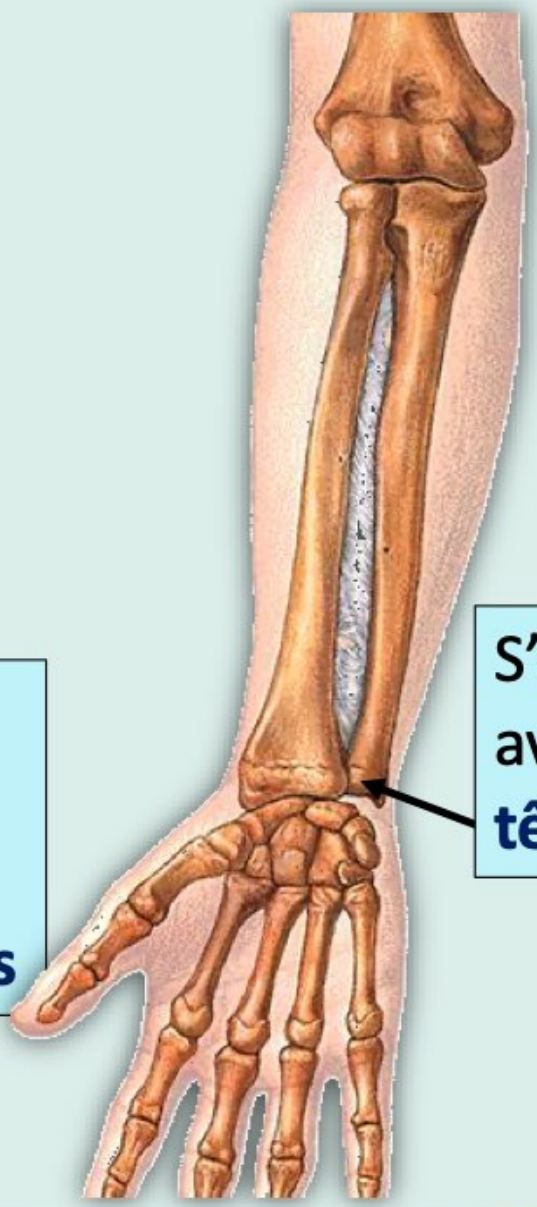
Creusée de: **2 sillons**
où passent: **7 tendons musculaires**



Sa face médiale:

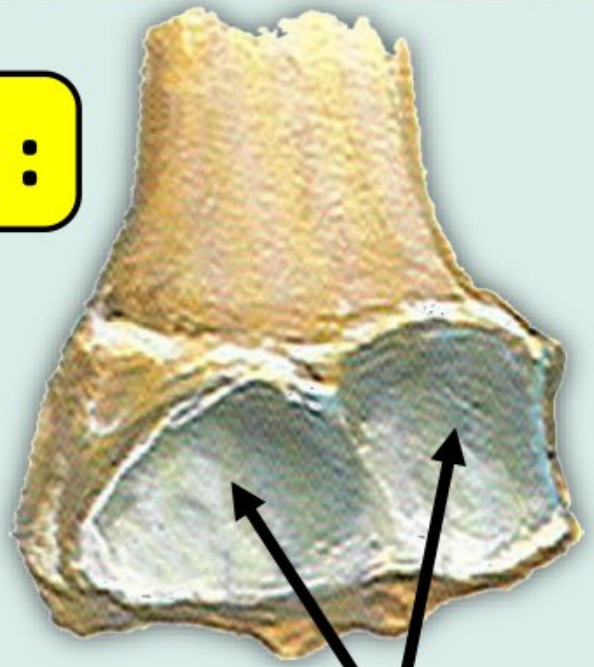


Porte:
incisure ulnaire du radius



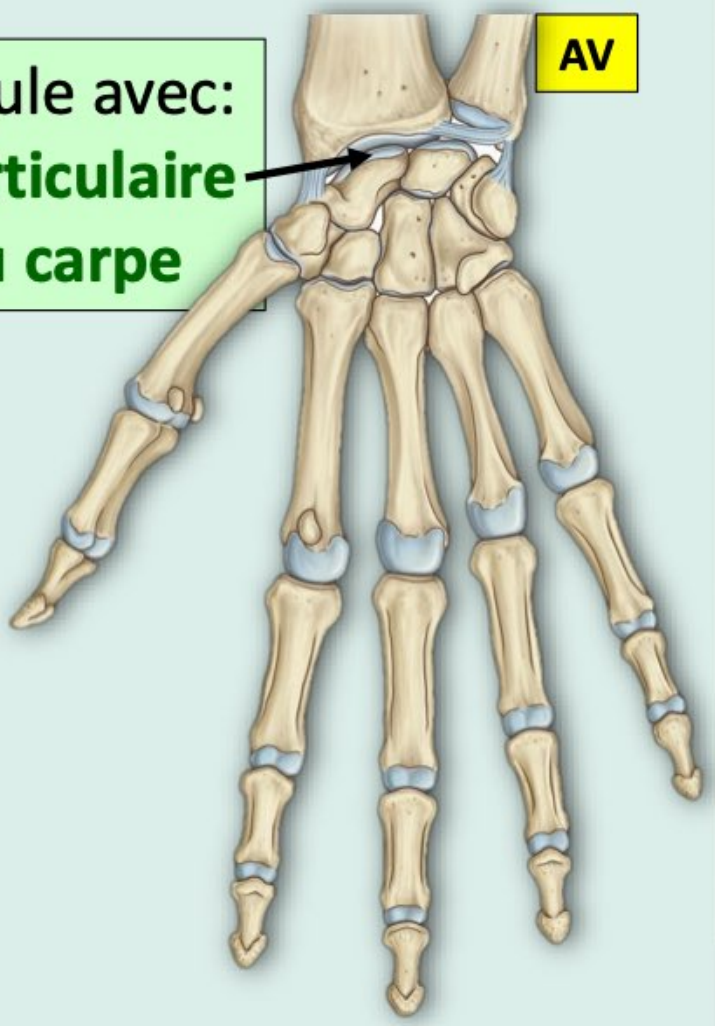
S'articule avec:
tête ulnaire

Base :



Porte: **surface articulaire carpienne**

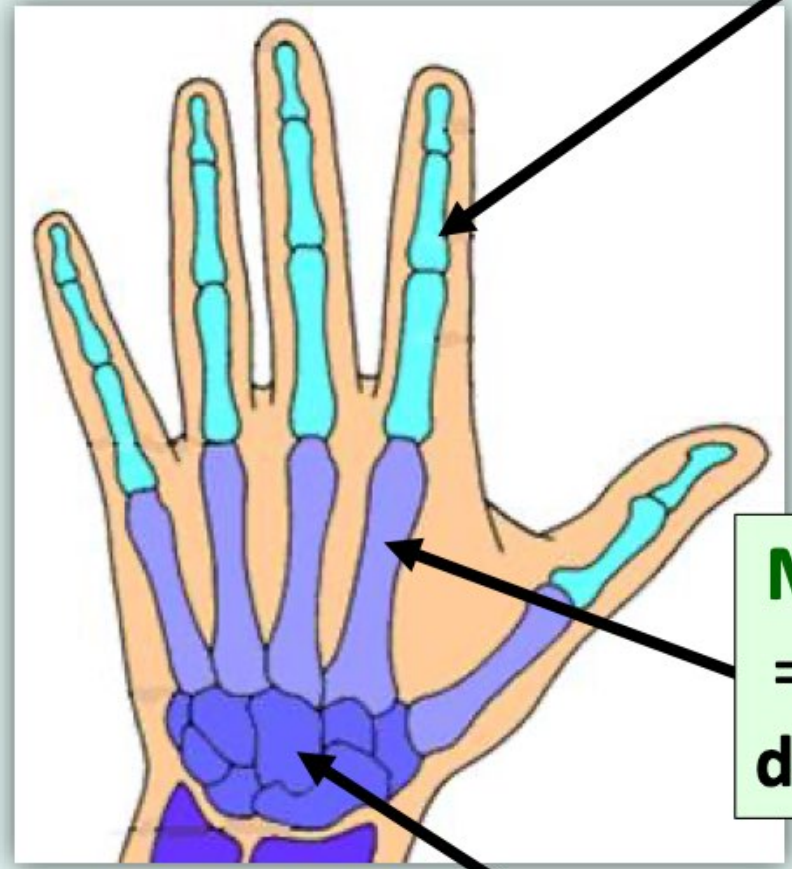
qui s'articule avec: **surface articulaire radiale du carpe**

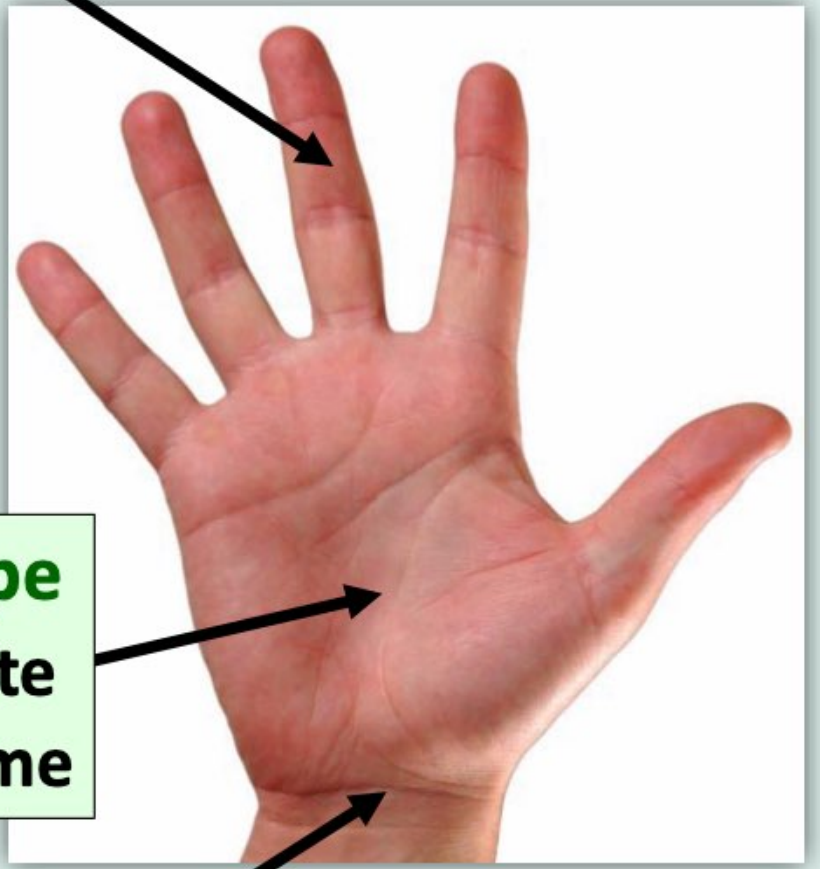


Squelette de la main

Formé de 3 parties …

Phalanges
= squelette des doigts





Métacarpe
= squelette de la paume

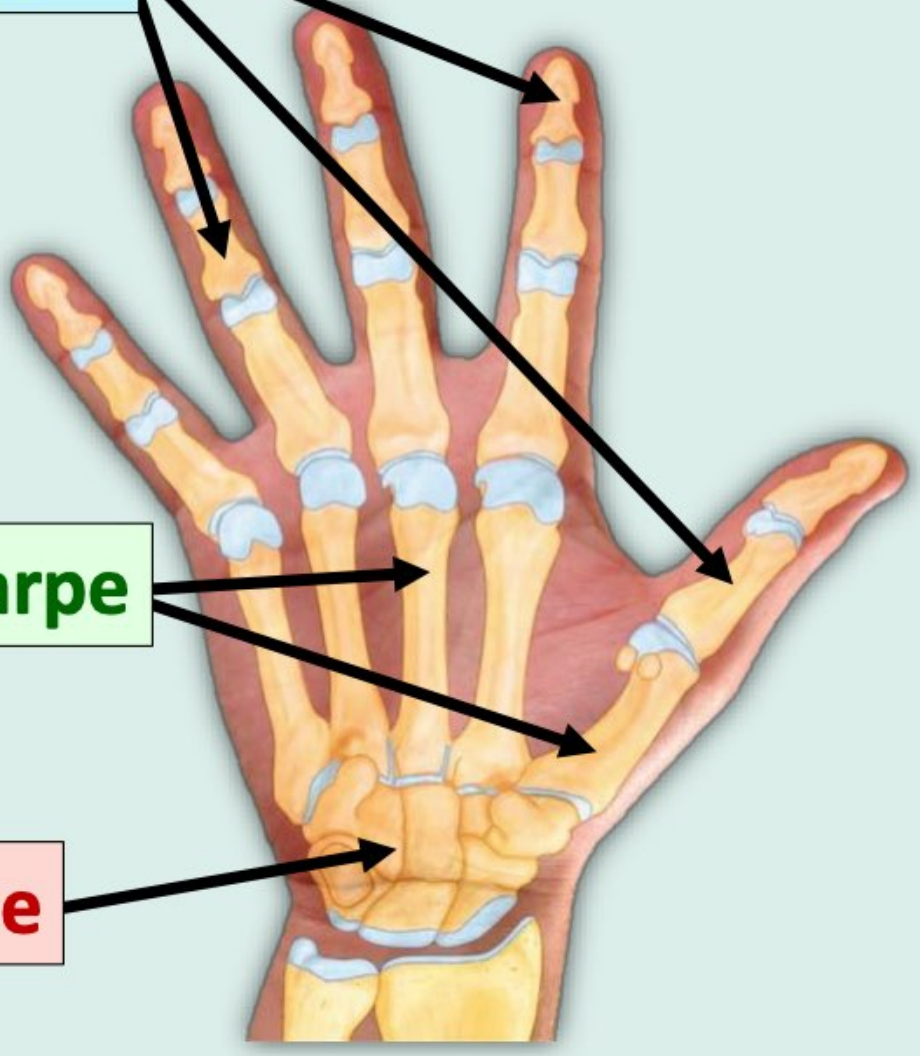
Carpe = poignet



Phalanges

Métacarpe

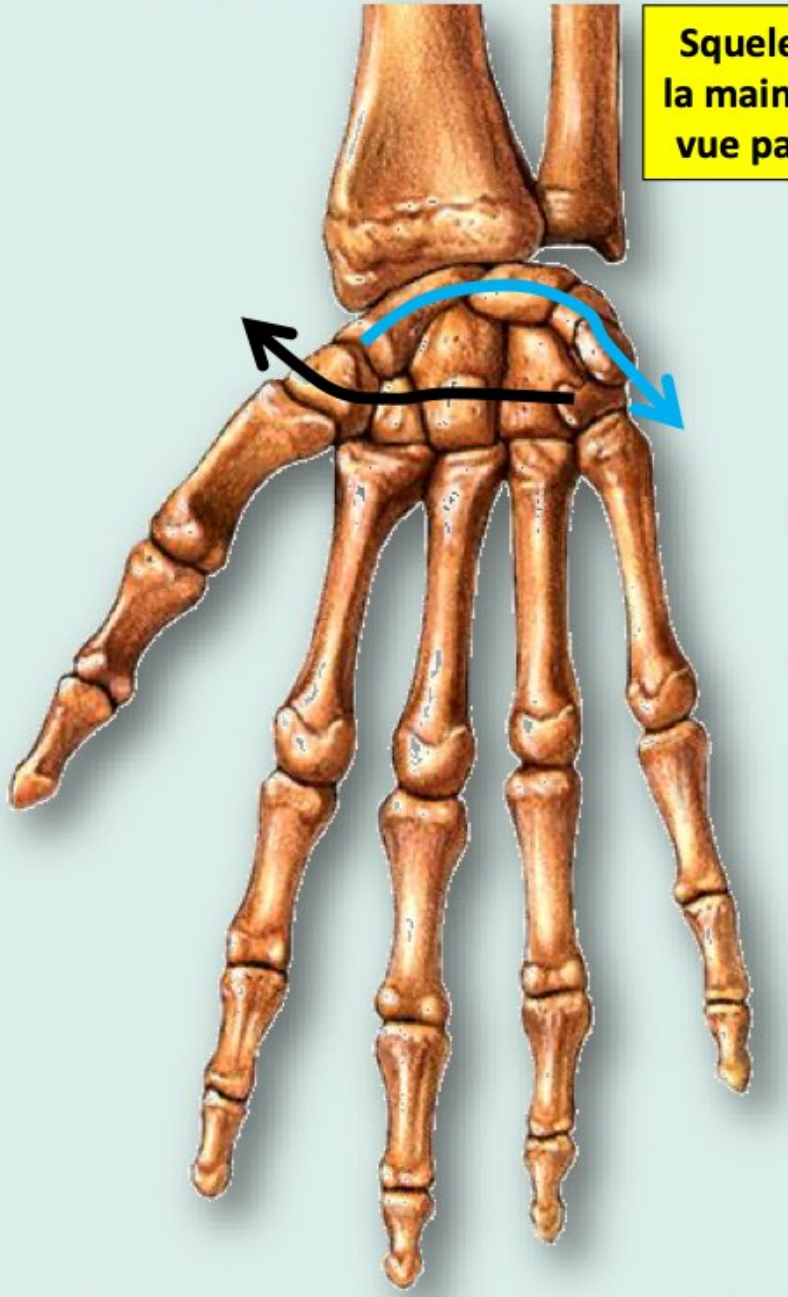
Carpe



Carpe

Métacarpe

Phalanges

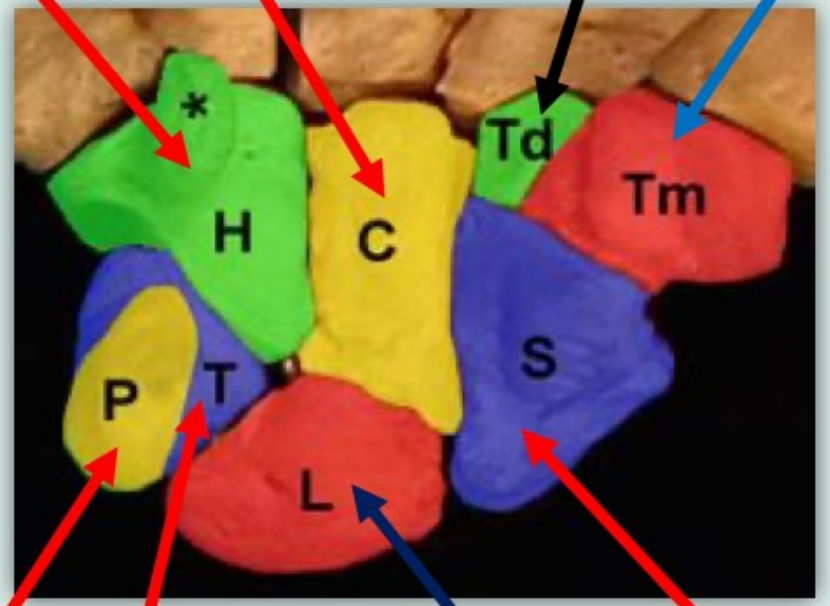


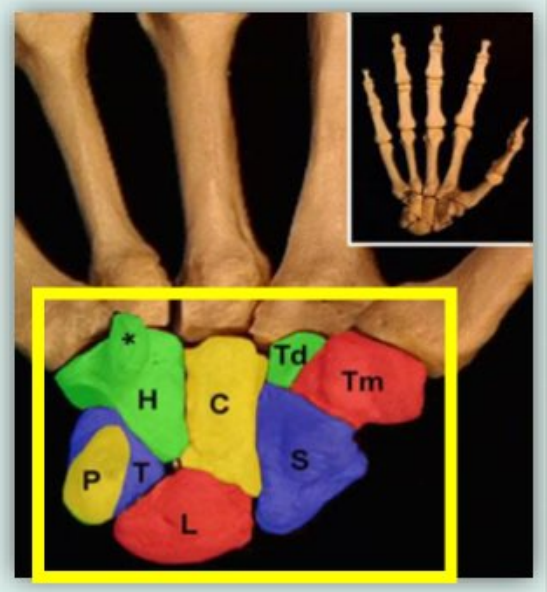
Squelette de la main droite, vue palmaire

- Formé de: **8 os**
- Disposés en: **2 rangées**

Rangée distale

Hamatum | Capitatum | Trapézoïde | Trapèze



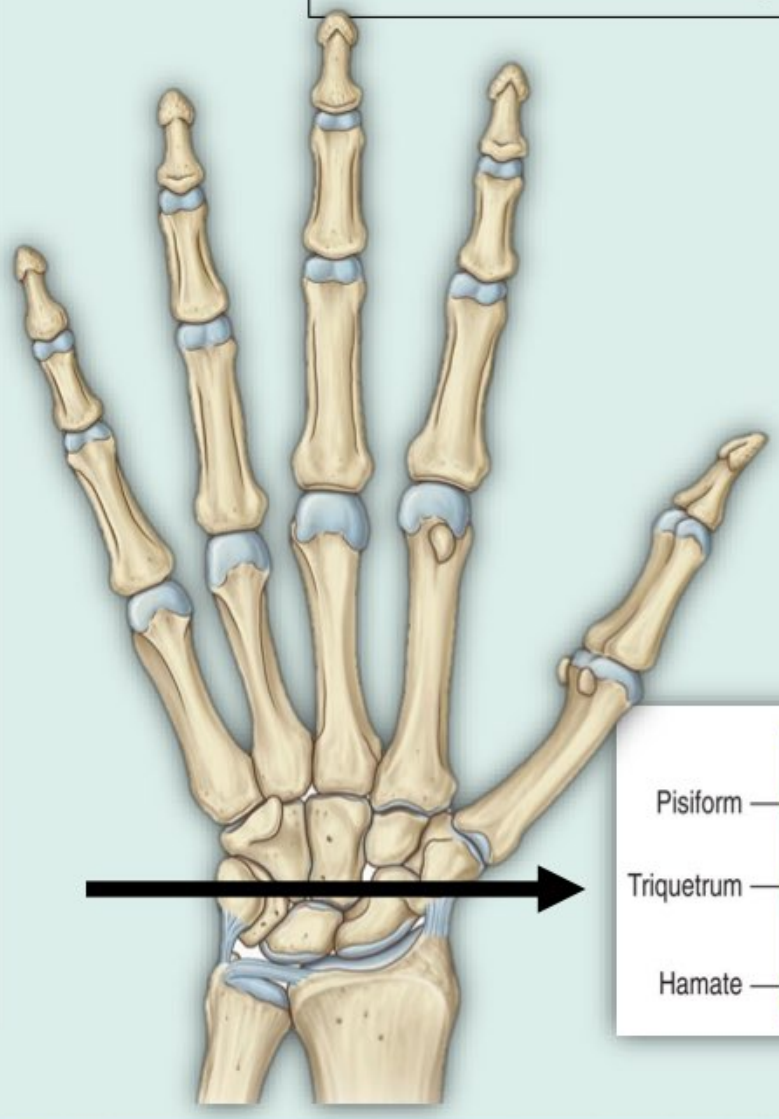


➔de dehors en dedans:

Pisiforme | Triquétrum | Lunatum | Scaphoïde

Rangée proximale

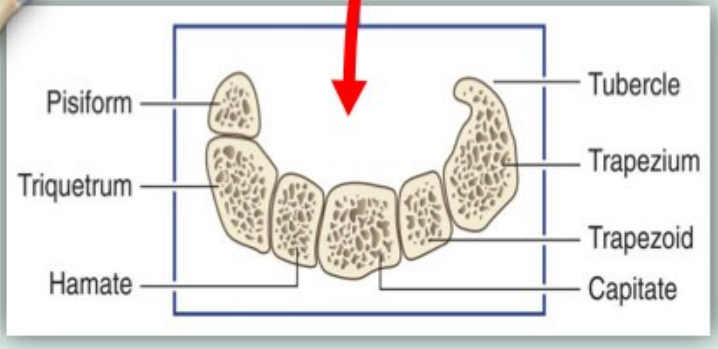
Remarque: Canal carpien



Face antérieure du carpe

est concave, formant

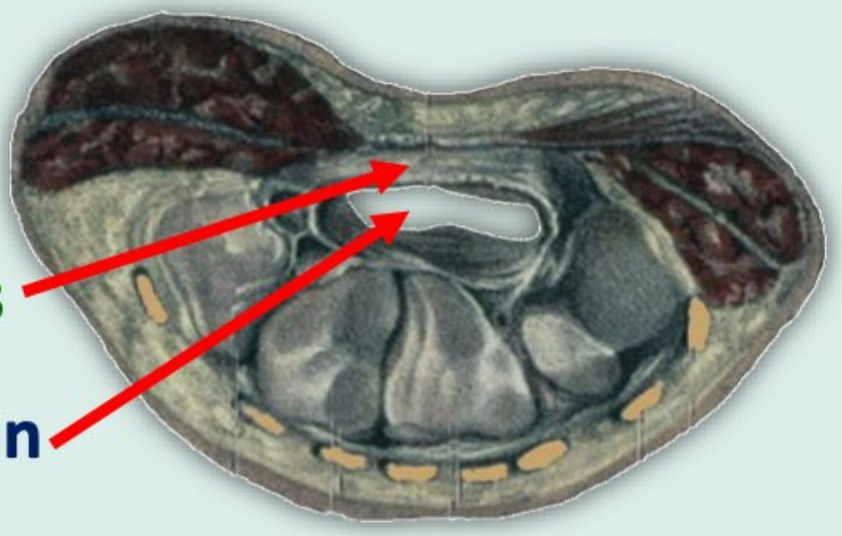
➔ **sillon carpien**

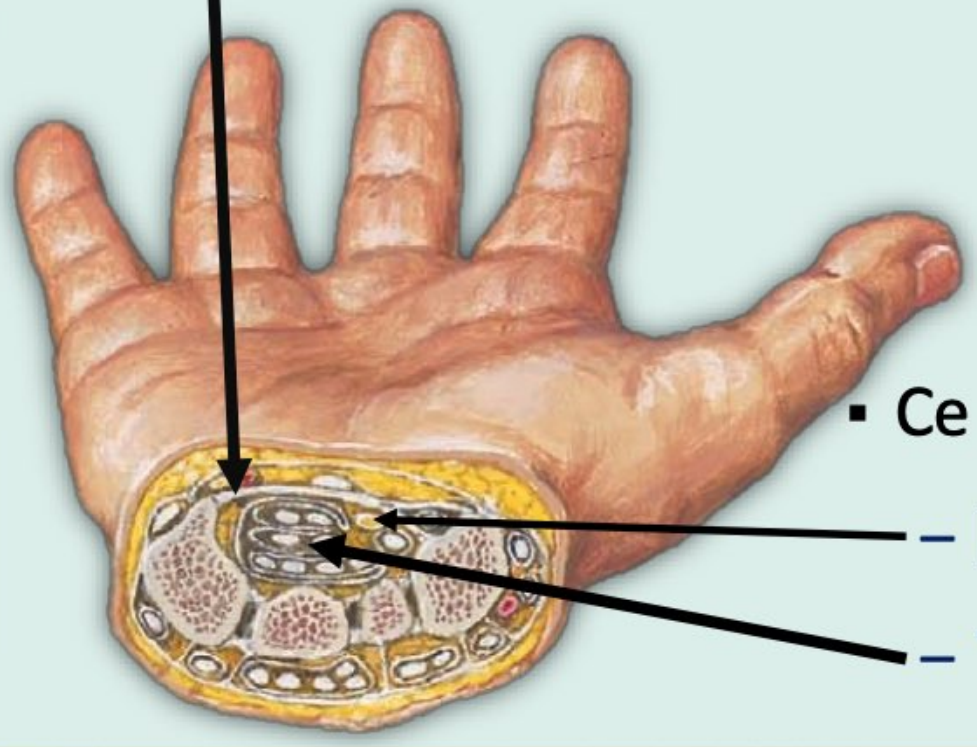




Sillon carpien complété par:
rétinaculum des fléchisseurs

➔ formant: **Canal carpien**





- Ce canal livre passage aux:
  - **Nerf médian**
  - **Tendons musculaires**

Carpe

Métacarpe

Phalanges

- Composé de:
  5 os métacarpiens

- Numérotés
  de « **I** » à « **V** »,
  en partant **de dehors**





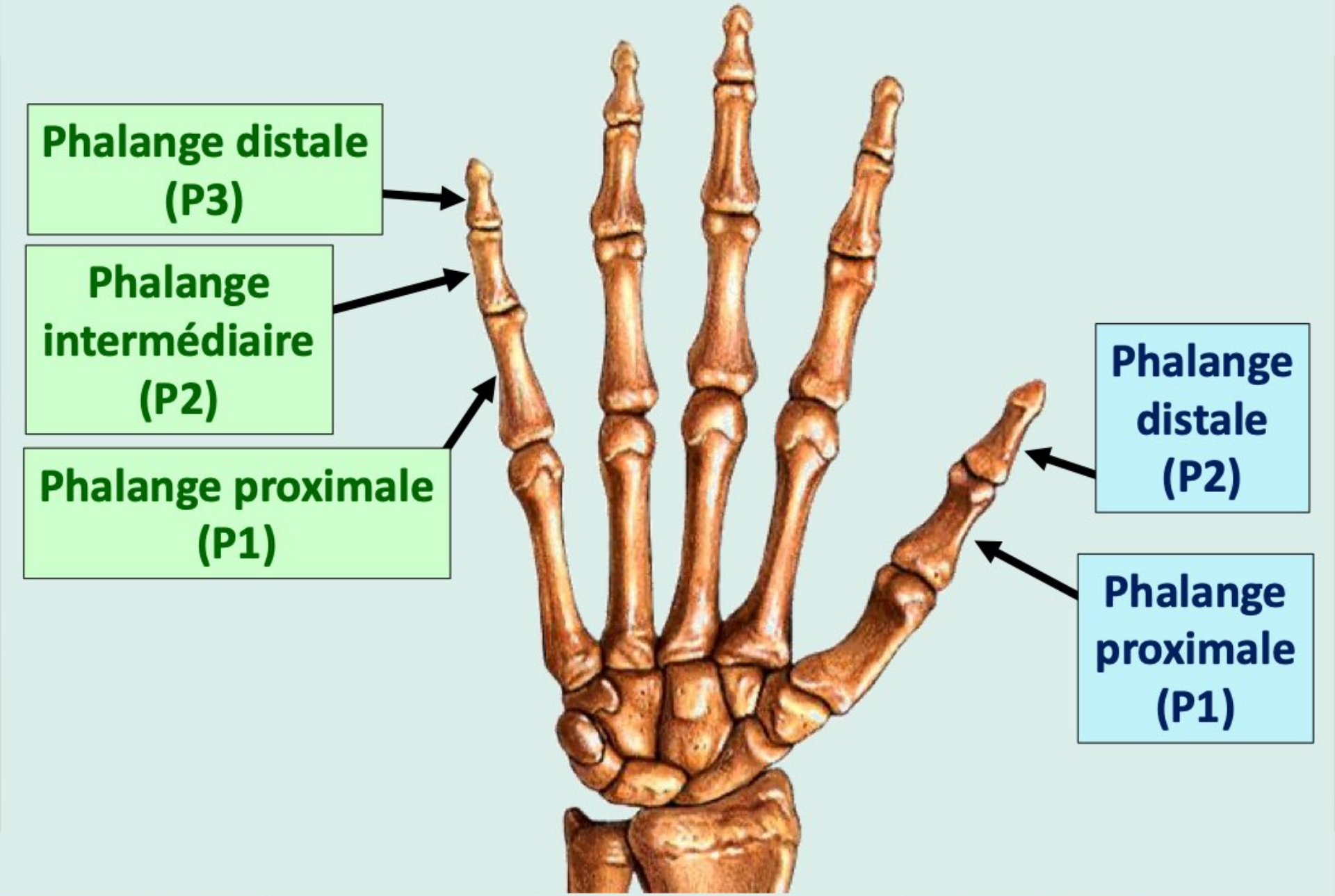
Carpe

Métacarpe

Phalanges

- Nombre = **14**
- **3 phalanges** / **doigt**
- **2 phalanges** / **pouce**




Phalange distale (P3)
Phalange intermédiaire (P2)
Phalange proximale (P1)
Phalange distale (P2)
Phalange proximale (P1)

3 remarques

(1) Os sésamoïdes

= **osselets** situés au niveau de la **face palmaire des articulations digitales**



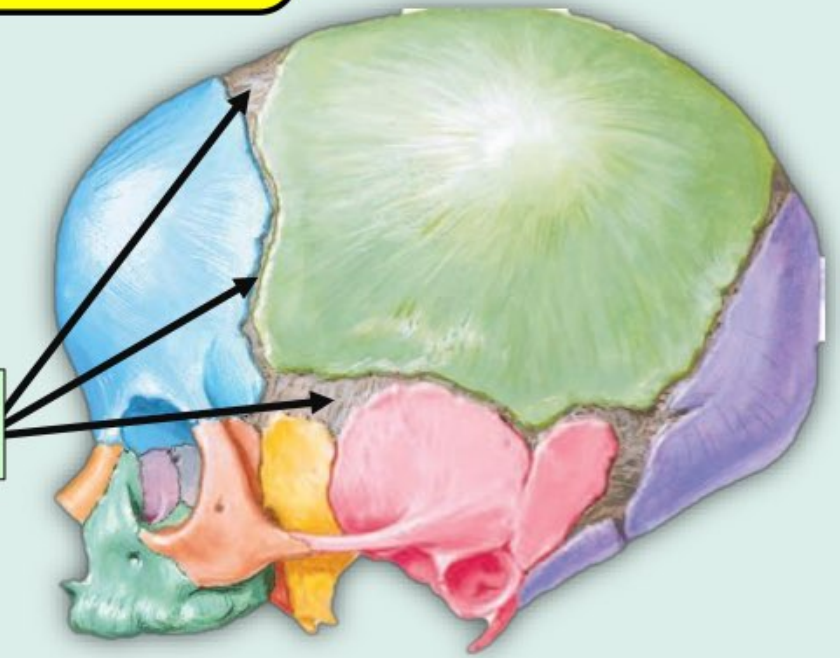
(2) Développement des os

Par **ossification**:



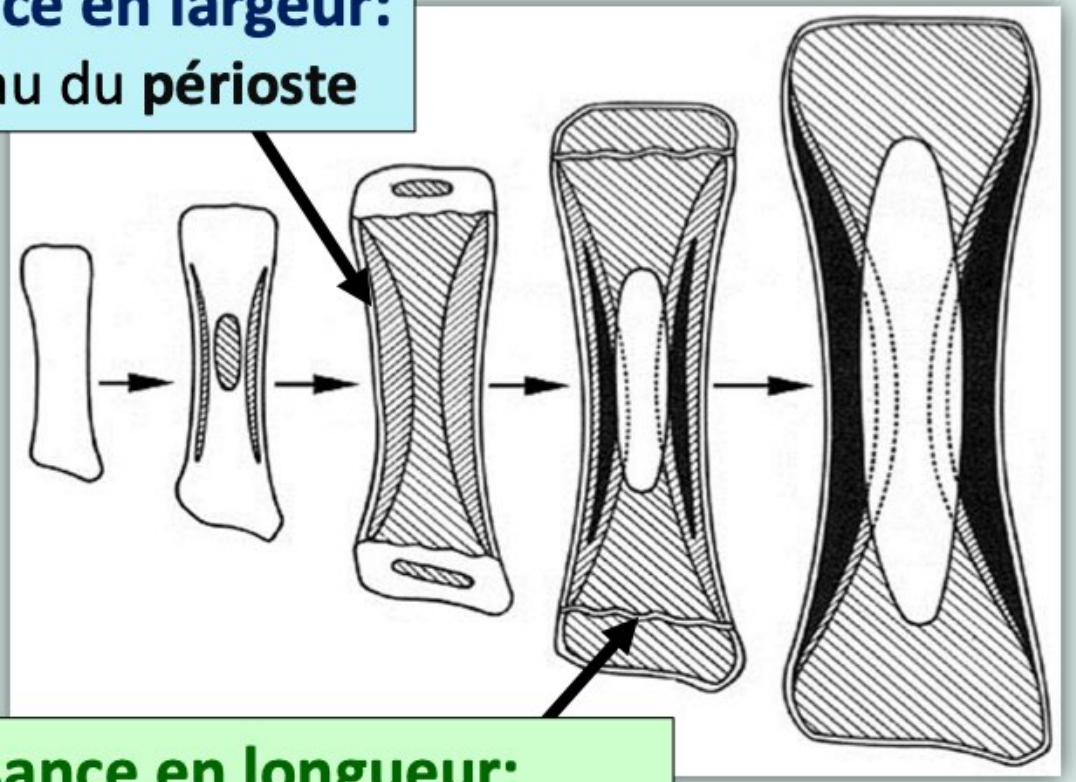
Cartilagineuse

Membraneuse



Après la naissance, l'os subit **2 types de croissance:**

Croissance en largeur:
au niveau du **périoste**



Croissance en longueur:
au niveau du **cartilage de croissance**
(ou de conjugaison)



Cartilage de conjugaison (flèches)

(3) Age osseux (chez enfant et adolescent)

- Apprécié sur: **radiographie du poignet et de la main gauches**
- En rapport avec: **ossification des os du carpe** et **des épiphyses des os longs du poignet et de la main**



Nouveau-né	3 mois	3 ans	8 ans	18 ans


Bonne
révision